DESIGN BIBLE AND TECHNICAL ARCHITECTURE

The Mansion

A NeoForge 1.21.1 interpretation of Kane Parsons' Complex



Version 0.8, 4 October 2026. Rebuilt to fit the project's standing rules
Source of truth for canon is The Complex Codex
By UnderwaterMarch. Unofficial fan work, not affiliated with Kane Parsons, A24 or Async Pictures

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How to Read This Document

If you're here to build the mod, start with chapter 0. It's written for someone with no other context, and together with Part E it holds everything needed to build the Mansion as intended.

This is the blueprint the mod will be built from. Every mechanic in it traces back to something the Codex records as canon, to one of the ten laws of the Codex's working theory, to one of your standing rules for the project, or to a design decision stated openly as ours. If a feature can't name its source, it doesn't ship.

Version 0.1 of this bible was a strong first draft, but it broke several of your rules and leaned on hand-made content. This version keeps its engine where the engine was sound, removes everything authored, removes every feature that was decoration rather than law, fits all storage inside level.dat and chunk NBT, and reorders the work so that the question every earlier attempt failed on, whether the place feels like the Complex, is answered first. Appendix F lists every change.

The labels

Each claim carries a label saying where it comes from.

Label	Meaning
[MUST]	The human-made mandate. Above every law and every other rule, always
[CANON]	Shown in the web series or the film, or said by Kane, as recorded in the Codex
[LAW]	Follows from one of the ten laws of the Codex's working theory. The law's number is given in the text
[THEORY]	A theory the Codex rates open. Used only where it contradicts nothing, and only so that removing it later would break nothing
[YOUR RULE]	One of your standing rules for the project. Rules beat design decisions every time
[DESIGN]	Our own decision where canon is silent, with the reason
[TECH]	A fact or limit of Minecraft 1.21.1 or NeoForge 21.1
[RISK]	Something that could fail technically and needs a prototype before anything depends on it

The single sentence that governs everything

[CANON] Kane describes the Complex as a domain that "recognizes, interprets, and stores" the physical structure of our world. The Mansion is that sentence turned into software. The Cache recognises and stores. The Remembering Engine interprets. Genesis lets the result grow. Every other system serves those three.

The single rule that governs the build

[DESIGN] Feel before infrastructure. Every version before this one was built whole and then judged whole, and the verdict was always the same, that it isn't the Complex. That verdict can't be acted on. So this bible starts with a small slice that can be judged against canon frames in an afternoon, and nothing else is built until that slice feels right. Chapter 32 sets out the order.

What changed since version 0.1, in short

- **NeoForge stays.** The mod targets NeoForge 1.21.1, with NeoForge as its only dependency.
- **No extra files.** Everything is stored in level.dat and in chunk NBT. No cache folder, no journal, no server config file.
- **Nothing authored.** No Founder rooms, no Canon Sites, no procedural building generators. Every room in the Mansion is a remembered copy of something in your world. The only fixed input is a set of numbers taken from canon, which is configurable down to zero.
- **No paging and no stateless chunks.** The dimension is one ordinary, saved level. Infinite vertical travel moves to a research track.
- **Decorations removed.** No compass wobble, stick test or torch shimmer. No aurora. No scripted sixth test. No voice recording.
- **Budgets tightened.** Display entities cut by a factor of ten, lighting treated as the main cost.
- **Feel first.** The first milestone is a slice judged against canon frames, not infrastructure.

New in version 0.3

- **Sky exposure.** Whatever faced the sky is streamlined, whatever was enclosed is faithful, measured from Minecraft's own sky light (chapter 9). Darkness doesn't matter, only the sky.
- **Outdoor places kept rare.** Openings form only inside, the sky drains the capture, outdoor parts live only on fresh islands, and a test fails the build above 2 per cent.
- **Re-expression.** Places are rebuilt in the Mansion's image from their own rooms, materials and objects, from generation 1 (operator R10).
- **Glass passes the capture, and lights keep their captured state.** From the three copies of Clark's store.
- **Reference frames.** Canon frames, one Minecraft build and a fan project, included for reference only.

New in version 0.4

- **The light of the moment.** An outdoor copy's sunlight is replaced by fittings that keep its colour and direction at capture. Midday gives panels, dusk gives red floodlights low on one side under stars (chapter 10).
- **Real floodlights.** Banks of lamps, vanilla's invisible light blocks for the throw, and a tint for the colour.
- **Coloured lamps.** A planned module with your crafting recipe, no new block, and exact coloured light (chapter 34).

New in version 0.5

- **Coloured lamps, detachable.** Built as their own mod inside the Mansion's jar, with their own namespace and one small API, so they can be published alone at any time without touching saves (chapter 34).

New in version 0.6

- **A stricter outdoor test.** All former outdoor space counts, even under a plain ceiling.
- **Rule O5.** Re-expression turns yards and streets into rooms with the ground as their floor, like Mary's yard in canon.

New in version 0.7

- **Substitutes that fit.** Same shape, inert, closest colour, then the flattest texture, so a lawn becomes flat green concrete, never a carpet that changes the floor's height (chapter 10).
- **Monotony and mods.** A floor that would match its walls takes the next close block from stock, installed mods or vanilla, or merges if nothing fits.
- **Removed mods leave no holes.** Fingerprints in level.dat let the Mansion replace a removed mod's blocks before chunks load (chapter 6).
- **Copies keep their biome colours.** Grass, leaves and water keep the colour of where they were captured.

New in version 0.8

- **The mandate.** Everything genuinely human-made, no AI anywhere, no comments or notes in code, nothing vibe-coded. Above every law, enforced by the build (chapters 0, 35 and 36).
- **Start Here.** Chapter 0 lets someone with no other context build the mod, with canon in brief, every rule, the history of failures and where everything is.
- **Part E.** The verified toolchain and build files, the data formats byte by byte, every algorithm with every number, the NeoForge entry points checked against the sources, and step-by-step milestones with stop points.
- **Corrections.** R1's numbers, the M0 floodlight tint, lamps keeping their state, determinism and storage, each found while checking the bible against the game.

0. Start Here

This chapter is for whoever picks up this bible to build the mod, starting cold, with no memory of the conversations that produced it. Read it whole before anything else. Everything a builder needs is in this document. Where it points to the Codex or to folders on the project machine, it says exactly where, and what to do if they can't be reached.

The mandate above every law

> **MUST. Everything is genuinely human-made**
>
> This is the first rule of the project and it outranks everything else in this bible, every law, every canon fact, every design decision and every deadline. It is never relaxed, never traded away, never "fixed later". Work that breaks it isn't finished work, however well it runs.
>
> - **No AI anywhere.** Nothing in the project may show any sign of having been produced by AI or by a code generator. Not in the code, the build files, the resources, the docs, the README, the commit messages, the mod metadata, the in-game text, the images or the file metadata. No tool credits, no co-author trailers, no "generated with" lines, no assistant phrasing, ever.
> - **No comments, no inline notes, no notes of any kind in code.** Not one. No line comments, no block comments, no Javadoc, no TODO, FIXME or XXX, no section banners, no commented-out code, no explanatory string constants standing in for comments, no license headers on source files. This covers main code, tests, the lab harness, mixins, Gradle scripts, mods.toml and every other file that has a comment syntax. The code explains itself through its names and its structure, or it gets rewritten until it does.
> - **It must not read as vibe-coded.** Every file looks like one careful person wrote it by hand, over time, for this project and nothing else. Chapter 36 lists the tells that give generated work away and forbids each one.
> - **Prose reads as a person wrote it.** No em dashes or en dashes, colons only where they're genuinely needed, no stock phrases, no filler, no emoji.
> - **Enforced by the build.** The `checkHuman` task in chapter 35 scans every source and resource file and fails the build on any comment, dash or forbidden tell. It runs on every build and it is never disabled, skipped or weakened. Test T-H.1 asserts it ran clean.
> - **Checked before anything leaves your hands.** Before every report, every commit and every file you hand over, read it as the user would and look for anything a person wouldn't have written. If you find one, fix it first.

What you're being asked to build

A Minecraft mod called The Mansion, for NeoForge 1.21.1, by UnderwaterMarch. It adds one dimension that behaves the way the Complex behaves in Kane Parsons' Backrooms. The dimension isn't designed. It's grown by a set of laws from copies of the player's own world, copies that get worse every time they're copied again, until the oldest of them have decayed into something like the classic yellow rooms.

- It adds no blocks, items, textures, models or sound files to the dimension. Everything in it is a vanilla block, another installed mod's block, or a block copied from the world.
- It contains nothing authored. No rooms, buildings, layouts, palettes or templates. The only fixed input is a short list of numbers measured from canon, the founder prior, and even that can be turned off.
- It writes no files of its own. Everything it keeps is in level.dat and in chunk NBT.
- It works in multiplayer, on any kind of world, inside strict performance budgets.
- Its first milestone isn't infrastructure. It's a small slice that the person you're building it for can stand in and judge against frames from canon. Nothing else gets built until that slice feels right.

Read in this order

1. This chapter.
2. Part A, chapters 1 to 4. The vision, the rules and the ten laws.
3. Chapters 8, 9 and 10. Capture, sky exposure and remembering. These are the heart of the mod and the part every earlier attempt got wrong.
4. Part E, chapters 35 to 43, before writing any code. They hold the exact setup, data formats, algorithms with every number, the verified NeoForge entry points, and the step-by-step plan for each milestone.
5. The rest of the bible as each milestone reaches it. Appendix B holds every default number in one place.

The order of authority

When two things disagree, the higher one wins.

0. **The human-made mandate.** Above everything, always, with no exceptions.
1. **The user's standing rules.** Chapter 2 and the list later in this chapter. Nothing overrides them.
2. **Canon, as recorded in the Codex.** If the bible contradicts the Codex, the Codex is right and the bible gets corrected.
3. **The ten laws.** Every mechanic must follow from one. A feature that contradicts a law loses.
4. **This bible's own decisions,** marked DESIGN.
5. **Your judgment,** only where everything above is silent. Write it down as a decision record (chapter 35) and say so in your next report.

Anything labelled LAW or YOUR RULE is never changed without asking first. Anything labelled DESIGN can be tuned, but a change that alters behaviour still gets a decision record and a mention.

The Complex in brief

[CANON] Kane Parsons began his Backrooms web series in 2022, and his feature film, Backrooms, followed from A24 in 2026. His version is one consistent vision, handled with the sober logic of hard science fiction, and it differs sharply from the community wiki's.

- **The place.** The research company Async found it in 1989 and called it the Complex. Their machine for entering it is the Threshold, opened in October 1989. Openings that form by themselves in the real world are null zones. Async's own building, seen from inside, is Standard. Nobody in canon says "noclip", "levels" or "Level 0".
- **How it looks.** Bright, washed out, mediocre and endless. Greyish yellow walls, stained beige carpet, a drop ceiling of speckled tiles, fluorescent fittings that never go out and hum at 120 Hz. It's cold. It goes up as well as along. Everything is connected by doors, gaps, holes, crawl spaces, chutes and ladders.
- **What it does.** It copies real places, physically and in precise detail, then copies its copies. Clark's line in the film is the key to everything. "The more times it remembers something, the less it does." Its errors are consistent. Wrong sizes, repetition, things sunk into floors, stretching, facades, mirrored signs, lost purpose (a door onto brick, a hood with no cooker, stairs with no steps), and things in the wrong context.
- **What lives there.** Only two kinds of creature, completely separate. The Lifeform, a tall, thin colony organism that grows from human remains, calls in a human voice and charges once seen. And the Still Lifes, copies of people who happened to be in a place when it was taken, remembered with their form mostly intact, too many eyes, wrong proportions, and almost no function.
- **Tone.** Kane didn't direct it as horror. No jumpscares, no darkness for mood, no music. Its pull is curiosity.

- **What isn't in it.** No numbered levels, no almond water, no wiki monsters, no exit signs, no sanity meters.

The Codex is the full record, 226 pages. If you have it, read chapter 11 (what the Complex is), chapters 12 to 15 (null zones, places, creatures, time and the green glow), and chapters 22a to 22c (the working theory, its tests, and answers to the user's questions). If you don't have it, ask the user for it before deciding anything about canon.

The ten laws on one page

These are the Codex's working theory, and the mod is built from them. Chapter 4 restates each as an engineering requirement.

Law	What it says	What it means in the mod
1. Capture	It copies only what an opening touched, a whole scene at a moment, people included as passengers	Every copy traces back to an opening in the Opening Ledger
2. Two fidelities	What a thing is copies exactly. Where it belongs, how big it is and what it's for drift	Block states, sign text and container contents are never edited. Positions, sizes and purposes are
3. Matter, not light	It can't copy the sky or sunlight. The sky drains the capture, so what faced the sky comes through streamlined and what was enclosed comes through faithful	No sky anywhere inside. Exposure from Minecraft's own sky light decides how streamlined each part starts
4. The open space is the centre	A copy keeps the open volume it was taken from and the faces bordering it. Fronts face inward	The capture is a flood that travels through air and dies within a few blocks of matter
5. Interpretation	Whatever is missing is filled from what it already holds, matched by look and role	Every substitute comes from a real source, keeps the exact shape of what it replaces, and is the flattest close colour
6. Every remembering is a worse copy	Copies are made of copies. Detail goes first, then shapes, then layout	Generation numbers drive a set of pure decay operators
7. The snowball	Its default is whatever it has repeated most, dominated by the earliest	Two Pólya urns, one for materials and one for memories
8. Continuation	It grows through its own openings, joining pieces at natural seams	Genesis grows territories that connect by construction
9. Structure before function	Form copies well, working insides and behaviour copy badly	Copied machines and Still Lifes keep less function the older they are
10. Reaching back	Once joined to a place, it changes our side too, only near openings	Reach effects happen only inside join radii

The person you're building for

The user is the project's designer and its only judge of whether it feels like the Complex. They know canon very well and have watched five earlier versions fail. Their rules are absolute. They're collected here in full, grouped, so nothing has to be remembered from elsewhere.

Code

- NeoForge 1.21.1 is the platform and NeoForge is the only dependency. No libraries, no required mods, no compile-time integrations. Test-only libraries such as JUnit are allowed because they never ship.
- **No comments in code at all.** No inline comments, no block comments, no Javadoc, no notes, not in tests, not in the lab harness, not in build scripts. Code explains itself through names. This is the rule the user has been angriest about.
- Plain, idiomatic Java with clear, consistent names. No generated boilerplate left in the repository.
- Test and harness tools live in their own source set or subproject and never ship.
- Mod author UnderwaterMarch. Package io.github.underwatermarch.mansion. Never put any other author name anywhere.

Writing

- Everything the project produces reads as written by a person. Code, docs, the README, commit messages, mods.toml descriptions, in-game text, image metadata.
- No em dashes or en dashes anywhere. Colons only where they're genuinely needed, in code, titles and times.
- No co-author trailers, tool credits or generated-by lines in commits, files or metadata.
- Commit messages are short and plain.

Content

- Nothing authored. No templates, presets, room lists, floor plans, structure files, palettes or shape catalogues. "Simplify into rectangles" counts as a template too.
- No vanilla generation machinery in the dimension. No structures, jigsaws, features, carvers, noise routers or aquifers.
- No custom blocks in the dimension.
- Laws acting on the world only. Canon's numbers may set how strongly a law acts, never what it produces.
- No gimmicks. A gimmick is anything with no canon source and no Minecraft need, anything that creates mood instead of behaviour, anything that fires on a timer or a dice roll for its own sake, a visible trick hiding missing engineering, or invented physics.
- It must work on any architecture. Never tune anything to one map. No map's coordinates, sizes, block lists or thresholds may appear in mod code. Validate on differently built worlds before claiming anything generalises.

Storage

- No files of the mod's own. level.dat and chunk NBT only.
- That rules out NeoForge's SavedData, NeoForge level attachments (they're saved to a data file), config files of any kind, disk caches, and logs of the mod's own beyond the game's normal log.

Process

- Feel first. The M0 slice is judged by the user before anything else is built.
- Ask before committing to git. Ask before changing anything labelled LAW or YOUR RULE.
- Never modify the user's real worlds or saves. Copy a world fresh for each test run and delete the copy afterwards. The C drive is nearly full, about 35 gigabytes free when this was written.
- Never kill the TLauncher Java process. Only stop processes whose command line contains the project folder's name and isn't the Gradle daemon.
- The old project folders are read-only. Never reuse their code. Their lessons are in this chapter.
- Give recommendations, not surveys. When a choice is genuinely the user's, ask with a recommended answer first.

Testing and honesty

- Multiplayer always. Test with several players spread far apart in all three axes.
- Walk tests use at least ten walker bots following the Pledge rule, heading in different directions in 3D through stairs, ladders, chutes, pits and doors. They spread out first, then regroup. This doubles as the multiplayer test.
- Every generation test produces full, uncropped plan images of every storey, styled like the fan maps of the Complex. The user looks at them before walking a build.
- Look at every screenshot and image yourself before sending it.
- Never report a pass from weak criteria. If something only half works, say so.
- When asked for a verdict, give a straight one, including that a plan is on the wrong track.
- Performance budgets are hard limits (chapter 30). No TPS drops, no FPS drops, no watchdog dumps.

Why every earlier version failed

[YOUR RULE] Five versions came before this one. Their code must not be reused, but what they taught was expensive, and every lesson below is measured.

Version	What it tried	Why it failed
V3, the original, Fabric 1.20.1	Raw voxels of Minecraft builds copied and stitched together	An incoherent mass. Players were stranded within about 20 blocks. Its one success was its floors, see below
V4 and V5	A window system generalising V3's floors	Never worked well. A 64-block painter painted the future layout into chunks players were standing in, so floors vanished under them
V6	Block-scale synthesis, then joining whole places	Patch quilting drifted or degenerated. Exact model synthesis stalled because rare faces had no continuation. Graph-cut synthesis misremembered 15 per cent of blocks and tiled copies of copies on a lattice. Every place-level join method left dead openings, blocked neighbours or runaway stacking
V7, Fabric 1.20.1	Moulding places onto a shared lattice in sealed bands	Riddled with outdoor places and painted skyboxes. It copied places faithfully at generation 0 and then bent them, but canon's copies are never faithful outside, even fresh. Left unfinished
The outside v0.1 bible	A strong engine with authored Founder rooms, canon sites, building generators and extra files	Broke the user's rules. This bible replaced it

[DESIGN] What those failures prove, and what this bible does about each.

1. **Copying places as they stand fails.** Minecraft builds are mostly outdoors, absurd and enormous, and a faithful copy of one is never the Complex. That's why chapter 9 streamlines everything that faced the sky and chapter 10 re-expresses places in the Complex's own image.
2. **Outdoor places fill every naive version.** Almost everything players build stands under the sky, the reverse of the real world. Rules O1 to O5 and test T-3.3 exist for this alone.
3. **Block-scale synthesis doesn't work on Minecraft architecture.** On a measured stress map, 95 per cent of architecture is unique at a radius of two blocks, even allowing rotations and mirrors, so block neighbourhoods barely repeat. Never try to synthesise geometry voxel by voxel from examples.
4. **Joining whole places to each other fails.** Genesis joins pieces through seam contracts and ports agreed in advance, and fills the gaps with the sea.
5. **Real maps aren't fully walkable either.** Only about 49 per cent of a real adventure map is two-way connected under vanilla movement. "No pocket you can enter and never leave" is the right rule, not "everything two-way".
6. **Light must travel with blocks.** Vanilla sends light-only section changes only to players at the edge of view, so any change to blocks must install its light in the same packet as the blocks.
7. **Never paint into chunks players are in.** Anything that rewrites the world rewrites it only where no player holds it (chapter 14).
8. **What worked.** V3's floors worked because each storey was sealed, so nothing outside it could ever be seen. The Mansion keeps that principle. It never shows an outside.
9. **Version mismatches are silent.** Blocks unknown to a game version load as air. Check a test world's version before using it.

Where everything is

What	Where
This bible, PDF	`C:\Users\MinhHuy\Pictures\Mod Project\Mansion Rebuilt\Complex Codex\The Mansion - Design Bible.pdf`
This bible, source	The `bible` folder beside it, one Markdown file per part. `python build_bible.py` rebuilds the PDF and must report 0 dashes and 0 colons
The Codex	`The Complex Codex.pdf` in the same folder, source in `chapters`, rebuilt with `python build.py`
Reference frames	`Complex Codex\images\user`. Canon frames supplied by the user, for comparison only, never shipped
The new mod	`C:\Users\MinhHuy\Pictures\Mod Project\Mansion`. It holds a copy of this bible and a docs folder, nothing else yet
Old versions, read-only	`Mansion Rebuilt\mansion` (V3), `mansion-complex` (V4), `mansion-complex-2` (V5), `project_mansion` (V7), `V7-2`
A working NeoForge 1.21.1 build, read-only	`Mod Project\CogitoErgoSum`. An unrelated mod whose Gradle files use exactly the toolchain in chapter 35
Test worlds, copy only	`Mansion Rebuilt\mansion\run\saves\Broville v11.4.2`, a long-lived server town and a 1.20.1 save that upgrades when opened. `Mod Project\Worlds For Testing`
Test mods	`Mod Project\Mods For Testing` holds Fabric 1.20.1 jars. None of them load on NeoForge. NeoForge 1.21.1 builds have to be downloaded, so ask first
Java	JDK 21 at `C:\Program Files\Java\jdk-21`
The user's launcher	TLauncher. Never stop its process

If you can't reach these paths, because you're working without access to the project machine, ask the user for the files you need rather than guessing.

How to work

- **Milestones in order.** Chapter 32 lists them, and chapters 40 and 41 break each into steps. A milestone isn't finished until its exit test passes, and the next one doesn't start until then.
- **Stop points.** The step lists mark the moments to stop and show the user what exists. Don't run past them.
- **Every session.** Say what you're about to do, do it, verify it, then report with evidence.
- **When the bible is wrong.** Don't quietly work around it. Write a decision record, tell the user what's wrong and what you propose. Once they agree, edit the bible's source, add a row to Appendix F, and rebuild the PDF.
- **When the bible is silent.** Use the defaults in chapter 43. If there are none, follow the order of authority. If it's still unclear, ask, with a recommendation.
- **Reports.** What was built, how it was tested, the numbers, the images, what failed, and what comes next. Chapter 42 has the template.

When a piece of work is done

- It does what the bible says, or the difference is recorded and agreed.
- Its tests exist, run, and pass on strong criteria.
- It has been checked in multiplayer and on more than one kind of world wherever that applies.
- It stays inside its budget from chapter 30, measured.
- It writes no file of its own. Test T-S.1 still passes.
- Its code has no comments, its text has no dashes, and nothing names anyone but UnderwaterMarch.
- The user has seen the evidence.

Your first session

1. Read this chapter, Part A, chapters 8 to 10 and Part E.
2. Check that JDK 21 is installed and that the reference build in CogitoErgoSum still runs `gradlew build` offline. If it does, every artifact chapter 35 needs is already cached.
3. Ask the user whether to make the mod folder a git repository, and confirm the commit author name, UnderwaterMarch.
4. Start M0 at step M0.1 in chapter 40.

PART A: Foundations

What the Mansion is, the rules it never breaks, and the ten laws as engineering requirements.

1. Vision, Pillars and the Anti-Gimmick Charter

What the Mansion is

The Mansion is a single Minecraft dimension that behaves the way Kane's Complex behaves. It doesn't contain a hand-built Backrooms. It's a machine that copies your world, badly, and keeps copying its copies until they decay into something like the yellow rooms.

Most Backrooms projects start from the image, with yellow wallpaper, a carpet texture and a grid of rooms. The Mansion starts from the mechanism. In the Codex's working theory the yellow rooms aren't a designed place at all. They're what copying ends up as, the most repeated memory of a system that copies its own output (Law 7). Build the mechanism faithfully and the yellow rooms appear on their own, and so do furniture sunk into floors, doors onto brick, mirrored signs, facades, shrinking repeated objects and copies of people with too many eyes. None of it is placed by hand.

The mechanism needs material, and here the Mansion is different from anything before it. The Complex copies what touches it (Law 1). In the Mansion, what touches it is your world. Your basement, your village, your friend standing next to an opening at the wrong moment, the blocks of every other mod in your pack. A player who falls through a null zone under their own house may find, deep inside, a near-perfect copy of that house. Walking away from it, they pass worse and worse copies of it, until it becomes wallpaper. And far enough out, the endless rooms themselves turn out to be the first room the Complex ever took from that world, remembered a thousand times.

One	dimension, one continuous place, no levels and no warps between levels
Zero	new blocks, items, textures, models or sound files
Zero	authored rooms, structures or templates. Every room is a remembered copy of your world
Zero	extra files. Everything lives in level.dat and chunk NBT
Ten	laws implemented as subsystems, each with an automated test that would show it broken

The pillars

Pillar	What it means in practice
1. Canon before invention	Kane's Complex is one place handled with the sober logic of hard science fiction. No community levels, no almond water, no wiki monsters. The only inhabitants are the ones canon shows, the Lifeform, the Still Lifes and things that fell in
2. Laws as code	Each of the ten laws is a named subsystem with an input, an output and a falsification test. If a feature contradicts a law, the feature loses
3. Nothing authored	No room, building, site or layout is made by hand or by a procedural building generator. Everything the player walks through is a remembered copy of something captured from their own world
4. Zero assets	No new blocks, items, textures, models or audio files. Environments are vanilla blocks, other mods' blocks and blocks copied from the world. Still Lifes are drawn from players' own skins. The hum is synthesised at runtime
5. Not a horror mod	Kane: "I didn't direct like a horror film." The place is bright, washed out, mediocre and endless. Threat is rare, physical and fair. No jumpscares, no scary music, no sanity meter, no darkness for mood. Its pull is curiosity. Kane has "always just wanted to run around in there"
6. No arbitrary decisions	Kane's rule is that if a socket moves an inch, the building shifted. Every oddity in the Mansion has a computable cause, a drift, a merge, a fill, a generation count. The debug tools can explain any block
7. Light on the machine	The mod must not change TPS or FPS measurably when nothing is happening, must never block the server thread, and must stay inside the budgets of chapter 30

The anti-gimmick charter

These are binding. A change that adds any of them is rejected.

> **We will never**
>
> - Use numbered levels, level names or "you entered" titles.
> - Generate rooms on any grid, visible or not, or place anything by a lattice the player could notice.
> - Use structure blocks, jigsaw pieces, template pools, structure placement, or any vanilla structure, feature, noise router, density function, carver or aquifer in the dimension.
> - Ship authored rooms, buildings, sites or layouts, or procedural generators of buildings, rooms or real places.
> - Add a custom wallpaper, carpet or any other block, or retexture a vanilla one.
> - Add a sanity meter, hallucination overlays, fake crashes, fourth-wall breaks, jumpscare stingers or ambient music.
> - Make the place dark and green like a haunted house, which Kane says is where other versions "miss the mark".
> - Spawn any creature canon doesn't show.
> - Put exit signs or win conditions in the Complex. Exits exist, but they're rare, unchosen and lead only back to the world.
> - Change anything in front of a group of players without the green glow.
> - Show a real sky, sunlight or a sun-cast shadow anywhere inside (Law 3).
> - Add "tells" canon doesn't show. Nothing wobbles, shimmers or glows to warn you of a null zone.
> - Record or replay players' voices.
> - Call the place "Level 0" or "the Backrooms" in the world. Async's words are the Complex, A-Space, Standard, null zone and the Threshold.

What is genuinely new here

As far as we know, no Minecraft mod and very few games combine these. A survey of earlier work is the first task of milestone M0.

1. **A generator whose only raw material is the player's own world.** Captures of real volumes of the world, modded blocks included, are the seeds of the whole dimension. Even the endless default rooms descend from them.
2. **Lossy, generational copying as the generation algorithm.** The dimension is a population of copies of memories, each at a generation number, and decay operators turn generation numbers into geometry.
3. **Error as emergence.** Sunk furniture, fused objects, facades and repeated shrinking objects aren't features. They're what's left when overlaps are resolved and relationships drift.
4. **A world that forgets when nobody is looking.** Saved chunks left unobserved are rewritten one generation worse before anyone comes back, your own base included, gently.
5. **Zero-asset rendering and synthesis.** Copies of people drawn from real skins, a hum synthesised per light cluster, walls that slide with vanilla display entities.

2. Your Standing Rules

These rules come from you, not from canon or from us. They outrank every design decision in this bible. Each one is listed with how the design meets it.

Rule	How the design meets it
MUST. Everything genuinely human-made. No AI anywhere, no comments, inline notes or notes of any kind in code, nothing that reads as vibe-coded	Outranks every law and every other rule, always. Chapter 0 states it in full, chapter 36 lists every forbidden tell, and the build fails on any breach through `checkHuman` (chapter 35) and test T-H.1
NeoForge 1.21.1, with NeoForge as the only dependency	No libraries, no required mods, no compile-time integrations with other mods. Compatibility only through tags, vanilla mechanisms and NeoForge events (chapter 28)
The mod stores no additional files. level.dat and chunk NBT are fine	Global records live in level.dat under a single compound. Per-chunk records live in chunk NBT as NeoForge chunk data attachments. Memories live in the chunk NBT of the place they were captured from. No world subfolders, no saved-data files, no journal, no server config file. Settings are gamerules, which level.dat already holds (chapter 6)
No custom blocks in the dimension	No blocks or items registered at all. Every block in the Mansion is vanilla, another mod's, or copied from the world
No presets, templates or gimmicks. Laws acting on the world only	No authored content of any kind. The only fixed input is the founder prior, a set of numbers measured from canon (materials by role, proportions, light spacing), and it can be set to zero for a Complex seeded purely by your world (chapter 12, Appendix C)
Works on any architecture	Nothing is tuned to one map. The test suite runs on very different worlds, from creative megabuilds to bare survival and heavy modpacks (chapter 29)
Multiplayer first	One shared world, one server-authoritative state, no per-player realities, no teleport tricks that split players (chapter 5)
Performance budgets	Heavy work off the server thread, server work time-sliced, no blocking waits on the server thread, measured budgets (chapter 30)
Human-written code	No comments in code, clear names, no generated boilerplate in the repository. Mod author UnderwaterMarch, package io.github.underwatermarch.mansion
Never touch real saves during development	All testing on copies of maps, never on the originals (chapter 29)
Consult the Codex	The Codex is the source of truth for canon. Where this bible and the Codex disagree, the Codex wins and the bible is corrected

3. Canon Compliance Charter

These are the facts the mod must never contradict. Each row is a statement recorded in the Codex and the rule it imposes.

Canon	Rule in the Mansion
[CANON] One place, not levels. Everything fans would call "Level 0" contains every kind of space, offices, houses, pools and red-lit streets. Pits lead to very different areas but aren't level changes	One dimension, one continuous geometry. Variety comes from what has been captured and how often it has been remembered, never from zones
[CANON] The only exits lead back to Earth, and nobody chooses when or where they come out	Exits are rare, unannounced null zones inside, depositing you somewhere in the world you didn't pick
[CANON] Everything tested is ordinary matter, with no rust and almost no wear, "possibly indefinite use"	Every block is ordinary and mineable. Copies never gain decay variants the original didn't have
[CANON] The lights never go out on their own, no supply can be found, the conduit stays live, the ballasts hum at 120 Hz	Light-role blocks keep the lit state they were captured in, without power, forever. A lamp captured on never goes out, a lamp captured off never comes on. Every lit cluster produces a synthesised 120 Hz hum
[CANON] Lights can switch on in response to people (the playground set, the Outpost's motion sensors)	Captured sensor-driven lighting keeps the form of its behaviour (Law 9)
[CANON] Coloured light is unusual. The exceptions are the red street, the Red City, the green glow and the Outpost's alarm	Default light is neutral white. Coloured closure light appears only in outdoor copies captured near sunrise or sunset, red at dusk and dawn. Green appears only at growth events. Players' own coloured lamps keep their colour when copied
[CANON] The Complex goes up as well as along. The space above the ceiling is "more than one floor"	Strata stacked through the dimension's height, with crawlable ceiling cavities
[CANON] Everything seen is connected, by doors, gaps, holes, crawl spaces, chutes and ladders	Genesis connects territories by construction. Walkability is inherited, not guaranteed. Some chutes are one-way
[CANON] It changes. Room 14C changed between visits, wallpaper stretched, walls moved with green light	Unobserved space is re-remembered. Live change only during growth events, always with green light
[CANON] Sound leaks. The real world can be heard through some walls, radios catch other years	Thin places carry sound from the world, live and from earlier
[CANON] Kane on food, water, toilets and bodies. Food probably exists but is hard to find and may be rotten. Water is easier. Toilets work. Bodies work normally	Food only where captured, drifting toward spoiled. Water where captured. Vanilla hunger, no thirst mechanic
[CANON] It's cold, though perhaps not everywhere	Temperature per territory, using vanilla freezing
[CANON] Null zones can be any size, grow when disturbed, are usually one-way, don't connect cleanly, and are tied to time slips and the green glow	Chapter 15, point by point
[CANON] People only ever come out later, never earlier	Every crossing moves forward in time only
[CANON] Two kinds of creature, completely separate (Kane), the Lifeform and the Still Lifes	Two entity families that share no path by which one becomes the other
[CANON] The Lifeform isn't a transformed human, but its life cycle involves human remains (Kane)	Lifeforms grow from remains as a substrate. Victims are never converted
[CANON] Kane: "I don't know if there is intelligence or intent behind it."	No system needs the Complex to have intent and none rules it out. No director that targets players
[CANON] Kane: it isn't picking moments out of people's minds	Captures are physical, block and entity state at a moment. Nothing reads chat, inventories for meaning, or "fears"
[CANON] Async's words are the Complex, A-Space, Standard, null zone, the Threshold. "Noclip" never appears	All player-facing text uses Async vocabulary

4. The Ten Laws as Engineering Requirements

The ten laws are the spine of the design. Each is restated as a requirement, with the subsystem that implements it and the test that would show it broken. The tests come from the Codex's own "what would break it" lines.

Law 1. Capture

Requirement. A fresh copy may exist only if an opening once touched its original. A copy is a whole scene at one moment, and entities inside it come along as passengers.

Subsystem. The Cache (chapter 8). Every memory names its opening in the Opening Ledger.

Test. Every memory has an Opening Ledger record. No Still Life is ever made of a player who was outside the capture volume of a recorded event.

Law 2. Two fidelities

Requirement. Identity is copied exactly, block state, block-entity data, sign text, banner patterns, container contents. Relationships may drift, size, position, orientation, count, adjacency, purpose.

Subsystem. Remembering operators R2, R4, R6 and R8. No operator may change identity.

Test. Over a hundred thousand fuzzed objects, every remembered object's identity hash equals its source's. Sign text is either identical or exactly reversed.

Law 3. Matter, not light

Requirement. No sky, sun, stars, horizon or sunlight inside. Every outdoor copy arrives without its sky and is closed with things the Complex has. The sky also drains the capture, so whatever faced the sky is streamlined and whatever was enclosed is faithful, and outdoor parts are rare, short and found only on fresh islands.

Subsystem. Capture marks the sky deficit and measures exposure from sky light (chapter 9). The starting generation, the diorama operator R0 and the sky drain apply it. Operator R7 closes the sky. Outdoor rules O1 to O4 keep it rare. The dimension has no skylight.

Test. Every air block in thousands of random generated columns has a block above it, and sky light is zero everywhere. Sky-facing faces are streamlined and enclosed ones are faithful (T-3.2). Outdoor-looking space stays under 2 per cent (T-3.3).

Law 4. The open space is the centre

Requirement. A capture keeps the open volume and the faces bordering it, fading with depth into matter. Facades face inward. Outward doors and windows end at walls. A building has a convincing interior only if the opening was inside it.

Subsystem. The transmittance flood (chapter 8).

Test. In outdoor captures, no back of a building survives unless it faced open space in the original. A building captured from outside never keeps a full interior beyond the fade depth.

Law 5. Interpretation

Requirement. Gaps, whether never capturable, lost to repetition or damaged, are filled from existing stock, matched by look or by role, nearest real source first. Substitutes keep the exact shape of what they replace, are inert, and are the flattest of the closest colours. Installed mods' blocks count as real.

Subsystem. The Role Lexicon, the stock urn, operators R5 and R7.

Test. Every filled block traces back to a real capture, an installed block or the founder prior. Fill rate never falls as generation rises.

Law 6. Every remembering is a worse copy

Requirement. Copies are made from copies. Fine detail goes first, then shapes, then layout. A copy never holds more of the original than its parent.

Subsystem. Operators R1 and R3 in generation order, and re-remembering of unobserved space (chapter 14).

Test. Agreement with the original, after best alignment, never rises with generation, for every memory tested.

Law 7. The snowball

Requirement. The default, the fill material and the drift target, is whatever has been repeated most, dominated by the earliest material. It must not shift quickly toward recent captures.

Subsystem. The Pólya-urn stock with the founder prior, and the memory urn that chooses what the sea is made of (chapter 12).

Test. Ten thousand simulated captures of one new material keep its share of any role under a configured bound.

Law 8. Continuation

Requirement. The Complex grows from its own openings. Pieces join at natural seams where possible and are forced together where not. Gaps are filled with the default. Everything is connected by construction, though not everything is easy to leave.

Subsystem. Genesis (chapter 11).

Test. In every territory a search over the walkable graph reaches every seam portal, and nearly all seams fall at doorway, corridor or pit ports. No sealed pocket exists that a player could enter but never leave.

Law 9. Structure before function

Requirement. Form copies well. Working insides and behaviour copy roughly, and best in the freshest copies.

Subsystem. Still Life fidelity (chapter 20) and mechanism fidelity of copied redstone and machines (chapter 10).

Test. A Still Life never has more behaviour than a fresher copy of the same template. No Still Life is affected by effects that need biology.

Law 10. Reaching back

Requirement. Once joined to a place, the Complex changes our side too, the right thing in the wrong placement, and only near openings.

Subsystem. Reach (chapter 16).

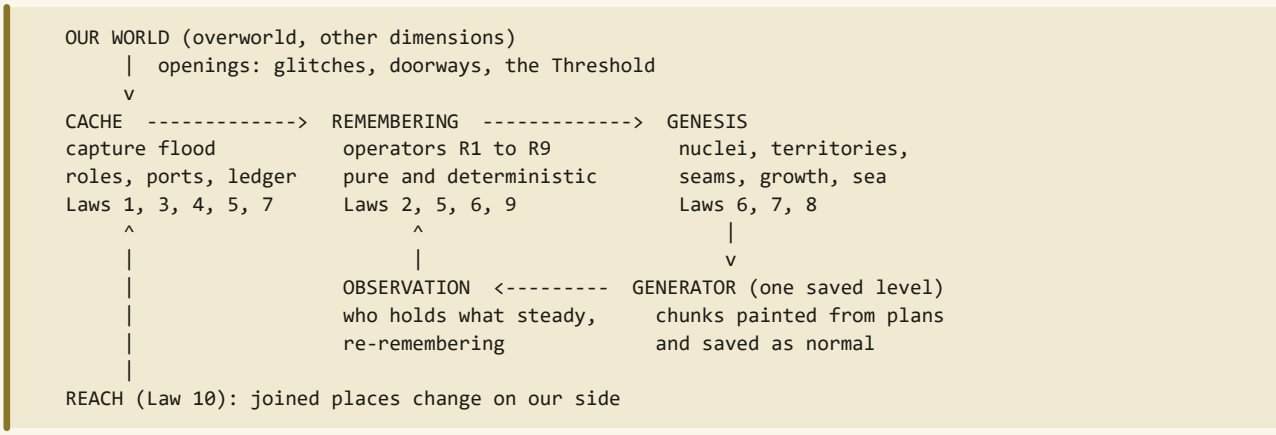
Test. No Reach effect ever happens outside the join radius of a registered opening.

PART B: The Engine

Capture, remembering and growth, the storage that holds them, and the generator that turns them into chunks.

5. System Architecture

The mod is three engines in a loop. Capture takes matter in, Remembering degrades it, and Genesis lays it out in space. Every other system feeds that loop or reads from it.



Modules

Package	Responsibility	Laws and canon
store	The level.dat record, chunk attachments, memory storage and reading	Your storage rule
cache	Memory format, capture flood, Role Lexicon, features, ports, Opening Ledger	1, 3, 4, 5
remember	Pure operators R1 to R9 that turn a memory, a generation and a seed into a remembered copy	2, 5, 6, 9
genesis	Nucleation, territories, seam contracts, growth, sea fill, connectivity repair	6, 7, 8
lineage	The Lineage Field, the material urn, the memory urn, epochs	6, 7
stage	The chunk generator and rasteriser	engine
observe	Hold radii, re-remembering, change behind a lone player's back	6, your stability hypothesis
opening	Null potential, glitches, doorways, crossings, exits	1, 8, canon
reach	Join field, power drift, crooked copies	10
chrono	Forward-only crossings, rare slips, sound leaks	canon
growth	Live growth events, green glow, sliding walls	8, canon
atmos	Self-sustaining light, hum synthesis, cold	3, canon
stilllife	Templates, distortion renderer, behaviour echo, edibility	1, 2, 9
lifeform	Colonies, life cycle, body, hearing	canon
devtools	Provenance inspector, lineage overlay, plan viewer, capture and remember commands	Pillar 6

The three rules every module obeys

1. **Purity at first generation.** Anything that decides a block when a chunk is first generated is a pure function of the world seed, the pinned urn epoch, the memory and opening ids involved and the coordinates. No shared mutable state, no random generator shared between threads, and generation order doesn't matter.
2. **Provenance everywhere.** Every generated block can answer why it's there, with a chain from instance to memory to capture to opening, plus the operators that touched it.
3. **Server-authoritative, client-cosmetic.** Geometry, captures and behaviour live on the server. The client only renders the hum, Still Life distortion, the Lifeform's body, the crossing effect and light

tints.

Multiplayer

[YOUR RULE] There is one Mansion level and one state. Two players standing in the same place see and touch the same blocks. Nothing is instanced, nothing is per player, and nobody is moved between copies of the world. Players who fall through the same opening may land far apart, which is canon (the father in Found Footage #3 falls through a wall right next to Ravi and doesn't land near him), but once inside, the place is shared.

6. Storage Without Extra Files

[YOUR RULE] The mod writes no files of its own. Everything it needs to remember is kept in the two places Minecraft already saves, level.dat and chunk NBT. This chapter shows that everything fits.

What lives where

Data	Where	Typical size
Opening Ledger, every opening that ever touched anything	level.dat, under the mansion compound	about 48 bytes per opening
Memory index, which chunk holds each memory, its size, provenance summary and urn weight	level.dat	about 40 bytes per memory
Material urn, current counts per role, and the epoch snapshots still referenced	level.dat	a few kilobytes per epoch, pruned
Territory pins, which urn epoch each realised territory was planned with	level.dat	about 16 bytes per territory
Island registry, landing sites of openings	level.dat	about 32 bytes per opening
Personal dates for forward slips, exclusion areas, world activity level	level.dat	small
Settings	Gamerules, which level.dat already stores	none extra
Material fingerprints, one for every modded block the Mansion has ever used, its colour, flatness, shape, properties and role	level.dat	about 24 bytes per material
Memory voxel data, block entities and passengers	Chunk NBT of the memory's home chunk, the chunk the opening was in, as a NeoForge chunk attachment	20 to 300 kilobytes compressed per memory
Null positions near an opening, a bitset per chunk	Chunk NBT of the overworld chunk	under a kilobyte
Realisation data of a Mansion chunk, territory, generation offset, last observed time	Chunk NBT of the Mansion chunk	under 100 bytes
Local memories made when a Mansion chunk is re-remembered	Chunk NBT of that Mansion chunk	a few kilobytes
Plant Still Life flags, copied plants that must never grow	Chunk NBT, as a bitset per section	under a kilobyte

The level.dat record

[TECH] level.dat is written by the world's primary level data. A small mixin adds one compound to what it writes and reads it back on load. That compound holds only indexes, counts and small records. Nothing large ever goes there, because the whole file is read at startup and rewritten on every save.

```
level.dat
  Data
    mansion
      version        int
      ledger         list of { id, dim, x, y, z, kind, gameTime, realTime, home }
      memories       list of { id, ledger, homeDim, homeX, homeZ, bytes, weight, flags }
      urn            { epoch, roles: { role: [ material, count ] } }
      epochs         list of { id, roles }      only epochs still pinned by a territory
      pins           list of { territory, epoch }
      islands        list of { ledger, cx, cy, cz }
      dates          list of { player, offsetTicks }
      exclusions     list of { dim, x1, z1, x2, z2 }
      activity       int
```

[DESIGN] Epochs no longer pinned by any territory are merged into the current one, so the record stays small over a long world. A busy world with a thousand openings and a few thousand territories keeps the compound under about half a megabyte.

Memories live where they were taken

[DESIGN] A memory is stored in the chunk NBT of its home chunk, the chunk the opening was in when the capture happened. That has a pleasing fit with canon. The Complex's copy of your basement is kept, quite literally, in your basement. It also means copying or backing up a world copies its Mansion, with nothing to forget.

[TECH] NeoForge chunk data attachments are written into the chunk's own NBT when the chunk saves. When a memory is needed and its home chunk is loaded, it's read from the attachment. When the home chunk isn't loaded, its NBT is read from the region file through the level's chunk storage on the IO worker, without loading or generating the chunk, and only the attachment is decoded. Decoded memories are kept in a memory-limited cache.

[TECH] Region files store a chunk larger than about one mebibyte in a separate file beside the region. That would break the storage rule, so each chunk's attachments are capped at 384 kibibytes. A memory larger than its home chunk can hold is split into bricks spread over the nearest chunks around the opening, and the index records which chunks hold which bricks.

[RISK] Reading chunk NBT for an unloaded chunk without loading it must be done without racing the game's own chunk saving. Spike S2 in milestone M1 proves the read path, the cap and the spill, and measures the cost.

When storage is lost

[LAW] If a home chunk is deleted by a world trimming tool, the matter is simply gone. The Mansion treats that the way the Complex treats any loss (Law 5). Instances of a lost memory are filled from the urn by role. The index keeps the memory's role summary, so even a lost memory degrades gracefully toward the default rather than leaving holes.

When a mod is removed

[TECH] When a mod is removed, vanilla turns its blocks into air as chunks load and logs a warning. Inside the Mansion that would punch holes in floors and walls, and by then the mod's textures and colours are gone, so nothing would know what to put back.

[DESIGN] Two things prevent it.

1. **Fingerprints.** The first time the Mansion uses a block from any mod, as captured matter or as a substitute, it records that block's fingerprint in level.dat, its colour, its flatness, its shape and

properties, and the role it played. Fingerprints are tiny and never removed.

2. **Rewriting before decoding.** When a Mansion chunk is read from disk, its block palette is checked before the game decodes it. Any block whose mod is no longer installed is replaced in the palette by the closest available block by fingerprint, same shape first, then colour, then flatness, through the substitution rules of chapter 10. The chunk then loads with no holes. Memories do the same when they're instantiated, since their palettes store block names.

[LAW] A block replaced this way stays replaced, even if the mod is installed again later. The Complex simply remembers it differently now, which is Law 6. Block entities of a removed mod, its machines and containers, can't be rebuilt from a fingerprint and are replaced by their nearest inert full-block match, their contents lost, as any removed mod's contents are.

[YOUR RULE] The overworld and every other dimension are left alone. Only the Mansion and its memories are rewritten.

7. The Dimension and Its Limits

One ordinary level

[TECH] The Mansion is one dimension with its own dimension type, biome and chunk generator. Its chunks are generated once and saved like any other level's, which keeps it compatible with the rest of the game and with other mods, and means lighting is computed once per chunk rather than every visit.

```
data/mansion/dimension_type/the_mansion.json
{ "has_skylight": false, "has_ceiling": true, "ultrawarm": false, "natural": false,
  "coordinate_scale": 1.0, "ambient_light": 0.0, "fixed_time": 6000,
  "bed_works": true, "respawn_anchor_works": false, "has_raids": false,
  "piglin_safe": false, "min_y": -512, "height": 1024, "logical_height": 1024,
  "infiniburn": "#minecraft:infiniburn_overworld", "effects": "mansion:complex",
  "monster_spawn_light_level": 0, "monster_spawn_block_light_limit": 0 }

data/mansion/worldgen/biome/complex.json
  no precipitation, no features, no carvers, no spawns, no music,
  no ambient or mood sounds, neutral fog
```

The honest limits

[TECH] A Minecraft level is bounded. Horizontally the world border sits at about thirty million blocks, which at a sprint takes more than two months of real time to cross, so for any practical purpose the Mansion is endless along. Vertically a dimension can be at most 4,064 blocks tall, and every chunk column carries every section, so height costs memory and lighting time.

[DESIGN] The default height is 1,024 blocks, from minus 512 to 511, enough for dozens of strata. It can be raised in the datapack before a world is created. The Mansion does not fake infinite height. Canon pits are long, but nothing in canon requires a fall that never ends, and a fall that loops or teleports would break the rule against tricks. The top and bottom of the level are simply the Complex's own fill.

[THEORY] Endless vertical travel stays on a research track (milestone R1). If it's ever built, it must keep one shared state for all players and must never move one player into a copy of a place another player is standing in.

8. The Cache: Capturing Matter

[CANON] Kane says the Complex builds "a cache of physical spatial information". So does the mod. The Cache holds every memory the Complex has taken in, and everything in the Mansion is drawn from it.

What a memory is

A memory is a captured volume of matter at one moment.

Field	Contents	Why
Voxel field	Sparse bricks of 8 by 8 by 8, each a palette of block states by name, plus a capture intensity per voxel	Laws 1 and 4
Block entities	Full NBT, signs with text, banners, lecterns with books, containers, other mods' machines	Law 2, identity
Passengers	Snapshots of entities present at the moment, players with their skin property, equipment and pose, villagers, animals, item frames, paintings, armour stands	Law 1, passengers
Open-space core	The air volume at the centre of the capture, and the faces bordering it	Law 4
Sky deficit	Every air voxel that had full sky light at capture, plus the time of day, moon phase, weather and the biome's sky colour	Law 3
Roles	One role per voxel, floor, wall, ceiling, light, door, window, furniture, plant, liquid, marking, container, mechanism, structure	Law 5
Features	A shell of floors, walls and ceilings, and objects, the connected pieces that aren't shell, each with a scale and a complexity score	Law 6
Ports	Where the open space meets the capture boundary, doorway, corridor end, gap, hole in floor or ceiling, chute, crawl space, window	Law 8
Provenance	Dimension, true coordinates, game time, real time, Opening Ledger id	Laws 1 and 7, Pillar 6
Biome colours	The grass, foliage and water colour of every 4 by 4 by 4 cell the capture covers	Laws 2 and 5

[DESIGN] Palettes store block names, never numeric ids. A block from a removed mod is replaced through its fingerprint (chapter 6).

When captures happen

[LAW] Capture comes in events, not as a continuous recording. In the film Clark spent weeks next to his doorway and was copied once. Events fire when

1. a null zone forms or widens (chapter 15),
2. a stable opening, a doorway or the Threshold, has a growth event (chapter 18), the Codex's best candidate trigger,
3. something passes through an opening, a player, a mob, a minecart, an item. The thing itself becomes a small object memory, like the car on the highway or the seagull,
4. a retrospective opening is drawn (chapter 23). Openings are scattered through time, so the Complex can hold places in your world it touched while nobody was watching.

The transmittance flood

[LAW] This is the core of capture, and it is Law 4 written as an algorithm. The Codex notes that facades are what you'd get if capture travelled freely through open space and was quickly absorbed by matter, like light under the Beer-Lambert law. The flood does exactly that.

```
capture(opening O, budget B)
  I[O] = 1.0
  frontier = max-heap ordered by I
  while frontier is not empty and captured < B
    v = pop(frontier)
    for each of the six neighbours n of v
      if n is not in a loaded chunk, skip
      mu  = open(n) ? MU_AIR + K_SKY * exposure(n) : (glass(n) ? MU_GLASS : muMatter(n))
      In  = I[v] * exp(-mu)
      if In > I[n] and In > I_MIN
        I[n] = In
        push(n)
  keep every voxel with I > I_MIN, its block state and its intensity
  for every air voxel with full sky light at capture, mark it SKY and never keep the sky
```

- MU_AIR is tiny, 0.004 per block, so a capture reaches far along halls, corridors and stairwells indoors. A 250-block hall keeps more than a third of its intensity at the far end.
- In open air the sky drains it. K_SKY adds 0.03 per block at full exposure, so a street open to the sky keeps only about 3 per cent after a hundred blocks (chapter 9).
- Glass passes it, at MU_GLASS of 0.15 per block, as light would, so a capture can look out through a window or leave through a glass door (chapter 9).
- muMatter is large, 0.9 per block scaled by hardness, so capture dies two to five blocks into walls. That's why the film's houses are cut off a few metres behind their fronts and the blue-box house ends at its fence line.
- Hollow structure counts as open space. Crawl spaces, roof voids and ducts are reached through gaps, which is how the Complex copies what no camera could see.
- The flood has no point of view. It keeps the original's own up, down and orientation.
- The flood only reaches loaded chunks. What isn't loaded is beyond its reach at that moment, a natural limit on size.
- Budget B scales with the opening, by default 32 thousand voxels for a glitch, 400 thousand for a doorway growth event and 2 million for the Threshold.
- Chapter 38 gives the complete flood, with how each kind of block is classed, I_MIN and the cap on outdoor air.

[LAW] Where the opening is decides what kind of copy results. An opening inside a building gives its rooms as the open space at the centre, so its interior comes through whole and its outward doors and windows end at the edge of the capture. An opening in a street or yard gives fronts that forget themselves as you go in. That's the difference between the house in the blue box and the red-street house the Codex describes.

[TECH] Reading the world happens on the server thread, time-sliced to at most two milliseconds per tick, by copying the paletted section containers of the chunks in reach, and their sky light layers, into an immutable snapshot. The flood and all analysis run off the server thread on that snapshot.

Passengers

[LAW] Any entity whose bounding box lies inside the captured open space at the moment becomes a passenger, "wearing what they wore and holding what they held".

- **Players.** Game profile with the signed textures property, so the skin can be drawn later without contacting Mojang. All equipment slots, pose, limb angles, crouch and swim state, and the last thirty seconds of a behaviour trace kept in memory (chapter 20). Inventory beyond equipped items isn't copied. A person is copied as a body in a place.
- **Villagers.** Profession, type, level and pose. They're Minecraft's ordinary people, so they become Still Lifes too.

- **Animals.** Kane asked what counts as an animal. They're captured as form-only Still Lifes.
- **Decor entities.** Item frames, paintings, armour stands and display entities are copied exactly, and their positions drift later.

[DESIGN] A captured chest keeps its contents, because what a thing is includes what's inside it. The original in the world is untouched, so copies could duplicate items. Copied container contents keep only a fraction of their stacks at capture, set by the gamerule mansionLootFidelity, with the rest filled from stock of the same role. Server owners can set it from zero, decor only, to one hundred, full canon.

The Role Lexicon

Law 5 needs to know what a block is for. Roles are assigned automatically, so blocks from any mod work without configuration.

Role	Assigned when
Floor	A solid top face directly under captured open space
Ceiling	A solid bottom face directly above open space that wasn't open to the sky
Wall	A solid vertical face bordering open space, part of the shell
Structure	Solid voxels behind shell faces, the material the Complex fills with
Light	Emits light and is attached to a shell face, or is lamp-like by tag
Door, gate, hatch	Vanilla door, fence gate and trapdoor tags and their modded equivalents
Window	A transparent or translucent block set into a wall face
Vertical passage	Ladders, scaffolding, climbable vines, stair runs, holes in floors and ceilings
Furniture	Pieces that aren't shell and have a partial collision shape, stairs used as chairs, fences as table legs, beds, modded furniture
Container	Has an item handler
Marking	Signs, banners, paintings, item frames, patterned carpets
Plant	Flower, leaf, sapling and crop tags, grass, moss
Liquid	Any fluid state
Mechanism	Redstone components, machines, anything that ticks
Sky substitute	Never captured. Produced only by operator R7

[DESIGN] Geometry decides first and tags second. Datapacks can override roles under data/namespace/mansion/roles.

Features, complexity and ports

1. **Shell.** The largest planar faces bordering the open space form the shell.
2. **Objects.** Connected pieces of non-shell, non-air voxels. A chair, a bookshelf run, a furnace with its chimney, a tree, a sign on its post.
3. **Scale score.** The base-two logarithm of an object's volume plus that of its extent, so a flower pot is low and a staircase high.
4. **Complexity score.** The compressed byte length of the object's voxels, the Codex's stand-in for Kolmogorov complexity. Faces and clutter score high, pillars and plain walls low, and repeated objects compress together.
5. **Ports.** Where open space meets the capture boundary, sorted by shape and facing. Doorway-like ports, one to three wide and two to four tall, are the preferred seams (Law 8).

The Opening Ledger

An append-only list in level.dat of every opening that ever touched anything, glitches, doorways, the Threshold, passing objects and retrospective openings. Every memory points to one entry. The Ledger is how Law 1 is proved in tests and how the debug tools explain why a copy of your house exists.

9. Sky Exposure and Streamlining

[LAW] [THEORY] This chapter is the answer to the failure that sank every earlier version, outdoor places appearing everywhere and Minecraft's wildest exteriors coming through intact. It turns one observation from canon into a mechanism that works on any Minecraft build. The Codex's working theory carries the full argument under Law 3. Here it becomes engineering.

> **The rule in one paragraph**
>
> Whatever faced the open sky when it was captured comes through streamlined, like a diorama or a stage set, with one flat colour per surface, simple shapes, no clutter and no wear. Whatever was enclosed comes through faithful. The sky drains the capture, the way it swallows light that would otherwise bounce around a room. In Minecraft the amount of sky a place sees is already measured, by sky light, so the rule costs almost nothing to apply.

The canon it comes from

[CANON] [YOURS] Every copy in canon that faced the open sky looks streamlined, and every fresh interior looks faithful. You noticed it first, in the house in the blue box. The frames below are included for reference only, to show what the mechanism has to reproduce. They are frames supplied by you, never distributed with the mod.



Reference only. Found Footage #3, the house in the blue box. Flat single-colour siding, hedges reduced to green boxes, no wear and no clutter, like a stage set. Its interior is near perfect.



Reference only. The film's suburban street. Every garage, fence and mailbox is toy-clean and new, even though this street was the centre of its capture. Behind each front, the wall of the room. Overhead, a drop ceiling and fluorescent panels.

Copy	What faced the sky	What was enclosed
The house in the blue box	Flat colour, boxy hedges, no wear, a stage-set front	A near-perfect interior, utensils, whiskey bottle, mounted fish, a shoe stuck in the floor
Clark's store	A plain box with a gable front, the sign and mascot	An accurate sales floor, signs, furniture, trusses, long tubes
The film's street	Toy-clean garages and fences, nothing incidental	Not shown
The red street	Boxy, simplified houses	The one house entered dissolves inward, a different effect (Law 4)
The Red City	Towers whose windows look like glowing pictures	Not shown
The playground set	Grass reduced to flat green carpet, a building that's only a front	Not applicable
The farm corridor	Foliage reduced to leaf-print wallpaper, a barn that's only a front	Not applicable



Reference only. The Red City through the skywalk glass. The towers are faces, their windows glowing panels.



Reference only. The red street. Simplified houses lit red from low on one side, under a dark surface with point lights for stars.

Clark's store, three copies from one doorway

[CANON] The film and Everything Must Go show three copies of Clark's store close together. The opening was in the store's basement, inside the building. The mechanism must reproduce all three.



Reference only. The accurate interior copy. Captured from inside, at full strength, with its lights on.



Reference only. The inside-out copy. The store's front, streamlined, facing a car park with half-sunk cars and real trees, closed overhead by unlit ceiling tiles. Captured through the glass front doors, weakly and in the dark.



Reference only. The dark maze copy, from Everything Must Go. The layout is gone and rebuilt as the Complex's own, but the wall design, the trusses and the signs remain, and the signs repeat down the corridor.

Copy	How the mechanism makes it
Accurate interior	The flood starts in the basement and fills the sales floor, which saw no sky. Starting generation near 0, so every sign, chair and truss comes through. Its glass front doors face outward, away from the captured open space, so in the copy they open onto the Complex's wall
Inside-out copy	The flood passes through the glass front doors, as light would, into the car park. The car park saw the whole sky, so it drains the capture and everything there starts at a high generation, the store front flattened to a box and its sign, the cars drifted and sunk, the trees kept as matter. The sky deficit is closed with ceiling tiles. Captured at night, so its lights are off
Dark maze	A remembered copy of a night capture, rebuilt by R10 re-expression. The rooms are the Complex's own, the materials, trusses and signs are the store's, the signs echo (R8)

[LAW] Two rules follow, and both go into the engine. **Glass passes the capture. Lights keep the state they were captured in.** A lamp captured on stays on forever, which is canon's "the lights never go out on their own". A lamp captured off stays off forever, which is why two of the store's copies are dark.

Why the sky streamlines, the bounce model

[YOURS] [OURS] Inside a room, light bounces between surfaces again and again, so every corner is lit from many directions. Outside, light heading for the sky never returns, so an outside wall is lit once, from one side, and most of the energy is lost. The Mansion treats capture the same way. Inside, the capture keeps bouncing and samples every surface many times, so detail survives. Outside, it drains into the sky and samples each surface once, weakly, so only large shapes and dominant colours survive.

[SCIENCE] The Codex's mathematics chapter works it through with the radiosity equation. With a typical wall reflectance of 0.7, an enclosed surface is sampled about 3.3 times as often as a surface that sees nothing but sky, and the share of its distinctiveness that survives the copy rises from about a half to about three quarters. Streamlining isn't a separate effect. It's the same pull toward the typical that remembering applies, made stronger wherever the capture saw less.

Measuring exposure in Minecraft

[TECH] Minecraft already computes how much sky every air block sees. Sky light runs from 15, open to the sky, down to 0, sealed away from it. It spreads in through doors, windows and holes and falls by one per block as it goes, which is close to how bounced daylight fades into a room. The mechanism reads it rather than inventing anything.

1. **At capture,** for every chunk section copied into the snapshot, the section's sky light layer is copied too. It's a small array of half-bytes per section, so copying it costs about as much as copying the blocks.
2. **Exposure of air.** For each captured air voxel, exposure e is its sky light divided by 15.
3. **Exposure of a surface.** For each solid face bordering captured air, e is the highest exposure of the air it touches.
4. **Exposure of an object.** For each object from feature decomposition, e is the area-weighted mean exposure of its faces.
5. **Exposure of the sky deficit.** Air at full sky light is the sky deficit (Law 3), as before.

[TECH] Where light data isn't available for a section, which can happen at the edge of loaded terrain, exposure is estimated inside the snapshot from the column heightmap, sky above any column with no opaque block over the voxel, and then spread sideways a block at a time, falling by one each step, the same way the game does it.

[DESIGN] Minecraft's own quirks follow canon's logic without special cases. Leaves let sky light through, so a roof of leaves is open sky and a treehouse comes through as an outdoor scene. Glass lets sky light through, so a glass-roofed conservatory is half outside. A room with tall windows has a bright, exposed band near the glass and a faithful core deeper in. None of that needs a rule of its own.

The starting generation

[LAW] Each part of a capture gets a starting generation from its exposure, before any remembering happens.

```
gStart(e) = G_SKY * e ^ 1.5           G_SKY 2.5

e = 0      sealed interior                gStart 0
e = 0.33   deep in a room with a window   gStart 0.5
e = 0.66   a porch, under eaves           gStart 1.3
e = 1      open sky                       gStart 2.5
```

[DESIGN] Every operator in chapter 10 already works on generation. Starting a sky-facing face at 2.5 means it has effectively been remembered two and a half times the moment it's taken. Its clutter is gone, its colours are pulled toward the typical, its small features dropped. The engine needs no separate streamlining code path for most of the effect, only the one operator below for the things remembering alone doesn't do.

R0. The diorama operator

[LAW] A few parts of streamlining aren't simply earlier remembering. They make an exterior look like a model, and they're applied to every face and object in proportion to its exposure at capture. In code they run first whenever a copy is made, on the raw memory, which gives the same result and keeps old captures useful while the operators are tuned (chapter 38).

1. **Wear removed.** Weathered blocks become their clean base. The chain is read from the game, never listed by hand. Copper's oxidation steps come from vanilla's weathering chain. Mossy and cracked variants are found through the recipe graph, the crafting or smelting recipe that turns a base block into its weathered form, read backwards. Other mods that add variants through recipes work automatically.
2. **One colour per face.** Each planar face region takes its own dominant material with probability rising with e. A wall built from four kinds of plank becomes one kind. This is the flat single blue of the house in the blue box.
3. **Clutter dropped.** Objects bordering the sky below a scale threshold that rises with e are dropped, torches, flower pots, buttons, lanterns, small decorations. Signs and banners are kept, because canon keeps signs, and keep their text exactly (Law 2).
4. **Plants simplified.** Plant objects that faced the sky become their simplified silhouette in their own dominant material, layer by layer, the outline of each layer replaced by its filled shape. Hedges become green boxes. Trees keep a tree's outline but lose their ragged edges.
5. **Edges straightened.** One-block notches, bumps and jagged steps along a face's outline are closed or removed, so the face reads as drawn with a ruler.

[YOUR RULE] Nothing in R0 is a list of blocks or a template. It reads exposure from the game's own light data and weathering from the game's own recipes, so it works on any build and with any mod's blocks.

The sky drain in the flood

[LAW] Exposure also changes how far a capture reaches. The flood of chapter 8 gains a drain term in sky-exposed air.

```
mu(air voxel)   = MU_AIR + K_SKY * e        MU_AIR 0.004, K_SKY 0.03
mu(glass)       = MU_GLASS                  0.15, glass passes the capture
mu(other block) = muMatter(block)           0.9 per block, scaled by hardness
```

A hall indoors keeps two thirds of its strength after a hundred blocks. A street in the open keeps about three per cent after a hundred blocks. So outdoor captures are weaker and shorter by physics, and that is most of why they're rare (Law 7 does the rest).

Darkness doesn't matter, only the sky

[LAW] The capture is not visible light, and the bounce model doesn't mean a place has to be lit to be copied. The capture behaves like light in exactly two ways. It moves freely through enclosed open space, bouncing between surfaces, and the open sky swallows it. Darkness doesn't weaken it at all.

[CANON] That's why the Lighting and Tile Survey found its most precise detail in the darkest places. The ductwork above the tiles, the insulation on the floor above, the serial number on a ballast inside a sealed fitting. Those are the most enclosed spaces in the building, so under this model they are the best copied, and canon agrees.

[TECH] In Minecraft the measure is therefore sky light, never block light. A sealed ceiling cavity, a chest, the inside of a wall, a pitch-black cellar all have sky light 0, so they're copied faithfully however dark they are. A sunlit lawn and a moonlit lawn both have sky light 15, so both are streamlined. Time of day changes how a copy is lit, through the captured state of its lamps, and how its sky is closed by R7, but never how faithful it is.

Why this doesn't fill the Mansion with outdoor places

[LAW] Streamlining decides how an outdoor part looks when there is one. It does nothing to make outdoor parts rare, and on its own it would give exactly what V7 had, outdoor places everywhere, only blander. Rarity is enforced separately, by four things that multiply.

1. **Where openings can form (O1).** A glitch forms only on a surface whose air has exposure below 0.4. In Minecraft that excludes every field, roof, lawn, street and open courtyard. Openings form in rooms, basements and garages.
2. **How far the capture reaches outdoors.** The sky drain cuts an outdoor run to a few per cent of its strength within a hundred blocks, and any capture's outdoor air is capped at a quarter of its open space. A capture from a living room may bring back a porch and a bit of street, never a town.
3. **Where outdoor parts can sit (O2).** Only on the island at the opening's landing site. Each step of the halo raises generation, R7's sky closure becomes the default ceiling by generation 4, and the sea starts at 9. Two or three rooms away from a fresh capture, nothing outdoor remains.
4. **What the test allows (O4).** If all of that still let former outdoor space exceed about 2 per cent of walkable cells, the build fails. It counts every former outdoor space, whatever now covers it, so a street under a plain ceiling counts as much as one under painted sky.

[DESIGN] As a rough check, islands make up a few per cent of territories, at most a quarter of an island is outdoor, and much of that is under a plain ceiling within a few steps, so outdoor-looking space should settle around 1 per cent. That's in line with canon, where seven outdoor-looking places appear among the dozens of places the series and film show, and every one of them is a long way from any entrance.

[YOUR RULE] And because no outdoor part is ever open to a sky (O3), even the 1 per cent reads as a street inside a hall, never as the overworld. V7's skyboxes are gone for good.

Outdoor rules, as hard limits

[YOUR RULE] [LAW] Outdoor places appeared everywhere in every earlier version because Minecraft is the reverse of the real world. Almost everything people build in Minecraft stands outdoors. These rules keep the Mansion as interior as canon's Complex is.

- **O1. Captures start inside matter.** A glitch can form only on a surface whose bordering air has exposure below 0.4, so openings form in basements, rooms and garages, never in open fields. Retrospective openings obey the same rule, so a retrospective capture of a village is a house's interior, not its street. Outdoor air enters a capture only by escaping through a gap or through glass, and it may make up at most a quarter of any capture's open space.
- **O2. Outdoor copies live only on islands.** Sky-facing parts start at generation 2.5, the sky closure of R7 climbs with generation, and sea instances sit at generation 9 or more. So by the time anything reaches the sea its sky is long gone. The sea never contains an outdoor scene.

- **O3. No sky, ever.** No skybox, no textured sky, no clouds, no sun. Sky closure is made of blocks from stock, as R7 describes, and is crude on purpose.
- **O4. Measured.** Test T-3.3 samples generated territories and measures the share of walkable cells that are former outdoor space, cells whose air had exposure of 0.5 or more when captured and that haven't yet been re-expressed into rooms. Whatever closes them counts, a painted surface, a dark one or a plain ceiling, because the film's street proves a plain ceiling doesn't stop a street reading as outdoors. It fails above 2 per cent, about canon's proportion of outdoor-looking places among all the places it shows.
- **O5. Re-expression turns outdoor space into rooms.** From generation 1 to 3, R10 rebuilds former outdoor space as a room of the Complex (chapter 10). That's what actually eats the outdoors, two or three rooms away from a fresh capture.

[TECH] To make O4 and O5 possible, every memory keeps its air exposure, quantised to four levels, as part of its voxel field, and every instance carries it through remembering until re-expression clears it.

A worked example, the build players actually make



Reference only. A Minecraft build of the kind players really make, a stacked castle town hanging between the arches of a giant aqueduct. This is the input every earlier version broke on.

[DESIGN] Suppose a glitch forms in this build. Under O1 it can only form inside one of its rooms, say a lantern-lit hall on the fifth level, sky light 2.

1. The flood fills the hall and the rooms joined to it at full strength. Those rooms start near generation 0 and come through faithful, every lantern, screen, chest and stair.
2. Through the hall's windows and doors the flood leaks out onto balconies and roofs, which see the sky. It drains fast there, and what it does reach starts at about 2.5. A few balcony faces, simplified to flat

colour, a roof edge, a lantern dropped.

3. The towering silhouette, the aqueduct arches, the floating mass, the gondola cable, are never reached at all. A silhouette isn't something capture takes. Only faces bordering the captured open space are kept.
4. The sky deficit over the balconies is closed by R7 from stock.
5. In the Mansion, this place appears at its landing site as a run of faithful Japanese-style rooms, opening onto a short, streamlined, ceilinged balcony. Its halo then rebuilds it in the Mansion's image, R10 re-expression, and the sea eventually swallows it.

Nothing of the build's absurdity survives except what was inside, and inside is the part of a Minecraft build that's most like a real building.

What the mechanism must never do

- Copy a sky-facing face with full detail at capture. That breaks the rule.
- Copy an enclosed room as a diorama at capture. That breaks it from the other side.
- Change the text on a sign, the contents of a container or the lit state of a lamp. That breaks Law 2.
- Use a list of blocks, a list of builds or a template to decide what's streamlined. That breaks your rule.

Tests

Test	Asserts
T-3.2	On a captured face with exposure 1, no weathered variant survives, the face has at most two materials, and no object below the clutter threshold remains. On a face with exposure 0, every block is identical to the source
T-3.3	Across a thousand generated territories, walkable cells of former outdoor space not yet re-expressed stay under 2 per cent, whatever closes them
T-3.7	Every re-expressed instance contains no former outdoor space. All of it has become rooms with walls and a ceiling
T-3.4	A lamp's lit state is the same in every copy as in its capture, across every generation
T-3.5	A capture started in a basement with no gap to the outside contains no sky deficit at all
T-3.6	Glass between two rooms lets the capture through. A solid wall of the same thickness doesn't

10. The Remembering Engine

[CANON] "The more times it remembers something, the less it does." The Remembering Engine turns a memory and a generation number into a remembered copy. It's a set of pure operators whose combined effect should reproduce the teaser's eight stages.

The contract

```
remembered = R(memory M, generation g, seed s, urn epoch e)
  pure        same inputs give the same output on any thread in any order
  lazy        evaluated per object and per brick, never as a whole-world pass
  monotone    agreement(M, R(M, g + 1)) <= agreement(M, R(M, g))
  faithful    no operator ever edits block identity, block-entity data or text
```

[DESIGN] Generation is a real number, not an integer, so freshness can vary smoothly across space.

R1. Spectral forgetting (Law 6)

[LAW] Under a tightening budget, features smaller than a rising "water level" are dropped entirely, the reverse water-filling of rate-distortion theory.

```
theta(g) = THETA0 + A * g                THETA0 -1.0, A 1.6
keep(object o)  if  scale(o) - theta(g) + noise(o) > 0       noise uniform within 0.6, seeded per object
shell faces are exempt until g > G_SHELL (9); windows get a boosted score until g > 4
```

The order of loss falls out of the scores. Clutter goes first, then furniture, then windows, then partitions, and walls, floors and ceilings last. That's stages 1 to 5 of the teaser.

R2. Relational drift (Law 2)

[LAW] Objects keep their identity while their relationships drift. The spread between two points joined by k joints after n rememberings grows as sigma times the square root of k times n.

```
for each object o
  k        = joints between o and the open-space core
  dpos     ~ Normal(0, SIGMA^2 * g * k) per axis, rounded        SIGMA 0.35 blocks
  dy       biased downward by BETA * g * SIGMA                   things settle into floors
  mirror   with probability 1 - (1 - 0.004)^g
  turn     a quarter turn about the vertical with probability 1 - (1 - 0.01)^g
  rescale  with probability 1 - (1 - 0.015)^g, by 0.5, 0.75, 1.5 or 2
```

Mirroring a sign reverses its text and keeps every letter, which gives the POTS stop sign. Wrong sizes come from resampling. Stretching repeats a wall's own pattern along one axis, as in the Static Dead End.

R3. Prior pull (Law 6)

[LAW] Each remembering keeps only a fraction k of what made a thing different from typical. After n rememberings about k to the power n is left.

- **Materials by role.** A shell voxel keeps its captured material with probability KMAT to the power g, 0.82 by default. Otherwise it's replaced by a draw from the urn for its role (chapter 12). Low-frequency noise per face keeps the repainting coherent rather than speckled.
- **Proportions.** Room dimensions are pulled toward the prior room shape, which leans larger, because in the teaser the room grows.

- **Light.** Captured light is pulled toward the prior light arrangement (chapter 11). Daylight goes first, lamps stay, the default arrangement takes over.

R4. Merge resolution (Law 2)

[LAW] Solid can't share space with solid. After drift, every voxel claimed by more than one solid source keeps one, by fixed priority.

1. Liquid never displaces solids, so pools stay in their basins.
2. At low generation the shell beats objects. At higher generation objects win more often, which is how a football ends up half inside the carpet.
3. Between two objects, the more complex wins at the contact surface and the loser's voxels inside it vanish. Touching things weld, the way Pirate Clark's hat grew into his head and the chair leg became his leg.
4. Containers are never deleted by a merge. A losing container moves to the nearest free voxel of its own object.

R5. Fill and R6. Lost purpose (Laws 5 and 9)

[LAW] **R5 fill.** Every gap, a dropped shell voxel, missing modded matter, damage or an unclosed sky, is filled by role with the Codex's three honesty tests built in.

1. The substitute comes from somewhere real, a captured material, any installed block or the founder prior.
2. It matches by role first, then by the shape, colour and flatness rules in the next section.
3. Substitution grows with generation, because R1 and R3 drive it.

The draw looks first in the instance's own memory, then in its territory's memories, and only then in the whole urn. The chair leg strapped to Clark becomes his peg. The green carpet next door becomes the playground's grass.

Choosing a substitute in Minecraft

[DESIGN] Every substitution in the Mansion, by R3, R5, R7 or R10, goes through the same rules.

1. **Same shape.** A substitute has exactly the collision shape of what it replaces, and keeps its properties wherever it has them, facing, half, axis, waterlogging. A full block is replaced only by a full block, a slab by a slab of the same type and half, a stair by a stair, a carpet by a carpet. Substitution can never raise or lower a floor, add a lip, take headroom away, or give mobs a new partial block to stumble over. Geometry changes only through drift, merging and re-expression, whose results are always checked by the walkability tests.
2. **Inert.** A substitute has no behaviour of its own, no block entity, no random ticks, no gravity, no light, no redstone function, and it's opaque if what it replaces was. Concrete passes. Grass, sand, concrete powder, leaves and glowstone don't.
3. **Colour.** Candidates are ranked by colour distance in CIELAB from what's being replaced. For grass, leaves, vines and water, the colour is the one captured with the memory for that cell, never the raw texture, which is grey. Where no texture colour is known, the block's map colour, which every block has on the server, is used instead.
4. **Flatness.** Among close colours, the flattest texture wins. Flatness is how well the block's texture compresses, the same measure of complexity the Codex uses for "simplicity survives". Concrete's near-flat texture compresses to almost nothing, wool and planks don't. So substitutes look cheap, plain and stage-set flat, the way canon's streamlined outsides do.

5. **Where candidates come from.** First the instance's own memory, then its territory's memories, then the whole stock, then every eligible block of every installed mod, then vanilla. An installed mod's blocks are as real as vanilla's. They're part of the world's matter, and the Mansion copies modded places anyway.
6. **Breaking monotony.** If the best candidate is the same block as a neighbouring role in the same room, a floor that would match its walls, the next candidate of similar colour and flatness is taken instead, in the same order of sources. If nothing is close enough, the two roles merge into one material. Loss is part of the Complex, so a merged room is allowed.

[TECH] A dedicated server never loads vanilla's textures, so it can't measure them. The build measures every vanilla block's mean colour and flatness once and ships them with the mod as a small table of numbers, measured from the game, never chosen by hand. Mod jars carry their own textures even on a server, so modded blocks are measured at server start from their own block models. Results are cached in memory and never written to disk.

[LAW] Inside the Mansion, copied grass, leaves and water would otherwise be drawn in the Mansion biome's colours. A small client hook gives them the colour captured with their memory instead, through the same colour handlers vanilla uses, so it works with Sodium and Embeddium. Everything else falls back to vanilla's own colouring.

A patch of grass through the generations

[DESIGN] A plains lawn that faced the sky, in a green concrete house.

Generation	What the lawn is
0, capture	Grass blocks in their captured plains green. Flowers and tall grass dropped, grass, dirt and path evened to the dominant block, edges straightened (R0)
1 to 3	Still grass in its captured green, with a growing chance per block of becoming the flattest close green. Green concrete would match the house's walls, so lime concrete, or a mod's flat green block if one is closer
4 to 8	Mostly flat green blocks now, some grass left. The playground stage, cheap and stage-set flat
9 and up	Pulled toward the sea's floor material by R3

[LAW] **R6 lost purpose.** Purpose is a relationship, so it breaks like one, with probability rising in g.

Pair	Broken form
Door and the passage behind it	The passage is filled. A door onto brick
Hood and cooker	The cooker is dropped and the hood kept
Stair and step	Stairs become slabs or full blocks in a slope. A staircase with no steps
Window and outside	Glass becomes wall inside a kept frame. The window is only a frame
Picture and wall	The painting or frame becomes an opening of the same size, "a rectangle on a wall became a rectangle in a wall"
Bed and frame	The bed splits, one half sunk
Lamp and switch	The switch survives, connected to nothing, and the lamp stays lit anyway
Machine and its inputs	The form is kept and processing is frozen

R7. Sky closure (Laws 3 and 5)

[LAW] Every sky deficit is closed with things the Complex has. Closure has two separate parts. The sky's appearance becomes surfaces. The sky's light becomes fittings. The Codex's working theory sets out the reasoning under Law 3, in its dusk theory.

The surfaces, by generation

[LAW] How much of the sky's look survives depends on how remembered the copy is. The rung is chosen by the generation of the instance, not by the starting generation of its sky-facing faces (chapter 9), so a fresh outdoor capture still gets canon's richest closures even though its fronts are streamlined.

Instance generation	Captured in daylight	Captured in darkness
0 to 1.5	An enclosing surface in the stock block closest in colour to the sky colour at capture, with patches of the closest white block in proportion to the cloud cover at capture. The blue box	A dark surface from the darkest wall stock, with light-role points for stars at a density from the moon phase
1.5 to 4	A plain surface in the sky-coloured stock, no patches	A plain dark surface with a few points
4 and up	The default ceiling and light arrangement	The default ceiling and light arrangement

The light, by the moment of capture

[LAW] The sun isn't matter, so it can't be copied. It's replaced by fittings matched by look, which keep the colour and direction of the light the sun was giving at the moment of capture. Lamps that were already in the scene, street lights, lit windows, lanterns, are matter and keep their own colour and lit state.

Moment of capture	Sunlight	Substitute
Midday	White, high	Bright light-role blocks in the closing surface overhead, evenly spread. The film's street
Morning or late afternoon	Warm, low	Floodlight banks low on the sun's side, warm
Dawn or dusk	Red, very low, stars appearing	Floodlight banks low on the sun's side, red, under a dark surface with stars. The red street and the Red City
Night	None	No substitute. Only the scene's own lamps and the star points

[TECH] All of this comes from the game's own data, never from a list.

- **The moment.** The capture stores the world's time of day and the moon phase.
- **Height.** The sun's height comes from the game's celestial angle at that time.
- **Direction.** In Minecraft the sun always rises in the east and sets in the west, so floodlight banks go low on the eastern side of the enclosure for a morning capture and the western side for an evening one.
- **Colour.** Vanilla tints the horizon at sunrise and sunset with a colour from a fixed function of the celestial angle. The core carries the same function, so the server can compute it at capture. Away from sunrise and sunset the light is white.

Floodlights in Minecraft

[TECH] No single block can flood a street. Block light in Minecraft is at most 15 and falls by one per block, so the brightest lamp reaches about fifteen blocks. The floodlight effect is built from three parts, all vanilla.

1. **Banks.** Visible fittings, a few light-role blocks from stock set low in the closing wall on the sun's side, each behind a housing of the stained glass nearest in colour to the sunlight's colour. This is what the player sees as the floodlights.
2. **Throw.** Vanilla's invisible light block, placed through the enclosure's open space so the light reaches as far as the sun's did, densest near the banks and thinning across the enclosure, with pools where the light falls on the ground. The level of each placed light block follows a falloff from the banks, so the light has a direction.
3. **Colour.** A tint volume over the enclosure colours block light toward the sunlight's colour, strongest beside the banks and fading toward the far side, so the scene's own white lamps on the other side still read white. In Mo this is the lightmap tint, which works with any renderer, but the lightmap is one table for the whole view, so the tint follows where the camera stands, strongest by the banks and

fading as the player walks away from them, and while it's on it colours every block light in view. Only once the coloured light module of chapter 34 exists are the throw's light blocks given the sunlight's colour, so that the colour varies from block to block and the scene's white lamps stay white.

[YOUR RULE] Every block here is vanilla. The substitutes are placed by the law from the moment of capture and from stock, never from a template.

R8. Instancing echoes (Law 2)

[LAW] How many there should be is a relationship too. With a small probability rising in g, an object repeats as a chain of instances of the same object, identical flaws and all, shrinking and sinking each step, as Async found with the closing-down signs. The chain follows the Codex's best-fitting rule, where the ratio itself shrinks by a constant factor.

```
width(1) = w0     ratio(1) = 0.785     ratio(i + 1) = ratio(i) * 0.744
width(i + 1) = width(i) * ratio(i)        1.44, 1.13, 0.66, 0.29 m
gap(i + 1)   = gap(i) * 0.939              3.47, 3.26, 3.06 m
each step sits a little lower; the last is barely clear of the floor
```

[DESIGN] An echo chain always runs toward the nearest fresher instance of the same memory, and may cross a wall. Following shrinking objects and cutting through where the sequence predicts the next one, as Async did, leads toward fresh islands.

R9. Convergence

[LAW] At high generation, by default above 7, a shell stops resembling its source and converges on the prior. Its materials are the urn's, its proportions the prior room shape's, its lights the prior arrangement. Where a room's span exceeds the prior span, partitions appear at the prior spacing, which is teaser stage 8. Where a partition crosses a doorway port, the opening takes the shape of the commonest arched or plain doorway in the stock, which is how an arch "that wasn't there" appears, stage 7, without anyone drawing an arch.

R10. Re-expression (Laws 5 and 6)

[YOURS] [CANON] This is your idea. Instead of keeping a place's geometry and bending it, the Mansion learns its materials, its general shape and how its rooms connect, then rebuilds it from the ground up in its own image, so it fits with everything else and loses the incomprehensible parts of the original. Canon shows exactly this in the dark maze copy of Clark's store, where the layout is the Complex's own but the wall design, trusses and signs are the store's (chapter 9).

[LAW] Re-expression is how most of the Mansion is made. A fresh island at generation 0 keeps its geometry. From generation 1 to 3 an instance is re-expressed with probability rising from 0 to 1, decided once per instance by its seed. Above 3 it always is. Sky-facing parts start at 2.5 (chapter 9), so exteriors are almost always re-expressed even on fresh islands, and the absurd outside of a Minecraft build almost never survives in its own geometry.

What is learned

[DESIGN] Re-expression never has to understand a build. It only measures what the capture already produced.

1. **Rooms.** The captured open space is split into rooms at its constrictions. A distance transform of the open space finds where it narrows to doorway width, three blocks or less, and those narrows become cuts. What lies between cuts is a room.

2. **Each room's measures.** Floor area, the longest and shortest horizontal spans, height, and which way it's elongated.
3. **Connections.** Which rooms meet at which cuts, and what kind of opening each cut is, a doorway, an arch, a gap, a stair, a hole.
4. **Materials by role.** The dominant floor, wall, ceiling, structure and light materials of each room, from the Role Lexicon.
5. **Objects.** Every object of each room, with its role, its size, whether it stood against a wall or in the open, and the order the wall objects came in going round the room.
6. **Ports.** The openings where the capture's open space met its boundary, kept so Genesis can join the re-expressed place to its neighbours.

How it's rebuilt

```
for each room r, in the connection graph's order from the open-space core
  span    = lerp(measured(r), prior(r), w(g))          w rises from 0 at g 1 to 1 at g 3
  height  = lerp(measured height, prior height, w(g))
  shape   = rectangle of span, or an L where the measured footprint was clearly two-limbed
  walls, floor, ceiling = r's own dominant materials, then pulled by R3
  lights  = the prior light arrangement (chapter 11) in r's own light material
  wall objects   = put back against the walls in their original cyclic order, positions spread evenly
  open objects   = put back toward the room's middle, keeping their order along its long axis
  containers, signs, banners, paintings = always kept, with identity intact (Law 2)
  connections = an opening of the measured kind, shaped from stock, where r meets its neighbours
rooms are laid out by their measured relative positions, drifted by R2, never on a grid
```

[YOUR RULE] Room sizes are continuous and come from each room's own measures pulled toward the prior, and rooms are placed by their measured positions, so no lattice appears. The founder prior supplies numbers, never shapes.

Outdoor space becomes a room

[LAW] When R10 meets former outdoor space, a yard, a car park, a street, it doesn't keep it outdoors. Each connected region of exposed air is treated as one room and rebuilt like any other.

1. **Walls.** The Complex's own, the default wall material pulled by R3, standing where the region's edge was.
2. **Ceiling.** At the prior height, or higher where the region was wide, with the default light arrangement. The sky closure is gone.
3. **Floor.** The ground is kept as the floor, the lawn, the paving, the road, in its own material and its captured colour.
4. **Fronts.** The faces of buildings that bordered the region are kept only as features on its walls, a door, a pair of windows, a porch rail, one block deep, against the wall.
5. **Things.** Trees, lamps, benches, signs and vehicles are kept as objects in the room, placed by the usual rules.

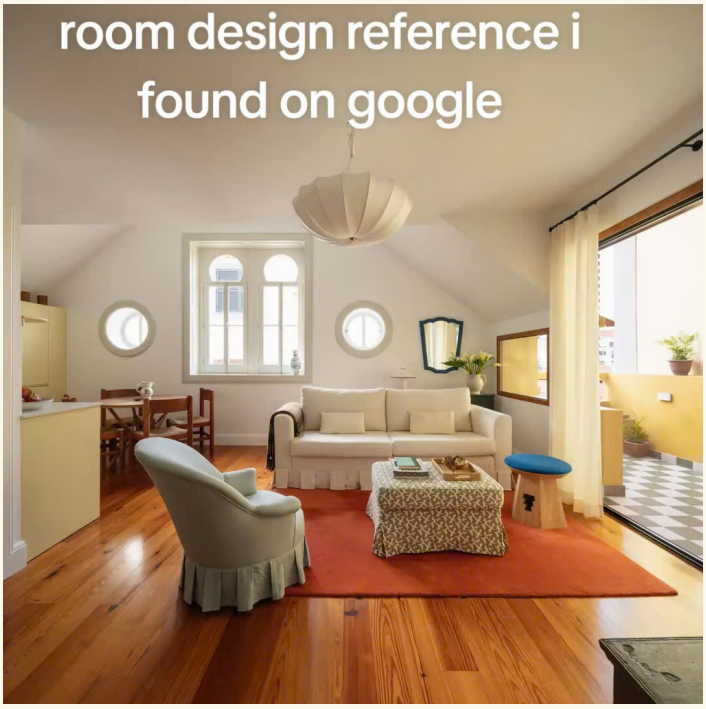
[CANON] That's exactly what canon's oldest-looking outdoor copies look like. Mary's yard is grass, paving and an old street lamp inside a yellow-walled room under fluorescent panels. The playground is a play area inside a room. A re-expressed yard is a yard inside a room, and it no longer counts as outdoor space.

What survives and what doesn't

Survives	Doesn't survive
The materials of the place, room by room	Its exact geometry
How many rooms, and roughly how big	Odd angles, curves, stacked overhangs, floating parts
Which rooms connect, and how	Connections that were only gaps in decoration

Survives	Doesn't survive
Its notable objects, signs, chests, paintings, beds, tables	Clutter below the R1 threshold
The order of things round a room	Their exact positions
Its ports to the outside	Its silhouette, which capture never takes anyway

[DESIGN] An insane build becomes rooms made of its own materials, holding its own things, joined the way its own rooms were joined. That's recognisably that place, remembered by something that has never understood it.



Reference only, and fan work, not canon. A real room photograph used by a fan project as its source.



Reference only, and fan work, not canon. The same fan project's version of that room as a copy of a copy. Clutter gone, wood floor turned to brown carpet, the pendant lamp become recessed lights, the round windows gone, colours washed out, and the window looking onto another room of the Complex instead of the outside. It's a good picture of R1, R3, R5, R6 and R10 together.

Two render paths

Most objects are voxels, with drift rounded to whole blocks. But canon's errors are often smaller than a block, the 1.27 by 1.14 metre tile, signs of 1.44, 1.13 and 0.66 metres, a switch "set crooked, at an angle". Those use vanilla display entities, which take any transform.

- An object takes the display path when its drift includes a scale that isn't whole, a rotation that isn't a right angle, or a sink depth that matters visually, or when it's an echo.
- Display objects get invisible collision where solid.
- [DESIGN] The budget is at most 32 display entities per chunk and 600 in view per player. Version 0.1 allowed ten times more, which would have cost both frame rate and tick rate. Over budget, objects fall back to whole-block rounding.

Mechanisms (Law 9)

[LAW] A copied block entity keeps ticking with probability f(g), e to the power of minus g over two, decided once per instance. At generation 0 a copied furnace still smelts. At generation 3 most copied machines are shells with their contents intact. Redstone dust keeps the power level it was captured with, so circuits stay in whatever state they were caught in.

11. Genesis: How the Complex Grows

[LAW] Law 8 says the Complex extends itself through its openings, and that connection isn't something it aims for but what growing from its own openings produces. Genesis is a growth process that can be evaluated anywhere, in any order, by any thread, and still give the same answer.

The problem

[TECH] Minecraft generates chunks in an unpredictable order, in parallel, and only near players. A true growth simulation, starting somewhere and attaching rooms at open ports, depends on order and is global, so it can't run that way. Grid-based generators dodge the problem by making every cell independent, which is exactly the visible lattice the charter bans. Genesis solves it with bounded locality. Space is divided into irregular territories around blue-noise nuclei. Each territory runs its own small growth simulation that depends only on its nucleus and on seam contracts agreed with its neighbours.

Nucleation without a grid

[DESIGN] Nuclei are points where the Complex once extended itself, spread as a Poisson-disk set, random but never too close, which gives an organic layout with no repeating period.

```
candidates  a hashed set of jittered points, several per hash cell, used as an index only
priority(c) = hash64(worldSeed, c)
radius(c)   = R0 * rho(c)                    rho from the Lineage Field; islands claim more room
distance    d^2 = dx^2 + dz^2 + (KAPPA * dy)^2   KAPPA 3.5, so territories are wide and flat
keep(c)     if priority(c) beats every candidate within the larger of the two radii
```

Whether a candidate survives depends only on candidates within its radius, so any thread can evaluate any nucleus and get the same set. Territories come out 90 to 180 blocks across and 18 to 40 blocks tall.

Territories and forced joins

A territory is the set of points closer to its nucleus than to any other, weighted by the nucleus's vigour. Fresh islands are more vigorous. Territory boundaries are never drawn as walls. They decide ownership only, and instances may extend across them, where R4 resolves overlaps.

[LAW] Two kinds of join come out of this, and both are canon. Where ports line up, because seam contracts put them there, you get a natural seam, a doorway, a corridor end, a pit. Where two territories' rooms collide, you get a forced join, a crack across a hallway floor where two kinds of room meet, a room half one material and half another, a sagging floor.

Seam contracts

For each pair of neighbouring territories, both sides compute the same contract from a hash of their two nuclei.

Contract field	Rule
Seam count	One to three between horizontal neighbours, zero to two between vertical ones. Every territory has at least one
Portal positions	Near the centre of the shared face, jittered, at the nearer territory's floor level
Portal type, horizontal	Doorway 62 per cent, open corridor 16, arch 8, crawl gap 6, broken wall 5, window gap 3
Portal type, vertical	Square pit 45 per cent, stair run 25, ladder shaft 15, one-way chute 10, collapsed floor 5
Keep-clear volume	Air reserved through the portal that no merge may fill, so a seam can't be sealed by accident

[DESIGN] The type shares are tuned to canon's record of transitions, doors most of all, then gaps, holes, crawl spaces, chutes and ladders. They're numbers, not shapes. The shape of each portal comes from the stock, the commonest captured doorway or arch, the commonest ladder material.

Growth inside a territory

1. **Seed.** Choose the nucleus memory from the memory urn (chapter 12). Place one instance with its open-space core on the nucleus, in one of eight orientations chosen by hash.
2. **Spines.** For each seam portal, route a path from the seed's nearest port to the portal with A star search over a coarse internal guide that's never visible. Costs favour straight runs, reuse of open space and little climbing.
3. **Port matching.** Walk each spine and place instances whose ports meet the frontier port, same position, opposite facing, compatible size. Candidates are descendants of the seed memory, at generation rising with each step from the seed, and sea memories at the territory's sea generation.
4. **Frontier fill.** Remaining open ports are taken from a seeded priority queue until the territory is 45 to 70 per cent open space. Instances may overlap by up to 30 per cent of their open volume, because overlap is a feature, merged by R4 into irregular architecture.
5. **Sea fill.** Leftover voids are packed with overlapping, rescaled sea instances, high-generation copies of real memories. Small leftover voids become solid structure fill, or rarely a half-formed room.
6. **Connectivity repair.** Build the walkable graph from open-space cores, ports and overlaps and find its components. Any component not joined to a portal is joined to the nearest one by a punched opening, the commonest doorway or the picture-rectangle of R6, through the thinnest wall between them.
7. **Character.** Space-syntax measures of the plan are nudged toward a character per territory, an open hall, a maze, a warren, a dead-end cluster. Async found most divisions were dead ends, so dead-end rooms target a quarter to two fifths.

[LAW] [YOUR RULE] Walkability is inherited, not guaranteed, but one rule is absolute. No pocket may exist that a player can enter and never leave. One-way chutes and pits are allowed, as in canon, but every one of them must land somewhere joined to the rest. That rule is a test that runs on every plan.

Verticality

- **Strata.** The flattened distance makes territories wide and flat, so the Mansion reads as an endless stack of floors, each one locally different.
- **Ceiling cavities.** Where strata are spaced far enough apart, the space above a ceiling is left hollow and crawlable, with structure fill above, so ceiling tiles can be removed and climbed through. "More than one floor, and I use that term loosely."
- **Square pits.** Vertical seams are square pits. Most pits in a cluster drop into a ceiling cavity and dead-end. Exactly one in each cluster is a true seam to the territory below, which is canon, since the room under Room 14D "has a ceiling where the other holes should come through". Clusters form where a territory has several vertical seams near each other. No grid of pits is drawn by hand.
- **Chutes.** One-way slopes of slabs and stairs at about one rise in two. You can't climb back, as Ravi couldn't.

The prior light arrangement

[CANON] [LAW] The default ceiling light isn't a model. It's a law with canon's numbers in it. Light-role blocks are set one block above the ceiling plane behind a window-role block that diffuses them, on a rhythm taken from the ceiling tile Async measured, 1.27 by 1.14 metres, which comes out as a line every five blocks one way and alternating four and five the other, so it never matches a standard two by two or two by four grid. Which light block and which diffuser comes from the stock and the founder prior, not from a fixed list.

Plans

A territory's growth result is a compact plan, a list of instances with memory, transform, generation, seed and merge priority, plus portals, keep-clear volumes, punched openings and character data, usually 2 to 20 kilobytes. Plans are pure functions of the seed, the pinned epoch and the island registry, so they're cached in memory and recomputed when needed, never stored.

12. The Lineage Field and the Snowball

[LAW] Deeper in the Complex means older, how many times something has been copied, not how far it is from a door.

Generation as a field

- **The sea.** Ordinary territories sit at a sea generation between 9 and 16, varying slowly through low-frequency noise. Everywhere, by default, is old.
- **Islands.** A territory seeded by a fresh capture has generation near 0 at its seed instance.
- **The echo halo.** Descendants of the island memory are placed along growth paths, generation rising by 0.6 to 1.6 per step, spilling into neighbouring territories through their seams. Walking away from a fresh copy of your house, you pass copies of copies, a maze of its materials, an inside-out version, rooms with its walls and nothing else, then the sea.

Two urns

[LAW] Law 7 is implemented twice, once for materials and once for memories, both as Pólya urns.

```
material urn, per role r
  P(material i) = (a_i + n_i) / (A + n)
  a_i  founder prior counts for role r, scaled by the gamerule mansionFounderWeight
  n_i  times material i has been realised in role r inside the Mansion

memory urn, for seeding sea nuclei
  P(memory m) = w_m / W
  w_m starts at 1 when m is captured and rises by 1 each time m seeds a sea nucleus
```

[LAW] The memory urn is where the snowball becomes visible. The first memories the Complex takes from a world get a head start that never fades, so the endless sea in each world is built out of that world's first captures, remembered a dozen times, with materials pulled toward the urn's. The room a player fell out of on their first night becomes, far out, the shape of the endless rooms. Each world gets its own first room.

[THEORY] That gives Wendigoon's theory, that the yellow wallpaper comes from the first thing the Complex touched in Ivan Beck's past, a working mechanism inside the mod without the mod ever authoring Beck's room. In every world, the first thing touched becomes the default.

[DESIGN] The founder prior is the one fixed input, and it is numbers, not content. It holds counts per role for a calibrated set of vanilla materials matching canon's look, a greyish yellow wall, a stained beige carpet, a speckled white ceiling, a white diffuser, plus canon's proportions and light rhythm (Appendix C). With mansionFounderWeight at its default, the deep sea reads as the classic yellow rooms within the first few hours. At zero, the default is grown entirely from the world, so a world of stone houses gets stone-coloured endless rooms. Appendix E leaves the choice to you.

[TECH] Because a world's urn changes over time but first generation must be reproducible, the urn is snapshotted into epochs, every 24 in-game hours or 64 captures. A territory is planned with the epoch current when it is first touched, and that pin is kept in level.dat until the territory is fully generated.

Variety that never dies

[LAW] Pure copying ends with one room repeated forever. Copying with mistakes settles at a steady variety. The mutation share is set so the deep sea is about 90 to 95 per cent default rooms, with a steady trickle of rarer memories at lower generation, whatever unusual places the world has supplied, a pool if someone built one, a chapel, a market. They always come from real captures.

Landing sites

[LAW] Copies are made at openings and attached at openings. Each opening gets a landing site in unrealised space when its ledger entry is created. Later captures from the same opening gather around the same site, so a doorway open for weeks builds up a branch of several copies of your house, like the film's Kelowna Branch with its three copies of Clark's store.

> **Invariant, islands never rewrite realised space**
>
> Islands are inserted only where nothing has been generated within two territory radii. Space that has been generated changes only by re-remembering (chapter 14). That keeps inserting your house into a partly explored world safe and deterministic.

13. The Generator and Determinism

The generator

[TECH] The dimension uses its own chunk generator, registered with its own codec. It extends the game's abstract chunk generator because that is the engine's contract, but uses none of vanilla's generation machinery. The biome source is a single fixed biome with no features, spawns or music.

Stage	What happens
Structures	Nothing. There are no structures
Fill from noise	The rasteriser. Every block is decided here
Surface and carvers	Nothing
Decoration	Writes block-entity NBT from memories
Mob spawning	Spawns passengers as Still Lifes and display-entity objects
Base height queries	Answered from the plan, the floor nearest the queried height

Rasterising a chunk

```
for chunk (cx, cz)
  territories = nuclei within reach of the chunk plus the largest instance radius
  plans       = territories mapped through the plan cache            asynchronous
  instances   = instances from plans that overlap the chunk
  for each 16-block section
    if no instance overlaps   fill with structure as a single-value container
    else for each voxel       resolve the remembered voxels of overlapping instances by R4
  write realisation data into the chunk attachment
```

[TECH] The fill stage returns a future, so generation waits on plan futures without blocking worker threads, and never blocks the server thread.

Saved like any other level

[DESIGN] Version 0.1 never saved the Mansion's chunks and regenerated them on every visit. That would have needed mixins on chunk storage, clashed with other mods, and recomputed lighting every time. Version 0.2 saves the Mansion's chunks normally. Player changes are just saved blocks. Re-remembering rewrites saved chunks instead of regenerating them (chapter 14).

Determinism

For any block at first generation, the result depends only on the world seed, the nuclei nearby, their plans, the memories, the pinned epoch, and the island registry. None of those depend on generation order or thread scheduling. Every random choice uses the project's own generator seeded from a hash of the world seed, a purpose and the ids involved. Floating-point arithmetic is strict on Java 21, so doubles are safe, provided every transcendental function comes from StrictMath (chapter 36). Test T-12.1 generates the same 512 chunks in four orders on eight threads and compares them byte for byte.

14. Observation, Stability and Re-Remembering

[THEORY] Your hypothesis, which the Codex says points the right way, is that the Complex is steadier where people are. Combined with Clark's line it becomes a mechanism. A place watched continuously is remembered once and kept. A place abandoned and revisited is remembered again, and each remembering is worse.

Who holds what steady

Observer	Holds steady
A player	Every section within 24 blocks, plus visible sections in view up to 64 blocks, with occlusion tested on the plan's room graph
A group within 32 blocks of each other	The radius grows by 60 per cent per extra player, up to five. Three people hold more than twice what one does, the three-person rule as a consequence
An Observer block placed by a player, facing into a room	A cone 16 blocks deep. Async's motion cameras, made from the vanilla block named for the job
A bed someone has slept in	A 6-block radius. People lived in there
Force-loaded chunks	Treated as observed

Forgetting

[DESIGN] Each Mansion chunk records in its attachment when it was last observed. When a player comes near a chunk that has gone unobserved for longer than mansionForgetDays, three in-game days by default, it's re-remembered before they arrive.

1. The chunk's saved NBT is read on the IO worker, before the chunk is loaded.
2. Its current blocks, player changes included, are captured as a local memory. Your changes are matter, so they're part of what gets remembered.
3. The chunk is rasterised again at one generation higher, or up to three for very long absences, with the current urn epoch, and its block entities and entities are carried across with drifted positions.
4. The new NBT is written back through the chunk storage, and the local memory is kept in the chunk's attachment.

[TECH] [RISK] Rewriting a chunk's NBT before it loads needs care, so the game's own loading never sees a half-written chunk. If a player arrives before the rewrite finishes, the old chunk loads unchanged and the rewrite waits for the next absence. Spike S4 proves the path.

The result is Room 14C. You come back and the walls are a different material, the proportions seem off, ceiling tiles have been moved, your chest is a block to the left, the item frames have sunk into the floor. Identity is kept, so every item survives (Law 2). Only placement drifts. Player-made objects drift more gently, by the gamerule mansionPlayerBuildDrift, so a base changes over weeks, not overnight.

> **Never in front of you**
>
> A section held by any observer is never changed, except during a growth event with green light (chapter 18). That's the Codex's falsifier, a room changing dramatically in front of a group with no green glow.

Change behind a lone player's back

[CANON] [THEORY] Canon's worst incidents, Tench and Marvin, both began with someone alone. A lone player's small hold radius leaves nearby rooms unheld. Rarely, a room the player has seen but isn't watching changes slightly, a door leads somewhere else, a corridor is a little longer, the Codex's "flexible spaces" result applied. It's limited to once in ten minutes per player, never touches chunks players have built in, never seals a portal, and is effectively impossible for a group of three, so the three-person rule is something you feel.

Building inside

[CANON] A-Space was pitched as unlimited storage and housing, and Kane says living there would be "a pretty tedious, difficult task". The Mansion delivers that trade. A base inside stays as you left it only if it stays observed, by cameras, visits or company. Server owners can set mansionPlayerBuildDrift to zero.

PART C

The World

Openings, the reach back into our world, time, growth, atmosphere, and everything that lives inside.

15. Openings

Openings are where our world and the Complex touch. They're how players get in, how memories are captured, and how the Complex reaches back.

The three kinds

Kind	Canon basis	Behaviour
Glitch	The garage floor, Ravi's basement wall, the car on the highway, the seagull, the father's wall	Natural, unstable and brief. Usually one-way. Any size, on any surface or rarely in mid-air. Widens when disturbed by a long object
Doorway	The crack in Clark's basement wall, which he could tape round and walk through at will	Natural, stable, two-way, forming slowly after the signs in chapter 16. Rare
Threshold	Async's Low-Proximity Magnetic Distortion System, Project KV31	Artificial, built by players. Stable and two-way. Raises null-zone activity across the world afterwards

Where glitches form

Each loaded chunk near a player has a null potential, recomputed every few minutes. A glitch can form only where it's high, and even then rarely.

```
potential = E * D * J * S
  E  enclosure, how roofed, walled and underground a candidate surface's surroundings are
  D  built density, the share of worked blocks (planks, bricks, concrete, glass, modded
     building blocks) against natural ones inside the enclosure
  J  join factor, rising near existing openings and joined places
  S  world activity, 1 before any Threshold opens, rising to between 3 and 6 afterwards

a surface can hold a glitch only if the air beside it has exposure below 0.4 (rule O1, chapter 9)
```

[CANON] Basements, garages and walls are where canon's openings form, and the missing-persons spike after the Threshold opened is S. The candidate surface is always matter, a floor, wall or ceiling, apart from very rare mid-air glitches like the seagull's. The baseline is tuned so that a player who builds basements meets roughly one glitch in ten to twenty hours of play before any Threshold exists, and a player who never builds underground almost never does. The gamerule mansionGlitchRate scales it.

What a glitch shows

[CANON] Nothing announces a glitch. The surface looks normal. Only what canon shows gives one away.

- **Things fall through.** Dropped items, sand and gravel placed on a floor glitch vanish, like the wood chips in the garage.
- **Long objects make it worse.** Pushing a stick, fishing rod, trident, spyglass, bamboo, blaze rod, end rod or anything tagged as a long object into a glitch makes it widen for a few seconds while the item shakes, then contract and seal. Anyone standing on or against it as it widens is pulled through. That's the tape measure in the garage.
- **Sound carries.** At a glitch with a live link, sounds from the far side can sometimes be heard through the surface.

[DESIGN] Version 0.1 added a wobbling compass, a stick test and a torch shimmer. None of them is shown in canon, so they're gone.

Crossing

[TECH] When an entity enters an opening's surface, its collision with that surface is switched off and it falls through. The block itself never changes. A mixin returns an empty collision shape for positions in the level's set of null positions, a bitset per chunk synced to nearby clients so movement prediction agrees.

1. **Capture.** The crossing fires a capture event at the opening, so the Complex takes in the room you just fell out of, with you in it.
2. **The fall.** One and a half to four seconds of falling, with picture and sound breaking up, as in every found-footage fall. It's a short client-side effect, horizontal tearing, colour bleed, dropouts and muffled audio, and it's the only screen effect the mod ever uses. Its strength follows the vanilla Distortion Effects accessibility slider, so a player who turns that to zero gets a plain fade instead.
3. **Arrival.** Usually at the opening's landing site, near the fresh copy of the place you fell from, through a ceiling hole, out of a cabinet like Ravi, or down a chute. A quarter of the time somewhere else entirely, because canon's openings don't connect cleanly.
4. **Time.** Arrival is forward in time only (chapter 17).

[DESIGN] Fall damage on arrival is real but capped, so an unarmoured player at full health survives.

Doorways

[CANON] A doorway forms only in a place that's already joined, after the signs of chapter 16 have played out. The lights lead toward one wall, and a vertical crack of fluorescent light opens in it. When the first entity reaches into it, the concrete seems to swallow them. After that it's a stable, two-way passage the size of a door. The wall block stays and only its collision is gone, so it can be marked round with carpets, signs or item frames, as Clark taped his.

[DESIGN] The crack is drawn with display entities of the brightest light-role block in the joined place's own stock, with a light block behind it for real illumination. Nothing new is rendered.

[LAW] A doorway has growth events every one to three in-game days, each capturing the room on our side, including whoever is standing in it. That's how the Complex can end up with a Still Life of you on an ordinary afternoon.

Exits

[CANON] The only exits lead back to Earth, and nobody chooses where. Exits are rare glitches inside, about one per forty territories, more common near fresh islands, since thin places cluster around fresh captures. An exit puts you back in the world at a place you didn't choose, within mansionExitScatter blocks of the nearest opening you've been through, sometimes high in the air like Kane's camera, always later than you went in. Height is limited so the fall is survivable from full health.

[CANON] An exit can take the form of the slope from the first Found Footage, a narrow gap onto a steep downward run that ends in daylight. It's the only daylight the Mansion ever shows, and only because you're leaving it.

The Threshold

[CANON] Canon fixes what the Threshold is. A magnetic machine built around a doorway in a chamber, with a control room that looks down onto it, tested again and again, failing before it works, and opening into the Main Chamber. After it opened, Async added an airlock, built an outpost on the far side, and missing persons climbed.

[DESIGN] The Mansion lets players build their own, recognised by code from vanilla blocks, never by a structure block. The exact block pattern is left open until milestone M9 (Appendix E), but some rules are

fixed now.

- It must be a magnet and coil arrangement around a doorway, inside an enclosure, with a window overlooking it, because Async was an MRI company and an MRI is a strong, even magnet with radio coils.
- Each activation is a test. Tests fail more often than they succeed, and success becomes more likely as distortion builds up over failed tests. Nothing is scripted to succeed on a particular test.
- When it opens, the far side is wherever Genesis grows from the Threshold's landing site. There's no authored Main Chamber. If the world's captures include a pillared hall, the halo around the Threshold may well look like one, because simple repeated structure survives remembering best.
- Afterwards world activity S rises for good, glitches start appearing in other players' basements, and the Complex starts copying the facility, so inside, sooner or later, you find Standard.

[DESIGN] Version 0.1 also had geomagnetic storms with a green aurora drawn in the overworld sky. They're gone. The Codex keeps the 1972 storm as a possible trigger, not a mechanism, and drawing a new sky was decoration.

16. Reaching Back

[LAW] Once the Complex is joined to a place, it starts to change our side of the join as well, and what it puts there carries its signature, the right thing in the wrong placement (Law 10).

The join field

Each opening has a join radius that grows while it exists, 8 blocks for a fresh glitch, fading after it seals, up to 48 for a doorway and 96 for a Threshold. Reach effects happen only inside join radii, which is Law 10's test enforced in code.

Before a doorway, in Clark's order

[CANON] A place with high null potential and repeated near-glitches becomes joined, and climbs this ladder over in-game days. These are the only warnings a player gets that a doorway is coming.

1. **Power drift.** At night, lamps and powered lights in the place go out on one floor, then on another floor the next night. "Upstairs one night and downstairs the next."
2. **The crooked switch.** A copy of one of the player's own controls, the nearest lever or button in the joined place, appears where no control could be, set at an angle on a surface it can't logically be attached to, connected to nothing. It's a display entity, so it can sit crooked. The nearest real source rule (Law 5) means it's always one of yours.
3. **Rerouting.** The copied controls become the ones that work. Flipping the crooked switch toggles the place's lamps. Flipping it off darkens the place, except for a faint fluorescent hum from behind one wall.
4. **Leading lights.** Lamps flicker in sequence toward one wall in the lowest room.
5. **The crack.** The doorway opens (chapter 15).

[DESIGN] Version 0.1 had a filled map that redrew itself as a map of the Complex, standing in for the television. Minecraft has no television, and a redrawn map isn't something canon shows, so it's gone.

The signature in our world

[LAW] Once a doorway or Threshold is established, its join radius occasionally receives small copies of things inside the radius in the wrong placement, a stair half sunk into the floor, a duplicate painting overlapping the original, a sign with your own text reversed. They appear only in air, never replace or remove a player's block, and never appear outside a join radius. The gamerule mansionReachIntensity scales them.

"Doors are opening everywhere"

[CANON] Phil's last line in the film is the late state of a world. Openings raise J around them and the Threshold raises S everywhere, so over a long-lived world openings multiply. The Codex suggests a feedback loop and the Mansion implements one, with a cap set by gamerule. It can't be stopped, as Phil says, but you can be careful where you dig basements.

17. Time

The rules

- [CANON] People only go forward. Every crossing, in either direction, moves forward in time only.
- [CANON] Slips are events, not a rate. Async teams go in for hours and come back the same day. Tench lost 68 days in half an hour.
- [CANON] Sound and signals leak across time far more often than bodies do.
- [LAW] Old things inside look normal. Only things from later look wrong, and Law 1's openings are scattered through time in both directions.

Ordinary crossings

[DESIGN] Leaving the Mansion returns you with a small forward offset, up to twenty minutes of world time, usually much less. In single-player the world clock moves. In multiplayer the server clock is left alone and the offset is recorded as the player's personal date in level.dat, shown only in their own dated records. The offset never depends on how long the fall was. The Codex rates the fan exponential curve as weak, because Tench jumped without falling at all.

Tench events

[CANON] [DESIGN] Rarely, at a thin place, a player who is alone and listening has a slip. A green flash, a moment of tape-break, and when they next leave the Mansion the world is weeks later, twenty to eighty days by default. In single-player the world clock really jumps. On servers the jump is recorded as the player's personal date unless the gamerule mansionTenchJumps allows a real one. From the player's side, as for Tench, a few hours at most have passed.

Leaks

Leak	Canon	In the Mansion
Live sound through walls	Ravi hears a television and talks to a father through a wall	A thin place is a wall in the Mansion whose instance came from a memory with a live opening. Sounds happening now within 16 blocks of that opening, doors, footsteps, villagers, jukeboxes, other players' actions, are replayed muffled from the matching wall
Sound from other times	Marvin's radio picks up 2008 and 2015	Thin places also replay sound events from earlier near the opening, from a rolling buffer kept in memory only, never written anywhere. Jukeboxes and note blocks inside near thin places drift between discs and tunes
Copies from later	The Heathrow courtyard in 1991, the 2022 flyer in 1995	When a region is re-remembered, newer captures can appear in it, so an old area can come to contain a copy of a house you built after you first explored it

[DESIGN] Two-way voice through a thin place in 0.1 depended on a voice chat mod and recorded players. It's gone, by the dependency rule and by the charter.

18. Growth Events and the Green Glow

[CANON] The green glow is canon's one reliable sign of the Complex actively changing, where walls move, rooms grow, or a door opens on a half-formed room. In the Mansion it's the visible face of Genesis and capture happening live.

When growth happens live

- At stable openings, every one to three in-game days, each event capturing the world side.
- At the frontier, rarely, when a player stands at the edge of generated space beside a territory whose plan includes a half-formed instance. [THEORY] Wendigoon floats the idea that the Complex generates as people approach its edges. It's used only for these rare, visible events, never as the general model.
- At re-remembering, when an observer happens to be present. This is the only way a held room ever changes, and it always shows green.

What players see

1. **Green light behind cracks.** For the length of the event, the space behind moving wall sections is lit by green light blocks seen through the gaps. It's the phenomenon, not matter the Complex copied, so it's removed when the event ends. A player quick with a pickaxe may keep one.
2. **Walls move.** Over four to twelve seconds, affected wall sections become display entities and slide or stretch to their new places with smooth interpolation, then become real blocks again. No geometry teleports.
3. **Stretching.** In stretching events, wall patterns lengthen horizontally, and markers a player has stuck to the wall, item frames, signs, paper in frames, stretch with it. Async's taped squares can be repeated.
4. **Half-formed rooms.** An event can end with a room whose objects are still sunk into the floor and whose floor sags, frozen at the generation where the event stopped, the Static Dead End.

[DESIGN] Growth events count against the display-entity budget, and large ones are split into stages so the budget is never exceeded.

19. Atmosphere

Lights keep their captured state

[CANON] [LAW] Every light-role block in the dimension is self-sustaining in the state it was captured in. A lamp captured on stays on unpowered forever, which is canon's "the lights never go out on their own". A lamp captured off stays off forever, which is why two of the three copies of Clark's store are dark (chapter 9). Copper bulbs and other mods' lamps behave the same way, and cutting wires changes nothing. Lights placed by the default light arrangement are always lit. Breaking a fitting gives you the ordinary item, "fully working, nothing strange". There's no darkness anywhere except where a fresh capture brought it, an unlit cellar, and even there the sea pulls it back toward lit. Hostile vanilla mobs never spawn, because the biome has no spawns and the light is always on.

The default light

[LAW] The default light arrangement is the law in chapter 11, not a model. Light-role blocks one above the ceiling behind a diffusing window-role block, on the rhythm of the 1.27 by 1.14 metre tile. Which blocks, the world's stock decides, pulled toward the founder prior. Anyone who climbs into a ceiling cavity with a pickaxe can repeat the Lighting and Tile Survey on whatever their world's Complex has made.

The hum

[CANON] "The endless hum of fluorescent lights turned all the way up." Every light cluster produces a synthesised 120 Hz hum with harmonics at 240, 360 and 480 Hz, a slight detune per cluster and a slow wander in loudness. Older fittings hum louder, as Async's report says loosening laminations make the hum worse, so loudness rises with generation. The hum is the soundscape. There's no music in the dimension. Its volume follows the vanilla sound category slider it belongs to, so no extra setting is needed.

Sound

[DESIGN] In this version sounds use vanilla attenuation. Room-graph acoustics, sound bending round corners and through doors, are a later improvement, built only if the core feels right without them.

Cold

[CANON] Each territory has a temperature from its provenance. The sea is cool with no effect. Old night-outdoor captures and deep strata can be cold, with vanilla freezing building slowly while you stand still and wearing off while you move. Leather armour protects, as in vanilla. Fresh captures keep the temperature of their original biome.

Food and water

- [CANON] Water is matter and copies as water. Bathrooms, kitchens and pools carry it. Copied cauldrons are filled.
- [CANON] Food appears only where it was captured, kitchens, pantries, chests. Under prior pull it drifts toward spoiled at higher generations, rotten flesh, poisonous potatoes. "Very hard to find, and might be rotten."
- [CANON] Still Lifes are edible. Interacting with an inert one while sneaking yields its filling, a vanilla bowl item with the food component, low in saturation. It's the reliable food of the Complex, and it's a copy of a person. The mod never comments on it.

20. The Still Lifes

[CANON] "It's more that it remembers them." A Still Life is a passenger of a capture, a person who happened to be in a place at the moment the Complex took it, remembered with their form mostly intact and their function mostly lost (Laws 1, 2 and 9).

Where they come from

- **Players** inside a capture at the moment of the event. Mary's copy is Mary in Async's interview room. Pirate Clark is Clark on commercial day. Yours is you at that moment, in whatever you were wearing. Usually it's an unremarkable moment, because the copy is about the place, not about you.
- **Villagers** captured in villages near openings or by retrospective capture. They're Minecraft's ordinary people and the main population of Still Lifes in most worlds.

[LAW] The rarity is canon. A person is copied only if they were in a place at the moment it was taken, and capture happens in events. In a typical world you'll meet dozens of villager Still Lifes, a handful of player ones, and very rarely yourself.

Drawn from the real skin

[LAW] Faces fail first, because a face packs the most detail into the least space. The renderer takes the template's real skin and the standard player model and applies seeded distortions that grow with generation. Fresh copies are "less distorted than most".

Distortion	Canon example	How it's drawn, with no new textures
Extra eyes	The bearded one, two on the cheeks and two on the forehead	Small extra quads on the head that sample the skin's own eye rows
Merged angles	The red-haired one, several angles of a portrait merged	The head drawn two or three times with small turns, inside one silhouette
Split face	Mary's, two noses, split down the middle	The head split in halves and offset, one half's face mirrored
Proportion drift	Pirate Clark, over eight feet and wire-thin	Per-limb scaling, thinner across, longer up, one arm too long or too thin
Extra digits	Six fingers on a hand	A small extra hand piece at the wrist using the arm's own texture
Fusion	The hat grown into the head, the chair leg as a real peg, ALS fused to his wheelchair	Equipment touching the body drawn as part of it at body scale. A held item can replace a limb. A seat captured under them becomes part of the entity
Half-sunk	The grey figure with a phone, half in the floor	Spawned with its feet below the floor by R2's vertical drift
Colour loss	The grey, statue-like figure	At very high generation the skin is desaturated toward one stone tone

Function

[LAW] Each Still Life gets a behaviour fidelity f, e to the power of minus g over 1.6, which unlocks behaviour in order. Fresh copies keep the most.

Fidelity	Behaviour	Canon
Below 0.15	None. Stands, sits, stares at its lap	The bearded one, who reacts to nothing

Fidelity	Behaviour	Canon
0.15 to 0.35	One habit, a single captured action looped without purpose	ALS and his lamp
0.35 to 0.6	Startle, a shriek and a lurch when surprised, then near stillness. Flees fresher, stronger Still Lifes	The red-haired one, scrabbling at a corner
0.6 to 0.85	Territory and pursuit, patrols a den, charges intruders, runs on all fours	The Speaker Room Still Life, fast and agile
Above 0.85	Hunting, knocking on doors, bleeding when hit, roaring when hurt	Pirate Clark

[CANON] Still Lifes have no real biology inside, so poison, wither, regeneration, healing and harming effects do nothing. They aren't invincible. They have ordinary health, and their filling spills when broken.

Behaviour as form

[LAW] Every player has a short rolling trace on the server, kept in memory only, of the last thirty seconds of discrete actions, using a door, interacting with a block, swinging, crouching, eating. If they're captured, the trace goes into the template, stored with the memory in chunk NBT. A Still Life replays fragments of it without the purpose.

- A player who used a door yields a Still Life that knocks, politely, then at higher fidelity breaks the door. It knows that people knock but not what a door is for.
- A player who toggled a lever keeps toggling things, like ALS's lamp.
- A player who was eating keeps raising an empty hand to its mouth.
- Villagers repeat their profession's motions at nothing.

Dens

[CANON] [DESIGN] Hostile Still Lifes, those above 0.6, make dens in rooms with soft fill, clothes piles from their own captured armour, wool, carpet, broken furniture. Canon plays the song "Still Life" in its dens and never explains why. The Mansion never reproduces that song. A den's speaker is a jukebox or note block from the stock that replays whatever disc or tune it held at capture, so the music always comes from somewhere real. If the stock has none, the den is silent.

Meeting yourself

[DESIGN] If a Still Life was made of you, the first time you see it nothing special happens. No sound, no message. It's just you at that moment, remembered slightly wrong. If it was made of a friend, it has their skin, their gear from that day, and fragments of what they were doing. That's the most powerful thing the mod can do, and the right way to do it is quietly.

Animals and plants

[CANON] [LAW] Kane asked what counts as an animal. Captured animals become form-only Still Lifes, a cow that never eats, a parrot that never flies. Animals that fell in alive are alive, and can starve. The film's seagull came through working.

[THEORY] Your hypothesis that copied plants are Still Lifes is followed. Captured plants never grow, wilt or drop leaves. A bitset in chunk NBT marks them, and a mixin suppresses their random ticks. Plants a player brings in alive grow normally under the fluorescent light, as real plants can.

21. The Lifeform

[CANON] Tall, stick-thin, wiry, far stronger than it looks. It hides just out of sight, calls in a human voice, deepens it and charges once seen. If it loses you it keeps scanning quietly. It grows from bacteria and human remains, but it is not a transformed person.

No spawners

[CANON] Lifeforms exist only where they've grown. No spawn rule, no spawn egg in survival, no "appears after too long". Each has an origin site you could find.

Life cycle

Stage	Duration	Form
Substrate	At once	Remains, where a player, Still Life, villager or mob died. A clothes pile and a dark stain. The young man in Missing Persons was overgrown with a mutated hay bacillus, his decay oddly halted
Film	One to three in-game days	Dark growth spreading from the remains along the edges where floor meets wall, sculk veins used purely as a shape, "something small and black growing out from under a skirting board"
Colony	Three to eight days	A room overgrown with dark, vine-like growth that moves, tendrils rooted at the remains, sensing vibration. The tentacled thing in Motion Detected is this stage seen on a camera
Walking	After that	The tall figure detaches and walks. The colony site stays as a stain and a clothes pile

[CANON] Your death matters. Die inside and your remains become a substrate. Come back days later and a trail of growth may lead to a room where something is growing. It isn't you. Kane was clear about that. It grew from what was left.

Senses and behaviour

- **Hearing.** It uses the vanilla vibration system that sculk sensors use, so footsteps, doors, breaking blocks and eating give you away, and sneaking and wool help.
- **Staying unseen.** While stalking it keeps out of every player's view, using occlusion along the room graph, waiting at corners and in holes at the ends of halls.
- **The lure.** [DESIGN] Canon's Lifeform calls "hello?" in a human voice. The Mansion won't record players' voices, so it mimics sounds it has heard players make, their footsteps, their doors, their eating, from out of sight. It's the honest version of the lure.
- **The charge.** Once a player has it in clear view for about half a second, it roars and charges, stomping despite its thin legs. A little faster than a sprint in open halls, slower through doorways and tight turns, so escape through cramped places is possible, as Marvin's was.
- **Temperament.** [THEORY] Individuals differ. One gives up at a pit's edge, as in Pitfalls, one never stops, as in Found Footage #2.
- **Damage.** Very high. It isn't meant to be fought, though it can be killed.
- **Green as it closes in.** In Found Footage #2 the hallway lights glow green as it approaches. Within eight blocks of a charging Lifeform, lights in the same room read greener through a client tint.

22. Async Without Authors

[DESIGN] Version 0.1 filled the Mansion with authored Async outposts, logs and paperwork. They were hand-made content, so they're gone. In a Minecraft world, Async is whoever explores. The players become Async, and the Complex remembers them doing it.

The player as Async

[LAW] Everything a player brings in is matter, and the Complex copies matter. Leads strung between fence posts as guide cables, Observer blocks set up as cameras, signs with survey notes, paper in item frames taped along a stretching wall, a camp with a bed. When those places are re-remembered or captured into new territories, their copies turn up elsewhere, drifting and sinking, with text intact. Players exploring a server's Mansion months later find the archaeology of the people who came before them, and it really is theirs.

Vanilla items that gain meaning

- **Observer block.** An Async motion camera that holds rooms steady (chapter 14) and still emits a pulse on movement, as in vanilla.
- **Lead and fence posts.** Guide cables.
- **Map.** Works, until re-remembering changes what it showed. "The floor plan changed again."
- **Bell.** An alarm that carries along the halls.
- **Any axe or pickaxe.** The saw Async used to cut through walls after the signs.

Text

[DESIGN] The mod writes no story text into the world. Kane's unrevealed story, Beck's goal and the main ending, is never written. Player-facing names of things use Async's vocabulary. Whether there are advancements, and whether they're written as dated Async records, is a decision in Appendix E.

23. Sources of Memory

[YOUR RULE] [LAW] Every memory in the Mansion comes from the player's own world. There is no other source of content. The founder prior adds numbers, never shapes.

Live captures

Captures at real openings as they happen, glitches, doorways, the Threshold and things passing through. These are the most valuable memories in the game, fresh copies of places players know.

Retrospective captures

[LAW] [THEORY] Openings are scattered through time, so the Complex can also hold places in your world it touched when nobody was watching. In the background, at low priority, the mod reads saved chunks of the overworld from the region files, never generating new terrain, finds built places by their built density score, creates a retrospective ledger entry there, and captures it.

- Villages, trail ruins, ancient cities, mineshafts, strongholds and every other structure the world generated count as built places. So even a brand new world has sources from the start, and a world nobody has built in still has a Complex.
- Players' own builds count too, an old outpost, a friend's tower.
- [YOUR RULE] A retrospective opening obeys rule O1 like any other. It forms only at a spot inside a built place whose air has exposure below 0.4, so a retrospective capture of a village is the inside of one of its houses, never its street. This is the fix for the mistake that filled earlier versions with outdoor places.
- The rate is limited by the gamerule mansionRetrospectiveRate, a few per real hour by default.
- Areas can be excluded with a command. Exclusions are kept in level.dat.
- Other dimensions are captured by the same rules if an opening forms there. Basalt and soul sand at generation 0, drifting toward the default.

[TECH] Retrospective captures read chunk NBT through the level's chunk storage on the IO worker at idle priority, exactly as memory loading does, so they cost nothing on the server thread.

The founder prior

[DESIGN] The founder prior is the one fixed input. It holds no rooms and no shapes, only numbers measured from canon, material counts per role, proportions and the light rhythm. Appendix C lists them. It decides how quickly a world's default turns toward the classic yellow rooms, and it can be set to zero.

24. Canon Situations Without Canon Sites

[DESIGN] Version 0.1 hand-built canon's places, the Main Chamber, Division 14, the Kelowna Branch, as rare authored sites. Under your rules they can't exist as authored content. What the laws can do is produce the same situations from whatever a world supplies. This table shows how.

Canon	How the laws produce it
Three copies of Clark's store, a maze of its materials, an accurate copy, an inside-out copy	A doorway in a player's building gathers several captures at one landing site, and the echo halo spreads worse copies around it
Room 14C, changed between visits	Re-remembering of an unobserved room
Room 14D's square pits, most leading nowhere	Clusters of vertical seams, where only one pit in each cluster is a true seam
The Static Dead End, half wallpaper and half tile, furniture sunk, floor sagging, wallpaper stretching	A forced join, or a growth event frozen part way
The Everything Must Go signs, shrinking and sinking	An R8 echo chain of a captured sign or banner
The red street and the Red City	A night capture of an outdoor place, closed by R7 with low floodlights
The playground with green carpet	A captured indoor play area in the wrong context, or a remembered outdoor one whose grass was filled from nearby green stock
The house in the blue box, complete inside, front door onto brick	A capture from inside a house, with the outside closed by a day sky, and the front door left as a seam between the inside and outside captures
The Main Chamber's grid of pillars	A captured pillared hall, which survives remembering better than anything else
The Christmas tree room, the shrine, the caveman in a hallway	Captured places with unusual things in them, set down in the wrong context by Genesis
Doors onto brick, staircases with no steps, a hood with no cooker	R6 lost purpose
Still Lifes at their moment	Passengers of captures

[DESIGN] Canon's places will never appear as themselves unless someone builds them in their world. That's the price of having nothing authored, and it's the right price. The situations appear, and the laws make them.

25. The Player's Arc

[DESIGN] There's no win condition, because canon has none for the people inside. There is a shape to a world's life, and it comes from the systems, not from a script.

Phase	What happens	Canon echo
Before	An ordinary world. Building basements slowly raises null potential. Retrospective captures quietly begin. Nothing visible happens	Before 17 October 1989
Glitches	A first fall, alone, through a basement floor, wall or garage, with no warning. You land near a fresh copy of the room you fell from, surrounded by the sea, which is made of your world's earliest memories. You learn the hum, the cold, the scarcity, the exits. If you get out, you come out somewhere you didn't choose, a little later	The found footage
Investigation	You learn what canon's tells are. You find a copy of your own house, then worse copies. You follow an echo chain through a wall. You find the guide cables and cameras of players before you, and maybe a Still Life of yourself. You learn that groups are safer and that what you leave unwatched drifts	Async, 1989 to 1990
Threshold	You build the machine. Test after test fails. Stable access, outposts, A-Space ambitions, then the costs. Activity rises, joined places change, doorways form in friends' basements, the Complex copies your facility. "Doors are opening everywhere"	A-Space and the film

Death and return

[DESIGN] Dying inside respawns you normally in the overworld. Your remains stay in the Mansion as a substrate with your dropped items. Getting back to them means finding a way in again, and the place may have been re-remembered by then.

PART D: Building It

NeoForge 1.21.1, rendering without assets, compatibility, tests, budgets, risks and the order of work.

26. NeoForge 1.21.1 Implementation

Toolchain

Item	Choice
Game and loader	Minecraft 1.21.1, NeoForge 21.1.250, pinned. Chapter 35 has the whole verified toolchain
Language	Java 21
Build	Gradle with NeoForge's ModDevGradle plugin, Parchment mappings for readable parameter names
Project shape	Four Gradle subprojects. colouredlamps is the coloured light module, a mod of its own that ships inside the Mansion (chapter 34). core is the engine, capture maths, remembering operators, Genesis, the urns and the memory format, in plain Java with no Minecraft dependency, written against a tiny block abstraction and tested at full speed. neoforge is the adapter, registries, generator, mixins, entities, rendering and networking. lab is a separate test mod that never ships
Dependencies	**[YOUR RULE]** NeoForge only. No libraries, no required mods, no compile-time integrations. Mixin as shipped with NeoForge
Maths	Doubles with StrictMath for every transcendental function, so results never differ between machines. The project's own generator seeded by 64-bit mixing hashes (chapter 36)
Identity	Mod id mansion, display name The Mansion, package io.github.underwatermarch.mansion, author UnderwaterMarch

What is registered

Registry	Entries
Blocks and items	None
Chunk generator codec	mansion:complex
Dimension type, level stem, biome	Data-driven JSON (chapter 7)
Dimension special effects	mansion:complex, registered on the client. No sky, clouds, sun, moon or stars, and black fog. NeoForge's special effects extension takes over sky, cloud and weather rendering and draws nothing, so no mixin is needed
Entity types	still_life, lifeform, lifeform_colony, remains
Data attachment types	Memory bricks, null positions, Mansion chunk realisation data, local memories, plant flags. All chunk attachments, all serialised into chunk NBT
Block colour handlers	A wrapper round vanilla's grass, foliage and water colour handlers, registered on the client, that gives copies in the Mansion their captured biome colour and defers to vanilla everywhere else
Shipped data	The table of vanilla block colours and flatness measured by the build, read from the mod's own resources
Gamerules	The settings in Appendix B, stored in level.dat with every other gamerule
Payload types	Chapter 26, below
Sound events	Only what the hum stream needs to exist, with no audio file (chapter 27)

Mixins, each justified

Mixin	Target	Why
X1	The primary level data's save and load	One compound in level.dat for the global record (chapter 6)
X2	Block state collision, both sides	Empty collision at null positions, so things fall through openings
X3	Block changes in the Mansion level, refusing a change that only flips `lit` or `powered` at a steady position	Lights keep their captured lit state, for vanilla and modded lamps alike

Mixin	Target	Why
X4	Retired. The dimension's special effects class adjusts the lightmap itself	Red floodlight and green glow tints, following the camera
X5	Random ticks on flagged plant positions	Plant Still Lifes never grow
X6	Chunk deserialisation, Mansion level only	Replace palette entries of removed mods by fingerprint before the chunk is decoded, so no holes appear

[TECH] Access to the level's chunk storage, for reading unloaded chunk NBT and writing re-remembered chunks back, goes through an access transformer rather than a mixin where one is enough. Everything else uses NeoForge events, level and chunk load and unload, block events, entity join, dimension change, server tick, payload registration, command registration, debug text, level render stages, dimension special effects and entity renderer registration.

Threading

- [TECH] Chunk generation runs on the game's worldgen workers. The fill stage returns a future, so the rasteriser composes on plan futures instead of blocking a worker.
- Plans and remembered-instance caches run on the mod's own bounded pool, cores minus two with a minimum of one, ordered by distance to the nearest player and by where players are heading.
- Captures read the world on the server thread, at most two milliseconds per tick, into immutable snapshots. Everything else about a capture runs off the server thread.
- Reading memories from unloaded chunks, retrospective captures and pre-load rewrites run through the chunk storage on the IO worker.
- [YOUR RULE] Nothing on the server thread ever waits on a future. If a result isn't ready, the server thread carries on and picks it up on a later tick.

Network payloads

Payload	Direction	Contents
NullPositions	Server to client	Bitset changes of null positions near the player, so movement prediction agrees with the server
TintVolumes	Server to client	Red floodlight and green glow volumes near the player
Crossing	Server to client	Start and stop the crossing effect
StillLifeTemplate	Server to client	Through the entity spawn data, the skin property, equipment, distortion seed and fidelity
LightClusters	Server to client	Light-role positions and generations near the player, for the hum

Settings as gamerules

[YOUR RULE] A server config file would be written into the world folder, which breaks the storage rule. Every server-side setting is a gamerule instead, and gamerules already live in level.dat. Client-side settings use vanilla's own options, the Distortion Effects slider for the crossing effect and the sound category sliders for the hum, so there's no client config file either.

Commands

```
/mansion where                          position, territory, instance, memory, generation, observation
/mansion enter                          debug entry for development worlds (operators only)
/mansion capture here [budget]          force a capture event at your position (operators)
/mansion remember <radius> [+n]         force re-remembering of nearby chunks (operators)
/mansion opening create glitch|doorway  make an opening (operators)
/mansion ledger [id]                    look up an opening
/mansion exclude add|remove|list        keep areas out of retrospective capture
/mansion debug plan|lineage|seams|roles|why
```

Devtools

[DESIGN] Pillar 6 made real. Looking at a block and running /mansion debug why prints its provenance, for example "smooth sandstone, instance 4417 of memory 12 from opening 3, generation 11.4, R3 replaced captured birch planks, urn epoch 3, territory 5e1f". A lineage overlay colours floors by generation. A plan viewer outlines territories, seams and keep-clear volumes. In development builds only, a remembering scrubber pastes a memory into a test area with a slider from generation 0 to 14, for tuning the operators against the teaser.

27. Rendering and Audio Without Assets

Still Life renderer

[TECH] A custom entity renderer built on the vanilla player model. The texture is the template's own skin, resolved from the stored signed textures property through the vanilla skin manager, falling back to the default skins in offline mode. Distortions are per-part transforms plus extra quads in the same pass, with texture coordinates pointing into the same skin, so no texture is generated or shipped. Villager Still Lifes use the vanilla villager model and textures with the same distortions.

The Lifeform's body

[DESIGN] A skeleton of linked nodes, a spine of about eight and four long limbs of six to nine, standing about three blocks tall, with a small featureless head. Procedural walking with planted feet and two-bone limb solving, heavy stomps on contact using vanilla step sounds pitched down. The mesh is thin tubes along each chain, textured with a dark, fibrous vanilla block sprite multiplied to near black with a red undertone, since the Codex notes it "isn't really black but a fleshy red". The colony stage is tendril chains rooted in the remains, swaying and reaching toward vibrations.

The hum

1. **Clustering, on the client.** Every second, light-role blocks within 24 blocks are grouped into at most eight clusters, each with a centre and an age from its generation.
2. **Synthesis.** Each cluster is a streaming source of 16-bit mono sound at 44.1 kHz, a 120 Hz fundamental with harmonics at 240, 360 and 480 Hz falling about 6 dB per harmonic, a detune of up to 0.3 Hz per cluster for slow beating, a little buzz modulated at 120 Hz, and loudness from the number of fittings and their age.
3. **Playback.** A custom sound instance supplies its own audio stream to the game's streaming path, positioned at the cluster centre with vanilla attenuation.

[RISK] The game's sound system expects sounds to come from resources. Spike S3 proves the cleanest NeoForge 21.1 way to return a synthesised stream and to register a sound event with no file without warnings. If it fails, the fallback is a short buffer generated by the build from code and packed in the jar. That's still synthesised, not recorded, but it is a sound file in the jar, so it's decision E-9.

Display entities

[TECH] Sliding walls, stretching wallpaper, crooked switches, sub-block echoes and mirrored objects are vanilla block, item and text display entities with transforms and interpolation, spawned by the server as ordinary entities, so they work with any renderer. The budget in chapter 10 applies, and display objects are turned back into blocks as soon as an animation ends.

The crossing effect

[TECH] A short post-processing pass, made of the vanilla post-effect format and a small shader program, which is code, not art. Horizontal displacement driven by a noisy tracking signal, colour offset, brief dropouts to black, and a muffled audio dip. It plays only during a fall through an opening. Its strength follows the vanilla Distortion Effects slider, and at zero it becomes a plain fade.

28. Compatibility

Mod or kind	Plan
Any block mod, including decoration and furniture packs	Works with no special support. Its blocks are captured, given roles by geometry and tags, remembered, and used as substitutes. Removing the mod later replaces its blocks through their fingerprints (chapter 6)
Chunk renderers for NeoForge, such as Sodium or Embeddium, and Iris	The mod's client features avoid chunk meshing entirely, entities, the lightmap tint and a post effect. When a shader pack is active, detected through the loaded mod list, the crossing effect falls back to the fade
Map mods	The Mansion is an ordinary saved level, so maps work. They simply show what was there, and re-remembering makes them wrong, which is canon
Pregeneration tools	The Mansion should be left out of pregeneration, because generating empty space ahead of players blocks islands from landing there. Documented, and warned about once in the log
Chunk-system optimisers	Supported as far as the pre-load rewrite allows. Spike S4 tests the common ones
Voice chat mods	No integration. The dependency rule forbids it, and the charter forbids recording voices

[YOUR RULE] No integration uses a compile-time dependency. Anything the mod knows about another mod, it learns from tags, from the loaded mod list, or from vanilla mechanisms.

29. Testing

The Codex's falsifiers as a test suite

[LAW] The Codex ends its theory with predictions that would break it. Every one becomes an automated test, and the build fails if the mod ever shows something the laws forbid.

Test	Asserts	Where
T-1.1	Every memory and Still Life template has an opening record and lay inside that opening's capture volume	core
T-2.1	No operator changes an object's identity hash, over a hundred thousand fuzzed objects. Sign text is identical or exactly reversed	core
T-3.1	Every air block in thousands of random generated columns has a block above it. Sky light is zero everywhere	game test
T-3.2	A captured face with exposure 1 keeps no weathered variant, at most two materials and no object below the clutter threshold. A face with exposure 0 is identical to its source	core
T-3.3	Across a thousand generated territories, walkable cells of former outdoor space not yet re-expressed stay under 2 per cent, whatever closes them	game test
T-3.7	Every re-expressed instance contains no former outdoor space	core
T-3.4	A lamp's lit state is the same in every copy as in its capture	core
T-3.5	A capture started in a basement with no gap to the outside contains no sky deficit	core
T-3.6	Glass between two rooms passes the capture, a solid wall of the same thickness doesn't	core
T-6.4	R10 never places rooms on a repeating lattice. The spread of room spans in a re-expressed territory is never narrower than the spread of the captured rooms it came from, pulled at most to the prior's spread	core
T-4.1	In outdoor captures no back of a building survives unless it faced open space. A building captured from outside never keeps a full interior beyond the fade depth	core
T-5.1	Every filled block traces to a real capture, an installed block or the founder prior	core
T-5.2	No substitution ever changes a block's collision shape	core
T-5.3	Every substitute is inert, no block entity, no random ticks, no gravity, no light	core
T-5.4	A substitute for a tinted block is matched to its captured colour, never to its grey texture	core
T-5.5	Monotony-breaking picks a different block when one is close enough, and otherwise allows the merge	core
T-5.6	After removing a mod whose blocks the Mansion used, no Mansion chunk has a hole where those blocks were, and every replacement has the same shape	game test
T-6.1	Agreement with the original never rises with generation, for five hundred memories	core
T-6.2	The teaser. A sunlit basement memory remembered to generations 0 to 8 meets the Codex's stage checklist, clutter gone by 1, window lowered or flattened by 2 or 3, only a frame by 4, an empty box by 5, bent walls or a step by 6, an arch by 7, partitions and the light rhythm by 8	core, with reference images
T-6.3	The signs. An echo chain seeded with a 1.44 metre sign gives 1.13, 0.66 and about 0.29 metres, with gaps of 3.47, 3.26 and about 3.06	core
T-7.1	Ten thousand captures of one new material keep its share of any role below the configured bound	core
T-8.1	In a thousand random territories a search reaches every seam portal, and nearly all seams are port-type	core
T-8.2	No sealed pocket. Every walkable cell can reach a portal, allowing one-way drops that land in joined space	core

Test	Asserts	Where
T-9.1	A Still Life never has more behaviour than a fresher copy of the same template, and ignores effects that need biology	game test
T-10.1	No Reach effect outside a join radius, over a hundred simulated hours	game test
T-12.1	The same 512 chunks generated in four orders on eight threads are identical byte for byte	game test
T-13.1	No held chunk ever changes without an active growth event, fuzzed with several bot players	game test
T-13.2	Re-remembering keeps every item in every container. Only positions change	game test
T-S.1	[YOUR RULE] A long session writes no new files. The world folder's file list before and after differs only in files vanilla itself creates	game test
T-S.2	No chunk's attachments exceed the cap, so no oversized-chunk file is ever written	game test
T-H.1	[YOUR RULE] The human check ran on every file of the build and found nothing. No comment in any code, no dash, no tool or assistant credit, no forbidden tell	build

Canon as the answer key

[DESIGN] Automated tests catch broken laws. They can't tell whether the place feels like the Complex. For that, canon gives matched pairs, a real original and the Complex's copy of it, and every pair is a playtest. Build the original in a test world, run the laws, and set the result beside the canon frame.

Original	Canon's copy	Law it judges
The teaser's sunlit basement	Its seven worse versions, ending in the yellow rooms	Law 6
A suburban street	The film's street inside the yellow hall	Laws 3 and 4
A house captured from inside	The house in the blue box	Law 4
A store with a doorway in its basement	The accurate copy, the maze and the inside-out copy	Laws 2, 4 and 8
A yard with grass and a lamp	Mary's yard under fluorescent panels	Laws 3 and 5
A plain corridor with only the hum and fitting light	The feeling of the classic rooms before anything is copied	Atmosphere
A store captured from its basement, with glass front doors	Clark's store's three copies, the accurate interior, the streamlined front facing a car park, the dark maze	Laws 2, 3, 4 and R10
A house with a front porch and hedges, captured from inside	The house in the blue box, faithful inside, a stage set outside	Law 3 streamlining
A messy, absurd Minecraft build	Rooms of its own materials holding its own things, with a short streamlined balcony at most	Chapter 9 and R10

When "it isn't the Complex" happens, these turn it into something that can be fixed, "stage four keeps the window too long", "the street's houses are too deep".

Worlds the tests run on

[YOUR RULE] Nothing is tuned to one map. Every milestone is checked on a set of very different worlds, always on copies, never the originals.

1. A flat creative town of ordinary houses.
2. A copy of a real, long-lived server map built by many hands.
3. A fresh survival world whose only builds are generated villages and structures.
4. A redstone-heavy technical base.
5. A heavily modded world full of decoration and furniture blocks.
6. A world of absurd builds, megastructures, pixel art, floating things, the strangest the testers can find.

7. An empty world with nothing built at all, to prove the Mansion still has a source.

30. Performance Budgets

[YOUR RULE] The mod must not change TPS or FPS measurably when nothing is happening, and must never stall the server thread.

Operation	Budget, on an eight-core desktop
Server thread, at rest	Under 0.2 milliseconds per tick for all of the mod's tick work combined
Capture reading	At most 2 milliseconds per tick, a doorway growth event finished within 5 seconds, including the sky light layers, which add about a quarter to the copy
Rasterise one new chunk column, 1,024 tall	Median 4 milliseconds, slowest one in a hundred 15, on a worldgen worker
Lighting a new chunk	[RISK] The real cost of a Mansion full of lights. Light blocks are placed only on the light rhythm, never as filler, and uniform structure sections are single-value. Measured from M0, with a hard target of no more than twice the lighting cost of an overworld chunk
One territory plan	Median 25 milliseconds, slowest one in a hundred 120, prefetched two territories ahead
One remembered instance	Up to 8 milliseconds, cached
Reading a memory from an unloaded chunk	Off-thread, cached, at most a few per second
Pre-load rewrite of one chunk	Off-thread, under 20 milliseconds
Observation update	Under 0.5 milliseconds every 10 ticks for 20 players
Display entities	At most 32 per chunk and 600 in view per player, block fallback beyond
Hum	At most 8 voices, under 0.3 milliseconds per audio buffer
Memory	Plan cache 64 megabytes, instance cache 128, memory cache 256, all configurable by gamerule
level.dat compound	Under half a megabyte for a long-lived busy world
Chunk attachments	Under 384 kibibytes per chunk

31. Risk Register

Risk	Impact	Mitigation
The slice doesn't feel like the Complex	Highest	That's why M0 exists. Find out first, with the answer-key playtests, before building anything else
Re-expression produces bland or broken rooms from absurd builds	High	R10 is in M0, tested on the world of absurd builds from the first week, tuned against the store and blue-box answer keys
Light data missing at capture	Low	The heightmap estimate of chapter 9
Streamlining feels too clean or too crude	Medium	G_SKY and the R0 thresholds are parameters, tuned against the blue-box house and the film's street
Lighting cost of light-dense chunks	High	Lights only on the rhythm, single-value sections, measured from M0
Reading unloaded chunk NBT races the game's saving	High	Spike S2, read only through the chunk storage on the IO worker, never touch loaded chunks this way
Pre-load rewrite conflicts	High	Spike S4. If it can't be made safe, re-remembering falls back to rewriting loaded chunks only while no player can see them
Chunk attachment size	Medium	The 384 kibibyte cap and spill to neighbouring chunks, test T-S.2
level.dat growth	Medium	Epoch pruning, compact records, measured in long simulated worlds
Synthetic audio stream	Medium	Spike S3, fallback is decision E-9
Display-entity load	Medium	The reduced budget and block fallback
Hidden nondeterminism	High	Fixed-point maths, no shared random generators, test T-12.1 on every build
Item duplication through captures	Medium	Loot fidelity gamerule and stock fill of container contents
Scope	High	The milestone order below. Nothing outside this bible without a written decision record
Anything looking machine-made	Highest	The mandate of chapter 0, the forbidden tells of chapter 36, `checkHuman` on every build, and reading every file through before it's handed over
Coloured light rendering	High	Its own module after M2, gated by spike C0. The Mansion's closures use the tint volume until then, so nothing in the core waits on it
Likeness and copyright	Medium	No ripped assets, no songs, no likenesses, nothing authored from the films. Clear fan-work credit

32. Roadmap

Ordered by what has to be known first. Each milestone has an exit test, and the next doesn't start until it passes. Chapters 40 and 41 break every milestone into steps.

Milestone	Delivers	Exit
M0 Feel slice	A plain Mansion dimension with no sky, a debug entry, the capture flood run by command on a loaded area with the sky drain and glass, exposure and the starting generation, the diorama operator R0, operators R1 to R7 and re-expression R10 on single memories, captured light state, sky closure, the default light arrangement and the hum	The answer-key playtests, the teaser basement, a store captured from its basement, a house captured from inside, a captured street, an absurd build, a plain corridor with the hum, judged against canon frames. You decide whether it feels right. If it doesn't, nothing else is built until it does
M1 Storage and capture	The level.dat record, chunk attachments, memory storage, spill and reading, the Opening Ledger, passengers, roles, ports. Spikes S1 level.dat, S2 unloaded chunk reads	T-1.1, T-4.1, T-S.1, T-S.2. A captured house survives a restart
M2 Genesis and lineage	Nuclei, territories, seams, growth, sea fill, repair, both urns, epochs, landing sites	T-7.1, T-8.1, T-8.2, T-12.1. A walkable sea made of a test world's own captures
M3 Openings and crossing	Null potential, glitches and their canon signs, crossing, arrival, exits	A first-fall playtest, live captures passing T-1.1
M4 Observation	Hold radii, Observer cameras, re-remembering, change behind a lone player. Spike S4 pre-load rewrite	T-13.1, T-13.2, a "not what you remembered" playtest
M5 Atmosphere	Light liveness, cold, food and water, sound leaks, the crossing effect. Spike S3 if not already done	T-3.1, a blind playtest against the original photograph
M6 Still Lifes	Templates, the distortion renderer, fidelity, behaviour echo, dens, edibility, plant and animal Still Lifes	T-9.1, a "meet yourself" playtest
M7 Reach and growth	The join field, the doorway ladder, doorways, growth events and the green glow, time slips	T-10.1, a full Clark-sequence playtest
M8 Lifeform	Colony life cycle, body, hearing, stalking, the charge	Fair-escape playtests
M9 Threshold	The player-built machine, decided in Appendix E, tests, consequences	A full Threshold playtest
M10 Polish and release	Performance pass on every test world, compatibility checks, documentation	Every budget in chapter 30 met
C Coloured light	Spike C0, then the module of chapter 34, the recipe, storage, the light field, the vanilla and Sodium hooks	T-C.1 to T-C.5, under 5 per cent of frame time. Only after M2
R1 Research	Endless vertical travel without splitting players	Only after M10, only if it can keep one shared state
R2 Research	Room-graph acoustics	Only if the place already feels right without it

33. Repository Layout

The mod lives in `C:\Users\MinhHuy\Pictures\Mod Project\Mansion`. Chapter 35 gives its full layout and every build file, with the four subprojects core, colouredlamps, neoforge and lab.

[DESIGN] The source of this bible stays with the Codex, in the bible folder beside the chapters, and is built with build_bible.py. A copy of the PDF sits in the mod's folder and is replaced whenever the bible changes.

34. The Coloured Light Module

[DESIGN] A planned module, built only after the core of the Mansion is proven (M0 to M2). It gives the Mansion's own closures exact coloured light, and it gives every builder in the world coloured lamps to decorate with.

Two users, one system

- **The Complex's closures.** Dusk copies lit red from low floodlights, warm afternoon copies, anything chapter 10's R7 colours. Before this module exists they use the lightmap tint volume, which is good enough for one colour across one enclosure.
- **Builders.** Coloured redstone lamps, anywhere in any dimension, in any colour of dye, side by side.

The recipe

[DESIGN] A redstone lamp in the middle of the crafting grid, surrounded by eight stained glass blocks of one colour, gives a coloured lamp of that colour. A coloured lamp surrounded by eight plain glass blocks gives back an ordinary lamp. Plain glass rather than white stained glass, because white is a colour of its own and would give a white lamp. The recipe doesn't clash with vanilla's stained glass recipe, which has a dye in the centre rather than a lamp.

Part of the Mansion, ready to stand alone

[DESIGN] The module ships as part of the Mansion for now, but it's built from the first line as a separate mod, so that splitting it out later is a matter of publishing a jar, not rewriting code or migrating saves.

- **Its own mod.** It lives in its own Gradle subproject, colouredlamps, with its own mod id, colouredlamps, its own display name, Coloured Lamps, its own mods.toml, its own mixin config, and its own package, io.github.underwatermarch.colouredlamps. The name is open decision E-16.
- **Shipped inside the Mansion's jar.** The Mansion carries the module's jar nested inside its own, through NeoForge's jar-in-jar mechanism, so players install one file and get both. It's still loaded as its own mod, and appears as such in the mod list.
- **Its own namespace from day one.** The data component on the lamp item, the chunk attachment that stores placed colours, its recipes and its gamerules all use the colouredlamps namespace, never mansion. A world that has used coloured lamps keeps every colour when the module is later installed on its own, with nothing to migrate.
- **It never knows about the Mansion.** The module imports nothing from the Mansion's code and depends only on NeoForge. Every builder in any world can use it, with or without a Complex.
- **The Mansion uses it through one small API.** The module exposes a tiny public interface in its api package, reading a placed colour at a position, writing one, and asking for a coloured light volume. The Mansion calls it only from a single bridge class, loaded only when the module is present. In the Mansion's mods.toml the module is an optional dependency.
- **The Mansion works without it.** If the module is missing, the Mansion's closures use the lightmap tint volume of chapter 10, and captured lamps keep their lit state without colour. Nothing in the Mansion's core ever waits on it.
- **Its own tests and build.** Tests T-C.1 to T-C.5 live in the module and run on their own. The subproject builds a standalone jar on every build, so the split is always one step away.

[YOUR RULE] Nesting our own module inside the Mansion's jar adds no outside dependency. NeoForge remains the only thing either mod needs.

Detaching it later

1. Publish the colouredlamps jar the build already produces.
2. In the Mansion's build, stop nesting it. Its mods.toml entry stays an optional dependency.
3. Nothing else changes. Namespaces, saves, recipes and the API are already the module's own.

[TECH] If a world drops the module entirely, placed lamps keep working as vanilla lamps and simply lose their colour, as with removing any mod.

No new block

[YOUR RULE] The coloured lamp is still a vanilla redstone lamp. The item carries its colour in a data component. When it's placed, the colour is written into the chunk's NBT as an attachment at that position. Breaking it drops the item with its colour. Pistons, explosions, silk touch and block swaps carry or clear the colour through NeoForge's block events. The lamp's behaviour is vanilla in every way, lit by redstone, brightness 15, off without power.

[LAW] When the Complex captures a coloured lamp, its colour is part of what it is, so it's copied exactly with its lit state (Law 2). A player's coloured builds turn up in the Mansion still coloured.

Coloured light

[TECH] Vanilla stores one light level per block, with no colour. The module adds a second light calculation on the client, three channels, red, green and blue, each falling by one per block like vanilla's, spread from coloured sources only and limited to their reach. White vanilla light and coloured light are mixed per vertex while terrain is built, so a red lamp beside a blue lamp gives red, blue and purple where they meet. The lamp's own faces are tinted its colour when lit.

[YOUR RULE] Floodlight power doesn't come from a stronger lamp. A coloured lamp is exactly as bright as a vanilla one. Lighting a whole street takes many of them, which is what the Complex's own floodlight banks are.

Where it's hard

[RISK] This is the riskiest rendering work in the project.

- Mixing colour per vertex means hooking terrain building. Vanilla's terrain builder and Sodium's or Embeddium's are entirely different, so each needs its own hook, written without compile-time dependencies, so no dependency is ever added.
- Shader packs replace lighting entirely. Under Iris with a pack active, the module turns itself off and lamps light white.
- The extra light field costs memory and rebuild time. It's recomputed only when a coloured source changes, and only within the source's reach.

[RISK] Spike Co builds the light field and the vanilla hook, measures the frame cost on a build with a hundred coloured lamps, and tries the Sodium and Embeddium hooks. The module goes ahead only if the cost stays under 5 per cent of frame time and both renderers can be supported.

Tests

Test	Asserts
T-C.1	Crafting with eight stained glass of a colour gives that colour, and with eight plain glass gives a plain lamp
T-C.2	A coloured lamp keeps its colour through placing, breaking, pistons and a world restart
T-C.3	A coloured lamp is never brighter than a vanilla lamp

Test	Asserts
T-C.4	A coloured lamp captured into the Mansion keeps its colour and lit state in every copy
T-C.5	With a shader pack active, the module is off and nothing breaks

PART E: The Builder's Handbook

The exact setup, the conventions, the data formats, every algorithm with every number, and the verified NeoForge entry points.

35. Project Setup

[TECH] Everything in this chapter was checked against the project machine on 4 October 2026. The toolchain below already builds a NeoForge 1.21.1 mod there, and every artifact it needs is in the local Gradle cache, except JUnit, which needs a network connection the first time.

The toolchain

Item	Version	Where it was verified
Minecraft	1.21.1	
NeoForge	21.1.250	Cached, `net.neoforged/neoforge/21.1.250`
ModDevGradle	2.0.147	Cached plugin
Parchment mappings	2024.11.17 for 1.21.1	Cached
Gradle wrapper	8.14.5	Used by `Mod Project\CogitoErgoSum`
Java	21	`C:\Program Files\Java\jdk-21`. The foojay resolver plugin 0.8.0 lets Gradle find it
JUnit Jupiter	5.11	Test only. Never shipped

[YOUR RULE] Pin these versions. Don't upgrade anything mid-milestone. An upgrade is a decision record and a full test run.

The repository

```
Mansion/
  The Mansion - Design Bible.pdf
  README.md
  settings.gradle
  build.gradle
  gradle.properties
  gradlew  gradlew.bat  gradle/wrapper/
  buildSrc/
    src/main/java/HumanCheck.java
  docs/
    adr/
  core/
    build.gradle
    src/main/java/io/github/underwatermarch/mansion/core/
    src/test/java/io/github/underwatermarch/mansion/core/
  colouredlamps/
    build.gradle
    src/main/java/io/github/underwatermarch/colouredlamps/
    src/main/resources/
  neoforge/
    build.gradle
    src/main/java/io/github/underwatermarch/mansion/
    src/main/resources/
  lab/
    build.gradle
    src/main/java/io/github/underwatermarch/mansionlab/
    src/main/resources/
    results/
```

[DESIGN] Four subprojects. **core** is the engine in plain Java with no Minecraft dependency, so its laws can be tested at full speed. **colouredlamps** is the coloured light module of chapter 34, a mod of its own. **neoforge** is the Mansion itself, the adapter between core and the game. **lab** is a separate mod,

`mansionlab`, holding everything used only for testing, fixtures, walker bots, plan image rendering, measurement tools and game tests. The lab is never published and never nested in the Mansion's jar. Its results folder is ignored by git.

settings.gradle

```
pluginManagement {
    repositories {
        mavenLocal()
        gradlePluginPortal()
        maven { url = 'https://maven.neoforged.net/releases' }
    }
}

plugins {
    id 'org.gradle.toolchains.foojay-resolver-convention' version '0.8.0'
    id 'net.neoforged.moddev' version '2.0.147' apply false
}

rootProject.name = 'mansion'
include 'core', 'colouredlamps', 'neoforge', 'lab'
```

gradle.properties

```
org.gradle.jvmargs=-Xmx3G
org.gradle.daemon=false
org.gradle.parallel=true
org.gradle.caching=true

minecraft_version=1.21.1
neo_version=21.1.250
parchment_version=2024.11.17
mod_version=0.1.0
mod_group_id=io.github.underwatermarch
mod_authors=UnderwaterMarch
```

build.gradle at the root

```
allprojects {
    group = mod_group_id
    version = mod_version
}

subprojects {
    apply plugin: 'java-library'
    java.toolchain.languageVersion = JavaLanguageVersion.of(21)
    repositories {
        mavenCentral()
    }
    tasks.withType(JavaCompile).configureEach {
        options.encoding = 'UTF-8'
    }
}

tasks.register('checkHuman', HumanCheck) {
    roots = files('core', 'colouredlamps', 'neoforge', 'lab', 'buildSrc', 'docs', 'README.md',
            'build.gradle', 'settings.gradle', 'gradle.properties')
}

subprojects {
    tasks.matching { it.name == 'build' }.configureEach {
        dependsOn rootProject.tasks.named('checkHuman')
    }
}
```

core/build.gradle

```
dependencies {
    testImplementation platform('org.junit:junit-bom:5.11.3')
    testImplementation 'org.junit.jupiter:junit-jupiter'
    testRuntimeOnly 'org.junit.platform:junit-platform-launcher'
}

test {
    useJUnitPlatform()
    maxHeapSize = '2G'
}
```

colouredlamps/build.gradle

```
plugins {
    id 'net.neoforged.moddev'
}

base {
    archivesName = 'colouredlamps'
}

neoForge {
    version = neo_version
    parchment {
        mappingsVersion = parchment_version
        minecraftVersion = minecraft_version
    }
    mods {
        colouredlamps {
            sourceSet sourceSets.main
        }
    }
}
```

neoforge/build.gradle

```
plugins {
    id 'net.neoforged.moddev'
}

base {
    archivesName = 'mansion'
}

dependencies {
    implementation project(':core')
    compileOnly project(':colouredlamps')
    jarJar project(':colouredlamps')
}

neoForge {
    version = neo_version
    parchment {
        mappingsVersion = parchment_version
        minecraftVersion = minecraft_version
    }
    runs {
        client {
            client()
            gameDirectory = project.file('run/client')
        }
        server {
            server()
            programArgument '--nogui'
            gameDirectory = project.file('run/server')
        }
    }
    mods {
        mansion {
            sourceSet sourceSets.main
            sourceSet project(':core').sourceSets.main
        }
        colouredlamps {
            sourceSet project(':colouredlamps').sourceSets.main
        }
    }
}

tasks.named('jar', Jar) {
    from project(':core').sourceSets.main.output
}
```

[RISK] ModDevGradle's `jarJar` configuration nests the module's jar inside the Mansion's. If it refuses a project dependency in this version, publish colouredlamps to the local Maven repository in the same build and nest that instead. Step M0.1 settles it, and the answer goes in a decision record.

lab/build.gradle

```
plugins {
    id 'net.neoforged.moddev'
}

dependencies {
    implementation project(':core')
    implementation project(':neoforge')
}

neoForge {
    version = neo_version
    parchment {
        mappingsVersion = parchment_version
        minecraftVersion = minecraft_version
    }
    runs {
        labClient {
            client()
            gameDirectory = project.file('run/client')
        }
        labServer {
            server()
            programArgument '--nogui'
            gameDirectory = project.file('run/server')
        }
        gameTestServer {
            type = 'gameTestServer'
            systemProperty 'neoforge.enabledGameTestNamespaces', 'mansion,mansionlab'
            gameDirectory = project.file('run/gametest')
        }
    }
    mods {
        mansionlab {
            sourceSet sourceSets.main
        }
        mansion {
            sourceSet project(':neoforge').sourceSets.main
            sourceSet project(':core').sourceSets.main
        }
        colouredlamps {
            sourceSet project(':colouredlamps').sourceSets.main
        }
    }
}
```

Mod metadata

[YOUR RULE] The Mansion's `neoforge.mods.toml` has no comments at all, which means deleting every comment the NeoForge template ships with.

```
modLoader="javafml"
loaderVersion="[4,)"
license="All rights reserved"

[[mods]]
modId="mansion"
version="0.1.0"
displayName="The Mansion"
authors="UnderwaterMarch"
description='''A dimension that copies your world, badly, and keeps copying its copies.'''

[[mixins]]
config="mansion.mixins.json"

[[dependencies.mansion]]
modId="neoforge"
type="required"
versionRange="[21.1.250,)"
ordering="NONE"
side="BOTH"

[[dependencies.mansion]]
modId="minecraft"
type="required"
versionRange="[1.21.1,1.21.2)"
ordering="NONE"
side="BOTH"

[[dependencies.mansion]]
modId="colouredlamps"
type="optional"
versionRange="[0.1,)"
ordering="NONE"
side="BOTH"
```

The licence line stays as it is until decision E-10 is made. Coloured Lamps has the same file with its own id, display name and mixin config, and only the neoforge and minecraft dependencies.

```
{
  "required": true,
  "minVersion": "0.8",
  "package": "io.github.underwatermarch.mansion.mixin",
  "compatibilityLevel": "JAVA_21",
  "mixins": [
    "PrimaryLevelDataMixin",
    "BlockCollisionMixin",
    "SteadyLightMixin",
    "StillPlantMixin",
    "ChunkSerializerMixin"
  ],
  "client": [],
  "injectors": {
    "defaultRequire": 1
  }
}
```

[TECH] No access transformer is needed for anything known so far. The chunk storage the Mansion reads and writes is reached through the public `chunkMap` field of the server chunk cache, and its `read` and `write` methods are public. Add an access transformer only when something genuinely needs one, and record why.

The human check

[YOUR RULE] `buildSrc/src/main/java/HumanCheck.java` is a Gradle task class that walks every file under its roots, skipping `build`, `run`, `.gradle` and `lab/results`, and fails the build with a list of file and line for every finding. It has no comments itself.

File kind	Fails on
Java and Gradle	Any `//` or `/*` outside string literals, character literals and text blocks. The scanner tracks those literals properly, so a URL inside a string is fine
TOML, properties, access transformer, mcfunction	Any `#` that starts a comment, at the start of a line or after the value outside quotes
JSON	Any key containing `comment`, `note` or `todo`, in any case
Every text file	The em dash and en dash characters. Any emoji. Any of the words TODO, FIXME, XXX or HACK. Any tool or assistant credit, `Co-Authored-By`, "generated with" or "generated by" followed by a tool's name, and the names of chat assistants and code assistants, kept as a list inside the task. Ordinary uses of "generate" for chunks and plans are fine
Markdown	All of the above, plus colons in prose by the same rule the bible's build uses
Git, if the folder is a repository	The last fifty commit messages, by the text-file rules

[YOUR RULE] The check runs before every build of every subproject and is never skipped, disabled or loosened to make a build pass. If it finds something, fix the file.

Decision records

[DESIGN] One short Markdown file per decision in `docs/adr`, named with a number and a few words, `0007-jarjar-from-maven-local.md`. Each says what was decided, why, what the bible said before, and whether the user agreed. Plain prose, no dashes, no headings beyond the title. Where a decision changes the bible, the bible's source is edited to match and Appendix F gets a row.

Git

[YOUR RULE] Ask before the first `git init` and before every commit. The author is UnderwaterMarch. Messages are short, plain, lower-key sentences a person would write, "capture flood and exposure", "fix seam keep-clear volumes", with no trailers of any kind. The `.gitignore` covers `build`, `run`, `.gradle`, `out`, `lab/results` and IDE folders.

36. Conventions in Code

What human-written code looks like here

[YOUR RULE] The mandate in chapter 0 is the first rule. This section turns it into concrete practice. Generated code has recognisable tells, and every one of them is forbidden.

Forbidden tells

- **Any comment.** Covered by the mandate and enforced by the build.
- **Names that narrate.** `processData`, `handleStuff`, `doTheThing`, `tempResult`, `myList`, `newValue2`, `isValidAndNotNull`. A name says what a thing is in this domain, `exposure`, `faceRegion`, `seamContract`.
- **Suffix spam.** Classes ending in Manager, Helper, Util, Utils, Handler, Service, Processor or Impl unless the word is genuinely the clearest name. `MemoryStore` beats `MemoryManager`. One `Hash` class beats a `HashUtils`.
- **Defensive noise.** Null checks on values that can't be null, try and catch around code that can't throw, catching and rethrowing, logging and continuing where failure should be loud, `Objects.requireNonNull` on every parameter.
- **Ceremony.** Interfaces with one implementation and no second in sight, builders for records with three fields, factories for objects nobody substitutes, getters and setters on plain data where a record does the job.
- **Uniform padding.** Every method the same shape, every class the same template, a blank line after every statement, the same log line at the start and end of every method.
- **Dead and placeholder code.** Unused parameters, unused imports, empty methods waiting to be filled, `throw new UnsupportedOperationException("not implemented")`, stub classes, `return null` placeholders.
- **Magic restated.** A constant named `FIVE = 5`. Constants are named for what they mean, `LIGHT_LINE_SPACING`, and live beside the code that uses them, or in the one parameters class if the bible lists them in Appendix B.
- **Overlong methods and god classes,** and their opposite, a dozen three-line classes that only forward calls.
- **Chatty logs.** Exclamation marks, emoji, "Successfully", "Starting to", "Done!". The mod logs rarely, plainly, once per real event.
- **Strings as notes.** Exception messages or constants written as explanations for the reader instead of for the player or the log.
- **Formatting tics.** Trailing whitespace, inconsistent indentation, mixed line endings, wildcard imports, fully qualified names written inline where an import belongs.

[DESIGN] What to do instead.

- Four-space indentation, braces on the same line, lines under about 120 characters, one top-level class per file, imports sorted and explicit.
- Records for plain data. Final fields wherever possible. Small, focused classes named for the domain thing they model.
- Methods short enough to read whole, named with verbs for actions, `capture`, `remember`, `rasterise`, and nouns for values, `exposure`, `startingGeneration`.
- Let the structure carry the meaning. If code needs a comment to be understood, rename, extract or reorder until it doesn't.
- Code written over time by one person isn't perfectly uniform. Let each class be shaped by what it does, not by a template.

The package map

Package	Holds
`core.space`	Voxel positions, bricks, transforms, the eight orientations, bounding boxes
`core.block`	The block table that stands between core and the game (chapter 37)
`core.math`	`Hash`, `Rng`, value noise, CIELAB conversion, distance transforms, fixed iteration helpers
`core.cache`	Snapshots, the flood, exposure, roles, face regions, objects, ports, the memory format
`core.remember`	Operators R0 to R10, the substitution rules, the remembered-copy format, agreement
`core.genesis`	Nuclei, territories, seam contracts, growth, sea fill, repair, plans
`core.lineage`	The lineage field, both urns, epochs
`core.store`	NBT-free codecs for memories and records, as byte layouts the adapter wraps in NBT
`mansion`	The mod class, registries, gamerules, payloads, commands
`mansion.stage`	The chunk generator, the rasteriser, the dimension's special effects
`mansion.world`	The block table built from the game, snapshots of the live world, chunk attachments, the level.dat record
`mansion.observe`, `opening`, `reach`, `chrono`, `growth`, `atmos`, `stilllife`, `lifeform`, `devtools`	As in chapter 5
`mansion.mixin`	The five mixins of chapter 26
`mansionlab`	Fixtures, walker bots, plan images, measurement, game tests

Determinism

[TECH] Since Java 17 all floating-point arithmetic is strict IEEE 754, so plain `double` arithmetic gives the same bits on every machine. What differs between machines is the `Math` library, whose transcendental functions may vary in the last bit. So core uses `double` freely, and uses `StrictMath` for every square root, logarithm, exponential, power and trigonometric function, never `Math`. This replaces the fixed-point rule of earlier versions of this bible.

[YOUR RULE] Core never uses `java.util.Random`, `ThreadLocalRandom`, `Math.random`, `SplittableRandom`, `HashMap` or `HashSet` iteration order, parallel streams, the system clock or thread identity in anything that decides an output. Collections whose order matters are sorted or insertion-ordered. Every random choice comes from an `Rng` seeded by `Hash.of`.

```
public final class Hash {
    private static final long GOLDEN = 0x9E3779B97F4A7C15L;

    private Hash() {
    }

    public static long mix(long z) {
        z = (z ^ (z >>> 30)) * 0xBF58476D1CE4E5B9L;
        z = (z ^ (z >>> 27)) * 0x94D049BB133111EBL;
        return z ^ (z >>> 31);
    }

    public static long of(long seed, long purpose, long... ids) {
        long h = mix(seed ^ GOLDEN);
        h = mix(h ^ purpose);
        for (long id : ids) {
            h = mix((h + GOLDEN) ^ id);
        }
        return h;
    }

    public static long purpose(String name) {
        long h = 0xCBF29CE484222325L;
        for (int i = 0; i < name.length(); i++) {
            h ^= name.charAt(i);
            h *= 0x100000001B3L;
        }
        return h;
    }
}
```

```
public final class Rng {
    private long state;

    public Rng(long seed) {
        state = seed;
    }

    public long nextLong() {
        state += 0x9E3779B97F4A7C15L;
        return Hash.mix(state);
    }

    public double nextDouble() {
        return (nextLong() >>> 11) * 0x1.0p-53;
    }

    public int nextInt(int bound) {
        return (int) Long.remainderUnsigned(nextLong(), bound);
    }

    public boolean chance(double p) {
        return nextDouble() < p;
    }

    public double nextGaussian() {
        double u = 1.0 - nextDouble();
        double v = nextDouble();
        return StrictMath.sqrt(-2.0 * StrictMath.log(u)) * StrictMath.cos(2.0 * StrictMath.PI * v);
    }
}
```

[DESIGN] Every purpose is a constant made once with `Hash.purpose`, named after what it decides, `DRIFT`, `MIRROR`, `RE_EXPRESS`, `SEAM`, `URN_DRAW`. A decision about one object uses `Hash.of(worldSeed, purpose, memoryId, instanceId, objectIndex)`, so it never depends on what else was decided first.

[TECH] Value noise, used by R3 and the lineage field, hashes the corners of its own internal cells and blends them with a smoothstep. That internal cell is never visible as geometry. It only decides which voxels are repainted.

Coordinates and transforms

- World positions are `BlockPos`-style integers. Memory-local positions start at the memory's bounding-box minimum, which is recorded in world coordinates as the memory's origin.
- An orientation is one of eight, a quarter-turn count 0 to 3 about the vertical axis, then an optional mirror across the x axis. Block states are rotated and mirrored by the game's own rotate and mirror methods through the block table, so stairs, doors and signs turn correctly. Mirroring a sign also reverses its text (R2).
- Generation is a `double`. Intensity is a `double` in memory and a byte on disk. Exposure is the sky light nibble, 0 to 15, on disk, and sky light divided by 15 in memory.
- Brick coordinates are voxel coordinates shifted right by three.

Failure

[DESIGN] The Mansion never crashes a server over its own data. A memory that won't decode is logged once, plainly, and treated as lost matter (chapter 6). A plan that fails to compute is retried once on a later tick, then its territory falls back to the sea and the failure is logged once. Everything else that should be impossible is a bug, so it throws.

37. Data Formats

The block table

[DESIGN] Core never sees a Minecraft class. It sees integer block ids and a `BlockTable` that answers every question the laws ask. The adapter builds it at server start from the block registry, the tags, the recipes, the data maps and the measured colour table, and keeps it for the session. Ids are assigned in sorted order of the full state string, so they're stable within a session. Nothing on disk stores these ids. Disk always stores state strings.

Field	Meaning	Where the adapter gets it
`state`	Full state string, `minecraft:oak_stairs[facing=east,half=bottom,shape=straight,waterlogged=false]`	The block state's own string form
`block`	Block name without properties	The block registry
`shape`	An integer key for the exact collision shape	Hash of the collision shape's boxes, rounded to sixteenths
`full`, `empty`	Collision is the whole cube, or nothing	The collision shape
`lightBlock`	0 to 15, how much light the block stops	`getLightBlock`
`emission`, `emissionLit`	Light emitted now, and when lit if it has a `lit` property	`getLightEmission` on the state and on its `lit=true` variant
`hardness`	Destroy speed, minus one for unbreakable	`getDestroySpeed`
`liquid`	A fluid block, not a waterlogged one	The fluid state and block class
`waterlogged`	Has the waterlogged property set	Block state properties
`blockEntity`, `randomTicks`, `gravity`, `signal`	Behaviour flags	`hasBlockEntity`, `isRandomlyTicking`, the block being `Fallable`, `isSignalSource` or having a power property
`tags`	A bit set of doors, trapdoors, fence gates, climbable, leaves, flowers, saplings, crops, signs, banners, carpets, beds, glass blocks, glass panes, stairs, slabs, fences, walls, plus anything in the datapack role overrides	Block tags, the common `c` tags for glass
`lab`	Mean colour in CIELAB, or unknown	The measured table for vanilla, or measured at start for mods
`tint`	None, grass, foliage, water or other	Whether its model has tinted faces, and which colour handler vanilla uses
`flatness`	0 to 1	The measured table
`mapColor`	The map colour as RGB	`getMapColor`
`family`	The id of the full block it's cut from, or its own id	The recipe graph, below
`clean`	The unweathered, unwaxed form, or its own id	NeoForge's oxidisation and waxing data maps, and the recipe graph
`rotate`, `mirror`	Ids after each of the eight orientations	The game's rotate and mirror methods

[DESIGN] **Families.** A block that isn't a full cube belongs to the family of the full block it's made from. For every crafting or stonecutting recipe whose output is that block and whose ingredients are all the same item, and whose ingredient's block is a full cube, that full block is the family. Oak stairs belong to oak planks, smooth sandstone slabs to smooth sandstone. Full blocks are their own family. Blocks with no such recipe are their own family. Materials in the urns are families, and the variant with the right shape is

found within the family when a material is placed.

[DESIGN] **Clean forms.** Weathered and waxed copper map to their clean form through NeoForge's oxidisation and waxing data maps and their inverses. Mossy and cracked variants map back through the recipe graph, the recipe that turns a base block into the variant, read in reverse. Other mods that register oxidisation data maps or such recipes work with no extra code.

A memory in memory

```
Memory
  id, ledger, dimension, origin, size
  gameTime, dayTime, realTime, moonPhase, weather, skyColour
  bricks          sorted map from brick key to Brick
  blockEntities   list of (local position, NBT)
  passengers      list of Passenger
  biomeColours    per 4 by 4 by 4 cell, grass, foliage and water RGB

Brick
  state[512]      block id, or NOT_CAPTURED
  intensity[512]  0 to 1
  sky[512]        0 to 15
```

[DESIGN] Roles, face regions, objects and ports aren't stored. They're computed from the voxels when a memory is first used and cached with it, which keeps storage small and means improving the analysis improves every old memory too.

Memory NBT

[DESIGN] The attachment `mansion:memories` on a chunk holds every memory part stored there. On disk everything is names and bytes, never session ids.

```
mansion:memories
  v          int        1
  parts      list
    id         long       memory id
    part       int        0 for the head part
    of         int        how many parts the memory has
    head       compound   only in part 0
      ledger     long
      dim        string
      origin     int[3]
      size       int[3]
      time       long       game time
      day        long       day time
      real       long       milliseconds since 1970
      moon       int
      weather    byte       0 clear, 1 rain, 2 thunder
      sky        int        sky colour, RGB
      colours    byte[]     deflated, 9 bytes per 4 by 4 by 4 cell in x, z, y order
    palette    list of string    state strings used in this part
    bricks     list
      at         int        brick x, y and z, 10 bits each, packed x | y << 10 | z << 20
      data       byte[]     deflated, 512 entries in x, z, y order
    entities   list
      at         int        local position, packed like bricks but in voxels
      nbt        compound   block entity data without its position
    riders     list
      type       string     entity type
      pos        double[3]  local position
      rot        float[2]   yaw and pitch
      nbt        compound   the entity's own saved data
      profile    compound   players only, name, id and the signed textures property
      trace      byte[]     players only, the behaviour trace of chapter 20
```

[DESIGN] Each brick entry in `data` is four bytes. Two bytes are the palette index plus one, so zero means not captured. One byte is intensity, quantised on a log scale so the faint edge of a capture keeps its precision, `round(255 times (1 + ln(I) / 4))`, clamped to 1 to 255 for any captured voxel. One byte holds the sky light in its high four bits and nothing in its low four. Bricks with no captured voxel aren't written.

[TECH] The cap of 384 kibibytes per chunk (chapter 6) is measured on the sum of the attachment's byte arrays and strings before the region file compresses it, because they're already deflated and won't shrink further. When a part would push a chunk over, it's moved to the nearest chunk around the opening with room, and the index in level.dat records where each part lives.

Chunk attachments

Id	Holder	Contents	Synced to clients
`mansion:memories`	Any chunk in any dimension	Memory parts, as above	No
`mansion:null`	Any chunk with an opening	Per section, a 4,096-bit set of null positions	Yes, by the NullPositions payload to players within view distance
`mansion:realised`	Mansion chunks	Territory id, epoch, generation offset, last observed game time, re-remembering count	No
`mansion:local`	Mansion chunks	Local memories from re-remembering, in the memory format	No
`mansion:still`	Mansion chunks	Per section, one bit set of still plants and one of steady lights	No

Id	Holder	Contents	Synced to clients
`mansion:colours`	Mansion chunks	Per 4 by 4 by 4 cell that holds copied tinted blocks, the captured grass, foliage and water colours	Yes, through the attachment's own sync
`colouredlamps:colours`	Any chunk	Position and RGB of each placed coloured lamp	Yes

[YOUR RULE] Never register a level attachment or a SavedData. Both write files of their own. Entity attachments on Still Lifes and Lifeforms are fine, because they're saved inside the entity's own NBT in the chunk.

The level.dat record

```
Data
  mansion
    v            int
    nextId       long
    ledger       list of { id long, dim string, pos int[3], kind byte, time long, real long, home int[2] }
    memories     list of { id long, ledger long, dim string, parts list of int[2], bytes int, weight int, flags int, roles
    urn          { epoch int, roles list of { role byte, families list of string, counts int[] } }
    epochs       list of { id int, roles same layout as urn }
    pins         list of { territory long, epoch int }
    islands      list of { ledger long, site int[3] }
    dates        list of { player int[4], offset long }
    exclusions   list of { dim string, box int[4] }
    fingerprints list of { state string, lab float[3], flat float, shape int, role byte, family string }
    activity     int
```

Kind	Byte
Glitch	0
Doorway	1
Threshold	2
Passing object	3
Retrospective	4
Command, development only	5

[DESIGN] `roles` in a memory's index entry counts its voxels by role, so a lost memory can still be filled by role (chapter 6). `parts` lists the chunk of each part. Player ids are stored as the four ints of a UUID.

A remembered copy

```
Copy
  memory, instance, generation, seed
  voxels         sparse map from local position to block id
  provenance     sparse map from local position to an operator bit set and a source tier
  blockEntities  list of (local position, NBT)
  displays       list of (object, transform with fractional scale, rotation or sink)
  passengers     list of (Passenger, local position, fidelity)
  lights         list of (local position, lit)
  outdoor        bit set of former outdoor air not yet re-expressed
```

[DESIGN] Copies are cached by memory, generation, seed and epoch, and evaluated brick by brick on demand (chapter 10's lazy rule).

A plan

```
Plan
  territory, nucleus, epoch, character
  instances      list of (memory, orientation, translation, generation, seed, priority)
  portals        list of (position, facing, type, width, height, keepClear box)
  punched        list of (position, facing, width, height)
```

[DESIGN] Plans live only in memory and are recomputed whenever needed, because they're pure functions of the seed, the pinned epoch and the island registry.

38. Algorithms in Full

[DESIGN] Chapters 8 to 14 say what each system does and why. This chapter says exactly how, with every number. Numbers marked as tuning starting points are tuned at M0 against the answer keys of chapter 29, and the tuned values replace these in Appendix B through a decision record. Nothing here may be tuned to a particular map.

The order of a remembering

[DESIGN] A memory is stored raw, exactly as captured. Every operator, R0 included, runs when a copy is made. That keeps old captures useful while the operators are tuned, and it means the same memory can be re-remembered under better operators later. Chapter 9 describes R0 as applied at capture. The effect is identical, because R0 depends only on the capture's own exposure.

```
remember(memory, g, seed, epoch)
  analysis  = analyse(memory)                          cached per memory
  copy      = raw voxels of memory
  R0  diorama, by exposure
  R1  forgetting, objects dropped
  R6  lost purpose, relationships broken
  R10 re-expression, room by room, where chosen
  R2  drift of surviving objects
  R8  echo chains
  R3  prior pull on materials
  R9  convergence above generation 7
  R4  merge resolution
  R5  fill of every gap, through the substitution rules
  R7  sky closure, surfaces and light
  lights and mechanisms set from their captured state
  return copy
```

Effective generation

[DESIGN] Every element of a copy, a face region, an object, a room, has its own effective generation, the instance's generation plus the starting generation from its own exposure.

```
gStart(e)        = G_SKY * StrictMath.pow(e, 1.5)           G_SKY 2.5
gEff(element, g) = g + gStart(e(element))
```

R1, R2, R3 and R6 use the element's effective generation. R7's choice of closure rung uses the instance's own generation, as chapter 10 says. R10 decides per room, using the room's effective generation.

The snapshot

1. A capture names a centre, the opening, and a kind. The snapshot box is a radius of 4 chunks and 48 blocks up and down for a glitch or passing object, 8 chunks and 96 blocks for a doorway growth event, 12 chunks and 128 blocks for the Threshold, clamped to the dimension's height.
2. On the server thread, at most 2 milliseconds per tick, each loaded chunk in the box has its section block containers copied, and the sky light data layer of each section. Sections that are a single value are stored as that value. Unloaded chunks are left out.
3. Block entity NBT is copied for positions inside the box. Entities whose bounding box lies inside the box are snapshotted with their saved data.
4. Biome colours are sampled once per 4 by 4 by 4 cell, the grass, foliage and water colour the client would use there, computed on the server from the biome's own colour data.

5. The finished snapshot is immutable and handed to the mod's worker pool. Nothing else in the capture touches the live world.

Exposure

```
sky(v)       = the copied sky light at v, or the estimate below when the layer is missing
e(air v)     = sky(v) / 15
e(face f)    = the highest e of the captured air touching f
e(object o)  = area-weighted mean e of o's faces
e(room r)    = mean e of r's air voxels
```

[TECH] The estimate, used only where a section's sky light layer is missing, starts from the column heightmap inside the snapshot, sky light 15 above the highest block that stops light and 0 below, then spreads sideways and down through open voxels a step at a time, falling by one per step, using `lightBlock` as the extra loss through each block, the way the game does it.

The flood

```
capture(snapshot, centre, budget)
  I[centre] = 1
  heap of voxels by intensity, ties broken by packed position, smaller first
  push centre
  exposedOpen = 0, shelteredOpen = 0, kept = 0
  while heap not empty and kept < budget
    v = pop
    for n in the six neighbours of v, in the fixed order -x, +x, -y, +y, -z, +z
      if n is outside the snapshot, skip
      b = block at n
      if open(b)
        if sky(n) >= 8 and exposedOpen >= shelteredOpen / 3, skip
        mu = MU_AIR + K_SKY * e(n)
      else if transmissive(b)
        mu = MU_GLASS + 0.06 * lightBlock(b)
      else
        mu = muMatter(b)
      In = I[v] * StrictMath.exp(-mu)
      if In > I[n] and In > I_MIN
        if n not yet kept, kept += 1, and count it as exposed or sheltered if open
        I[n] = In
        push n
  memory = every kept voxel with its state, intensity and sky light

open(b)         = collision is empty and b is not a liquid
transmissive(b) = b is a liquid, or lightBlock(b) < 15
muMatter(b)     = 0.9 * clamp(0.6 + 0.27 * hardness(b), 0.6, 2.0), and 1.8 when unbreakable
```

Constant	Value	Meaning
MU_AIR	0.004	Loss per block in sheltered air
K_SKY	0.03	Extra loss per block in air at full exposure
MU_GLASS	0.15	Loss through glass, water and other blocks that let light through
I_MIN	0.02	Below this nothing is kept. About four blocks into stone from full strength
Outdoor share	A quarter	Exposed open voxels are never more than a third of sheltered ones, so at most a quarter of open space

[DESIGN] Using the game's own light opacity makes the flood treat blocks the way chapter 9's bounce model says it should. Glass, panes, leaves, slabs, stairs, fences and doors all let some capture through, and solid opaque blocks stop it within a few blocks. A torch, a flower or a button has no collision, so the capture flows round it and keeps it.

Roles

[DESIGN] Roles are assigned in this order to every kept voxel that isn't air. The first rule that matches wins.

1. **Liquid.** A fluid block.
2. **Light.** `emissionLit` of 10 or more, or in the datapack's lamp tag.
3. **Door.** The doors, trapdoors or fence gates tag.
4. **Window.** The glass block or glass pane tags, or a full cube with `lightBlock` 0 that isn't leaves.
5. **Marking.** Signs and banners. Paintings and item frames are entities and get this role as passengers.
6. **Container.** The adapter finds an item handler capability on its block entity.
7. **Mechanism.** A signal source, a block with a power or powered property, or any other block entity that ticks.
8. **Plant.** Leaves, flowers, saplings, crops, grass blocks, moss, vines, and the datapack's plant tag.
9. **Vertical passage.** The climbable tag, or scaffolding.
10. **Floor.** Solid, with captured open air directly above.
11. **Ceiling.** Solid, with captured open air directly below whose sky light was below 15.
12. **Wall.** Solid, with captured open air beside it, in a face region of the shell.
13. **Furniture.** Collision that isn't a full cube, not in the shell.
14. **Structure.** Anything solid left.

[DESIGN] Datapacks can override any block's role under `data/<namespace>/mansion/roles/<role>.json`, as a block tag. Geometry still decides first for blocks with no override.

Face regions, the shell and objects

1. **Faces.** Every face between a kept solid voxel and kept open air.
2. **Face regions.** Faces with the same normal on the same plane, joined through shared edges. Each region records its area, its materials, its exposure and its role, floor, wall or ceiling.
3. **The shell.** Every floor and ceiling region, and every wall region of 6 faces or more.
4. **Objects.** The non-shell, non-air, non-structure voxels split into 6-connected components. Each is one object, with its role taken from the commonest role among its voxels, a block entity always counting as the object's role.
5. **Support.** An object rests on another if any of its lowest voxels sits directly on it. Its support depth k is 1 for an object on the shell, 2 for one resting on that, up to 4.

Scale and complexity

```
scale(o)      = log2(volume(o)) + log2(largest extent(o))
complexity(o) = length of the deflated byte form of o's voxels, state indices in x, z, y order
```

Object	Volume	Extent	Scale
Torch, button, flower pot	1	1	0
A bed	2	2	2
A table of four blocks	4	2	3
A run of six bookshelves	6	6	5.2
A staircase of twelve	12	6	6.2
A small tree	60	7	8.7

Ports

1. A boundary voxel is kept open air with a neighbour that wasn't kept, through budget, intensity or the snapshot's edge, and that wasn't matter.
2. Boundary voxels with the same facing are joined into patches.
3. Each patch is classified by its facing and the size of its cross-section.

Facing	Cross-section	Port
Up or down	Any	Hole in ceiling or floor, a chute if the patch slopes
Sideways	Width 1 to 3, height 2 to 4	Doorway
Sideways	Height 1	Crawl space
Sideways	Width 1 to 3, height 5 or more	Gap
Sideways	Width 4 or more	Corridor end
Sideways, bordered by window-role blocks	Any	Window

R0, the diorama, in numbers

Step	Applies to	Rule
Wear removed	Every face region and object	With probability e, every block in it becomes its `clean` form. One decision per region
One colour per face	Wall, floor and ceiling regions	With probability smoothstep(0.2, 0.8, e), the whole region takes its dominant family, in each block's own shape
Clutter dropped	Objects touching sky-facing faces	Dropped if scale is below 2.5 times e. Signs and banners are never dropped
Plants simplified	Plant objects with e of 0.5 or more	Layer by layer, every voxel inside the layer's outline that isn't reachable from outside the outline becomes the object's dominant plant block. At e of 0.75 or more, single voxels joined by only one face are removed
Edges straightened	Face regions with e of 0.5 or more	On the region's flat mask, a closing then an opening with a plus-shaped element one cell across, applied once

[DESIGN] R0 changes geometry only where the sky was, only by dropping, filling holes and straightening edges, and its results always go through the walkability tests.

R1, forgetting, in numbers

[DESIGN] Earlier versions of this bible gave THETA0 1.5 and A 0.55. With the scale defined above, those values would have dropped torches at generation 0 and kept beds at generation 5, contradicting test T-6.2. The corrected values follow, as tuning starting points.

```
R1 does nothing where gEff <= 0.05
theta(g) = THETA0 + A * g                THETA0 -1.0, A 1.6
noise(o) uniform in [-0.6, 0.6], seeded per object
keep o   if scale(o) + boost(o) - theta(gEff(o)) + noise(o) > 0
boost(o) = 2.0 for window objects while gEff <= 4, otherwise 0
shell regions are never dropped by R1 while gEff <= G_SHELL (9)
```

gEff	theta	What goes
1	0.6	Torches, buttons, pots, small clutter
2	2.2	Beds, chairs, small furniture
3	3.8	Tables, small runs
4	5.4	Large furniture. Windows lose their boost after this

gEff	theta	What goes
5	7.0	Staircases and most trees. An empty box

R2, drift, in numbers

```
for each surviving object o, with support depth k
  sigma2 = SIGMA^2 * gEff * k                      SIGMA 0.35
  dx, dz = round(nextGaussian * sqrt(sigma2))
  dy     = round(nextGaussian * sqrt(sigma2) - BETA * gEff * SIGMA)      BETA 0.5
  mirror with probability 1 - (1 - 0.004)^gEff
  quarter turn with probability 1 - (1 - 0.01)^gEff
  rescale with probability 1 - (1 - 0.015)^gEff, by 0.5, 0.75, 1.5 or 2, equally likely
```

[DESIGN] Rescaling by 2 or 0.5 resamples voxels. Rescaling by 0.75 or 1.5, any rotation that isn't a right angle, and any sink of a fraction of a block take the display path of chapter 10.

R3, prior pull, in numbers

```
for each face region f
  keep = KMAT^gEff(f)                              KMAT 0.82
  u(v) = value noise at v, wavelength 6 blocks, remapped so it's uniform on 0 to 1
  family = one urn draw for f's role, seeded by the region
  every voxel v of f with u(v) > keep becomes family, in v's own shape
```

[DESIGN] One draw per region, so a repainted patch is one material, never speckle.

R4, merge order

When two sources claim one voxel after drift, the first rule that decides wins.

1. A solid beats a liquid.
2. The keep-clear volume of a seam beats everything.
3. A container is never deleted. If it loses, it moves to the nearest free voxel of its own object.
4. Below gEff 3 the shell beats objects. From 3 up, an object beats the shell with probability (gEff minus 3) over 6, capped at 0.9.
5. Between objects, the higher complexity wins.
6. Otherwise the lower instance priority wins, then the lower memory id.

Substitution in full

[DESIGN] Every substitution, by R3, R5, R7 or R10, and every palette rewrite for a removed mod, goes through this.

```
substitute(target, role, room, sources)
  candidates = every block b in the block table where
      shape key of b, after copying target's facing, half, axis, type, shape
        and waterlogged properties where b has them, equals target's
      b is inert
      b is opaque exactly when target is
  cost(b) = deltaE(lab(b), colour(target)) + W_FLAT * (1 - flatness(b)) + W_SOURCE * tier(b)
  allowed = candidates not too close to a neighbouring role in the room
  if allowed has a member within cost(best of candidates) + 15
    return the cheapest of allowed
  return the cheapest of candidates
```

Term	Value
inert	No block entity, no random ticks, no gravity, no emission, not a signal source
colour(target)	Its lab colour, or for tinted blocks its texture colour times the captured tint of its cell, or its map colour when nothing else is known
deltaE	Euclidean distance in CIELAB, the 1976 formula
W_FLAT	10
W_SOURCE and tier	1.5, with tiers 0 for the instance's own memory, 1 its territory's memories, 2 the stock, 3 installed mods, 4 vanilla
Too close	Same family as, or deltaE under 10 from, the dominant family of any other role in the same room

[TECH] CIELAB from sRGB, using StrictMath throughout.

```
linear(c)   = c <= 0.04045 ? c / 12.92 : ((c + 0.055) / 1.055)^2.4
X = 0.4124564 R + 0.3575761 G + 0.1804375 B
Y = 0.2126729 R + 0.7151522 G + 0.0721750 B
Z = 0.0193339 R + 0.1191920 G + 0.9503041 B
f(t)        = t > 0.008856 ? cbrt(t) : 7.787 t + 16 / 116
L = 116 f(Y) - 16
a = 500 (f(X / 0.95047) - f(Y))
b = 200 (f(Y) - f(Z / 1.08883))
```

[TECH] **Measuring a block.** For each block's default state, read its blockstate file, resolve its model and every texture its faces use. A texture's colour is the alpha-weighted mean of its first animation frame in linear RGB. A block's colour is the area-weighted mean over its model's faces, converted to CIELAB. Faces with a tint index are averaged separately and multiplied by the tint when it's known. A texture's flatness is `clamp(1 - (c - cFlat) / (cRaw - cFlat), 0, 1)`, where c is the deflated length of its first frame's RGB bytes at level 9, cFlat the same for a single-colour image of the same size, and cRaw its uncompressed length. A block's flatness is the area-weighted mean of its faces' flatness.

[TECH] Vanilla is measured once by a lab task from the game's own client resources jar, which ModDevGradle places in `build/moddev/artifacts`, and written to `neoforge/src/main/resources/mansion/vanilla_measures.json`, committed so builds work offline. Modded blocks are measured at server start from their own jars through FML's mod file resource lookup, and kept in memory only.

R6, lost purpose, in Minecraft

[DESIGN] Each relationship found in a copy breaks with probability 1 minus 0.9 to the power gEff, decided once per pair.

Relationship	Found where	Broken form
Door and passage	A door with captured open air on both sides	The air on the side away from the open-space core is filled, two blocks deep, with the wall's family. A door onto brick
Stair and step	Stair blocks in a run	Each becomes the full block or slab of its own family, keeping the slope's outline
Window and outside	Window-role blocks with open air on both sides	The glass becomes the wall's family inside the kept frame
Picture and wall	A painting or item frame on a wall	The entity is removed and the wall behind it is cut to the same size, one block deep, or through if open air lies behind
Bed and frame	A bed	The foot half sinks one block
Lamp and switch	A lever or button within 8 blocks of a lamp	The switch stays, the lamp keeps its captured state whatever the switch does

Relationship	Found where	Broken form
Lost support	An object hanging on a wall above an appliance block entity	The appliance is dropped, the hanging object stays. A hood with no cooker

R7, closure, in Minecraft

[DESIGN] The closing surface goes in at the height of the highest kept block in each column of sky deficit plus one, or at the region's prior ceiling height where that's higher, so a closed yard is never cramped. The surface follows the rungs of chapter 10.

```
floodlight banks, for morning, afternoon, dawn and dusk captures
  side       = east for a morning capture, west for an evening one
  banks      = light-role blocks from stock, one every 4 blocks along the closing wall on that side,
               1 to 2 blocks above the region's floor, each behind the stained glass whose colour
               is nearest the sunlight colour at capture
  throw      = invisible light blocks, at most 64 per chunk, in open cells 2 blocks above the floor,
               3 blocks apart, at level round(15 * (1 - 0.6 * d / depth)), where d is the distance
               from the bank wall and depth the region's depth away from it, only where the
               existing light is lower
```

[TECH] The sunlight's colour and height come from the core's copy of the game's sun maths, for the captured day time.

```
timeOfDay(dayTime) = (d * 2 + (0.5 - cos(d * PI) / 2)) / 3,  d = frac(dayTime / 24000 - 0.25)
sunHeight          = cos(timeOfDay * 2 PI)
sunrise colour when -0.4 <= sunHeight <= 0.4, with s = sunHeight / 0.4 * 0.5 + 0.5
  red   = s * 0.3 + 0.7
  green = s * s * 0.7 + 0.2
  blue  = 0.2
  strength = (1 - (1 - sin(s * PI)) * 0.99)^2
outside that band the light is white
```

Moment	Rule
Midday	sunHeight of 0.7 or more. White light-role blocks evenly in the closing surface
Morning or afternoon	sunHeight between 0.4 and 0.7. Warm banks on the sun's side
Dawn or dusk	Inside the sunrise band. Red banks on the sun's side, a dark surface, star points
Night	sunHeight below minus 0.4. No substitute, star points by moon phase

R8, echoes, in numbers

[DESIGN] Each surviving object starts an echo chain with probability min(0.03, 0.002 times gEff). The chain runs toward the nearest fresher instance of the same memory, or along the room's long axis when there is none, using the ratios of chapter 10, and stops when the next width would fall below a quarter of a block. Echoes take the display path.

R9, convergence

[DESIGN] Above gEff 7, materials become urn draws for every region, proportions move fully to the prior, and lights to the prior arrangement. Where a room's span exceeds the prior span plus two spreads, partitions are placed every 5 blocks with gaps of 1 to 2, and where a partition crosses a doorway port the opening takes the shape of the commonest doorway in the stock.

R10, re-expression, in full

```
rooms(memory)
  for each layer of open voxels at standing height, one above each floor
    clearance(v) = chessboard distance to the nearest non-open voxel in the layer
  cores        = 6-connected components of open voxels with clearance >= 2, at least 6 floor cells
  grow every core through the remaining open voxels by breadth-first search, all cores at once
  where two cores' fronts meet, the meeting voxels are a cut between those rooms
  rooms smaller than 6 floor cells join the neighbour they share most cut area with
  each cut is classified by its cross-section, as ports are
```

```
reexpress(room r, g)
  chosen with probability clamp((gEff(r) - 1) / 2, 0, 1), seeded by instance and room index
  w       = clamp((gEff(r) - 1) / 2, 0, 1)
  spanX   = lerp(measured span along r's long axis, prior span, w)
  spanZ   = lerp(measured short span, prior span, w)
  height  = lerp(measured height, prior height, w), rounded, 3 or 4
  shape   = a rectangle, or an L where r's footprint fills less than 70 per cent of its box
            and splits cleanly into two rectangles
  floor, walls, ceiling = r's own dominant families
  lights  = the default arrangement in r's light family
  wall objects in their measured order round the walls, spaced evenly
  open objects toward the middle, in their measured order along the long axis
  containers, signs, banners and paintings always kept
  former outdoor air becomes part of the room, its ground kept as the floor
```

```
layout(rooms)
  start every room at its measured centroid, scaled by its span change
  repeat 32 times, in room index order
    push apart any two overlapping rooms along the axis of least overlap, half each
  for each measured connection
    if the two rooms share a wall, cut an opening of the measured kind through it
    otherwise join them with a corridor 2 or 3 wide at the prior height, a room of the Complex's own
  run the no-pocket check on the result, and punch the thinnest wall to fix any pocket
```

Prior	Value
Room span	7 blocks, spread 2, drawn per room
Height	3, or 4 where the measured height was 4 or more
Growth	Spans grow 10 per cent per remembering, up to 16 blocks
L threshold	Footprint under 70 per cent of its bounding box

The default light arrangement

```
for each room
  axis      = the room's long axis
  offset    = Hash.of(worldSeed, LIGHTS, roomId) picks an offset 0 to 4 on each axis
  lines     = every 5 blocks across the axis from the offset
  along a line, alternate gaps of 4 and 5 blocks from the offset
  at each point, the ceiling block becomes a diffuser, a window-role block from stock,
  and the block above it a light-role block, both chosen from stock and the founder prior
```

Agreement

```
agreement(memory M, copy C)
  S = M's non-air voxels with intensity at least 0.2
  best over the 8 orientations and every translation in [-4, 4] on each axis
    share of S whose state in C, at the transformed position, equals its state in M
```

[DESIGN] Agreement is a test measure only, used by T-6.1, and never runs in the game.

Rasterising, for M2

1. Gather the plans of every territory whose nucleus is within reach of the chunk plus the largest instance radius.
2. Collect the instances overlapping the chunk, evaluate their copies brick by brick, and resolve every voxel by R4 across instances.
3. Sections no instance touches become one structure block as a single-value container.
4. Place light blocks, block entities and the steady-light and still-plant flags.
5. Write the realisation attachment.

39. The NeoForge 1.21.1 Cookbook

[TECH] Every name below was checked against the decompiled 1.21.1 sources with NeoForge 21.1.250 on 4 October 2026. Parchment names are used. If a name ever fails to resolve, read the source in the ModDevGradle artifacts folder rather than guessing.

Where each need hooks in

Need	Entry point	Notes
Registering anything	`DeferredRegister.create(registryKey, "mansion")`, registered on the mod event bus in the mod's constructor	
The chunk generator	Subclass `ChunkGenerator`, register its `MapCodec` with `Registries.CHUNK_GENERATOR`	Implement `codec`, `applyCarvers`, `buildSurface`, `spawnOriginalMobs`, `getGenDepth`, `fillFromNoise`, `getSeaLevel`, `getMinY`, `getBaseHeight`, `getBaseColumn`, `addDebugScreenInfo`. `fillFromNoise` returns a `CompletableFuture`, which is how plans are awaited without blocking
Dimension, biome, level stem	JSON under `data/mansion`, below	
No sky	Subclass `DimensionSpecialEffects` with sky type `NONE`, register with `RegisterDimensionSpecialEffectsEvent.register` on the client	Override `renderSky`, `renderClouds`, `renderSnowAndRain` and `tickRain` to return true, which tells the game they're handled, and draw nothing
Light tints	`adjustLightmapColors` on the same class	The lightmap is one table per frame for the whole view, so a tint can follow only where the camera is. It can't vary from block to block. Chapter 10's per-block colour comes only with the coloured light module
Chunk attachments	`AttachmentType.builder(...)` with `serialize(IAttachmentSerializer)` and, where needed, `sync(StreamCodec)`, registered with `NeoForgeRegistries.Keys.ATTACHMENT_TYPES`	Read and write through `getData`, `setData` and `getExistingDataOrNull` on the chunk. Remember to mark the chunk unsaved after a change
Unloaded chunk NBT	`serverLevel.getChunkSource().chunkMap.read(chunkPos)`	Public, returns a future of the raw saved tag, served by the IO worker. Not upgraded by the data fixer, so check `DataVersion` and upgrade with `upgradeChunkTag` when it differs
Writing chunk NBT back	`chunkMap.write(chunkPos, tag)`	Only for chunks that aren't loaded, never while any player holds them (chapter 14)
The level.dat record	Mixin X1 on `PrimaryLevelData`, at the tail of the private `setTagData` to write, and at the return of the static `parse` to read the `Dynamic` it was given	
Falling through openings	Mixin X2 at the head of `BlockBehaviour.BlockStateBase.getCollisionShape(BlockGetter, BlockPos, CollisionContext)`	Movement passes the level itself as the getter. This method is very hot, so return early through a static flag that's false whenever no null positions are loaded on that side
Lights keep their state	Mixin X3 at the head of `Level.setBlock(BlockPos, BlockState, int, int)`, Mansion level only	Refuse a change that keeps the same block and only alters `lit` or `powered`, at a position flagged steady. Covers redstone lamps, copper bulbs and every modded lamp that uses those properties. Player placement clears the flag through `BlockEvent.EntityPlaceEvent`

Need	Entry point	Notes
Still plants	Mixin X5 at the head of `BlockBehaviour.BlockStateBase.randomTick`, Mansion level only	Cancel at flagged positions
Removed mods	Mixin X6 at the head of `ChunkSerializer.read`, Mansion level only	The tag is already upgraded at that point. Rewrite each section's `block_states` palette in place, using fingerprints
Settings	`GameRules.register(name, GameRules.Category.MISC, GameRules.IntegerValue.create(default))`, from the mod constructor	Booleans through `GameRules.BooleanValue.create`
Network	`RegisterPayloadHandlersEvent`, `event.registrar("1").playToClient(type, codec, handler)`	Send with `PacketDistributor.sendToPlayer` and `sendToPlayersNear`
Commands	`RegisterCommandsEvent` on the game event bus	
Ticking	`ServerTickEvent.Post`	Every tick handler checks its own time budget
Block changes	`BlockEvent.BreakEvent`, `BlockEvent.EntityPlaceEvent`	Steady-light flags, coloured lamp colours
Containers	`level.getCapability(Capabilities.ItemHandler.BLOCK, pos, null)`	For the container role
Weathering chains	`NeoForgeDataMaps.OXIDIZABLES` and `NeoForgeDataMaps.WAXABLES`, with their inverses through `DataMapHooks`	For Ro's clean forms
Invisible light	`Blocks.LIGHT` with `LightBlock.LEVEL`	For the floodlight throw
Map colour	`getMapColor(level, pos)` on the block state	The colour fallback
Tinted blocks	`RegisterColorHandlersEvent.Block`, `event.register(colour, blocks)`	The Mansion's handler for grass, leaves, vines and water reads `mansion:colours` in the Mansion and calls vanilla's `BiomeColors` functions everywhere else
A synthesised hum	Override `getStream(SoundBufferLibrary, Sound, boolean)` on a custom `SoundInstance`, returning a completed future of your own `AudioStream`	`AudioStream` has `getFormat`, `read(int)` and `close`. Spike S3 proves that a sound event with no file is accepted quietly
Game tests	`RegisterGameTestsEvent`, in the lab	
Mod jar resources on a server	`ModList.get().getModFiles()`, then `getFile().findResource(path)`	For measuring modded blocks

Data files

```
data/mansion/dimension_type/the_mansion.json
{
  "has_skylight": false, "has_ceiling": true, "ultrawarm": false, "natural": false,
  "coordinate_scale": 1.0, "ambient_light": 0.0, "fixed_time": 6000,
  "bed_works": true, "respawn_anchor_works": false, "has_raids": false, "piglin_safe": false,
  "min_y": -512, "height": 1024, "logical_height": 1024,
  "infiniburn": "#minecraft:infiniburn_overworld", "effects": "mansion:complex",
  "monster_spawn_light_level": 0, "monster_spawn_block_light_limit": 0
}

data/mansion/worldgen/biome/complex.json
{
  "has_precipitation": false, "temperature": 0.5, "downfall": 0.0,
  "effects": { "fog_color": 0, "sky_color": 0, "water_color": 4159204, "water_fog_color": 329011 },
  "spawners": {}, "spawn_costs": {}, "carvers": {}, "features": []
}

data/mansion/dimension/the_mansion.json
{
  "type": "mansion:the_mansion",
  "generator": {
    "type": "mansion:complex",
    "biome_source": { "type": "minecraft:fixed", "biome": "mansion:complex" }
  }
}
```

Skeletons

[DESIGN] These show the shape of the entry points, with no comments, as all code in the project must be written.

```
@Mod(Mansion.ID)
public final class Mansion {
    public static final String ID = "mansion";

    public Mansion(IEventBus modBus) {
        Registration.GENERATORS.register(modBus);
        Registration.ATTACHMENTS.register(modBus);
        Settings.register();
        modBus.addListener(Network::register);
        NeoForge.EVENT_BUS.addListener(Commands::register);
        NeoForge.EVENT_BUS.addListener(Ticker::onServerTick);
    }
}
```

```
public final class MansionGenerator extends ChunkGenerator {
    public static final MapCodec<MansionGenerator> CODEC = RecordCodecBuilder.mapCodec(instance -> instance
            .group(BiomeSource.CODEC.fieldOf("biome_source").forGetter(ChunkGenerator::getBiomeSource))
            .apply(instance, MansionGenerator::new));

    public MansionGenerator(BiomeSource biomeSource) {
        super(biomeSource);
    }

    @Override
    protected MapCodec<? extends ChunkGenerator> codec() {
        return CODEC;
    }

    @Override
    public CompletableFuture<ChunkAccess> fillFromNoise(Blender blender, RandomState random,
            StructureManager structures, ChunkAccess chunk) {
        return Stage.rasterise(chunk);
    }
}
```

```
@Mixin(PrimaryLevelData.class)
abstract class PrimaryLevelDataMixin {
    @Inject(method = "setTagData", at = @At("TAIL"))
    private void mansion$write(RegistryAccess registries, CompoundTag tag, CompoundTag player, CallbackInfo info) {
        tag.put("mansion", Record.current().save());
    }

    @Inject(method = "parse", at = @At("RETURN"))
    private static void mansion$read(Dynamic<?> tag, LevelSettings settings,
            PrimaryLevelData.SpecialWorldProperty special, WorldOptions options, Lifecycle lifecycle,
            CallbackInfoReturnable<PrimaryLevelData> info) {
        Record.load(tag.get("mansion").result().map(d -> d.convert(NbtOps.INSTANCE).getValue()).orElse(null));
    }
}
```

[TECH] The skeletons stop where the real work starts. `Record`, `Stage` and the rest are the project's own classes. The level.dat record is one per server, so `Record` is keyed by the running server and cleared on shutdown, never held in a static across worlds. `parse` runs before the server object exists, so what it reads is held as pending and claimed by the server in `ServerStartingEvent`.

40. Milestone M0, Step by Step

[DESIGN] M0 answers one question. Does a copy made by these laws feel like canon's copies? It builds only what that question needs. No storage, no Genesis, no openings, no creatures. Memories live in the server's memory for the session and are captured again after a restart, which writes nothing and keeps the storage rule.

Each step lists what to build, how to check it, and what to keep as evidence. **STOP** marks a point where you show the user what exists and wait for their answer.

M0.1 The skeleton

- Create the four subprojects, the build files, the mod metadata, the mixin configs and `HumanCheck`, exactly as chapter 35 gives them.
- Each mod class registers nothing yet except what the next step needs.
- `gradlew build` passes, `checkHuman` included. `gradlew :neoforge:runClient` opens the game, and the mod list shows The Mansion and Coloured Lamps.
- Settle whether `jarJar` takes the project dependency, and write the decision record.

M0.2 The empty dimension

- Register the chunk generator codec and add the dimension type, biome and level stem JSON of chapter 39.
- The generator fills every block with the structure material, the founder prior's first structure candidate (Appendix C), as single-value sections. No other block.
- Register the dimension's special effects with sky type `NONE`, black fog, and the four render overrides returning true.
- Add `/mansion enter` and `/mansion exit`, operators only, which teleport a player into a small open cell carved at 0, 64, 0 and back to where they came from.
- Check that the dimension loads, that there is no sky, cloud, sun, star or weather anywhere, that sky light is 0 everywhere, and that a new chunk column 1,024 blocks tall generates within its budget.
- **Evidence.** A screenshot looking up and down the carved cell, the debug screen showing sky light 0, and the time to generate a column.
- **STOP.** Show the user. This is the first thing they'll see of the new mod, and the right moment to catch anything wrong with the dimension itself.

M0.3 The block table and the measured colours

- Build the block table of chapter 37 at server start.
- Write the lab task that measures vanilla colours and flatness from the client resources jar, run it, check the numbers by eye on a dozen blocks (concrete flat, wool and planks less so, grass grey before its tint), and commit the table.
- Measure modded blocks at start the same way, in memory.
- Core tests T-5.2 and T-5.3 run against the real table, every substitute keeps shape and is inert.

M0.4 The snapshot, exposure and the flood

- `/mansion capture here [budget]` takes a snapshot round the player's feet, runs the flood off the server thread, and stores the memory in the session's memory list. It prints the memory id, kept voxels, exposed share of open space, sky deficit size and time taken.
- Core tests T-3.5 and T-3.6 pass, and a lab game test checks that capture reading stays within 2 milliseconds per tick.
- `/mansion debug exposure` pastes a view of the last capture in the Mansion with each voxel shown as a concrete colour by its sky light, for checking by eye. Development only.

M0.5 Analysis

- Roles, face regions, the shell, objects, support depth, scale, complexity and ports, as chapter 38 gives them.
- `/mansion debug roles` pastes a view with each role shown as one concrete colour. Development only.
- Check on three captures of different kinds of building that floors, walls, ceilings, furniture and lights come out where a person would say they are, and that ports sit at doorways.

M0.6 Remembering

- Operators R0 to R9 in the order of chapter 38, with R10 left for the next step, and the substitution rules in full.
- `/mansion paste <memory> [from] [to] [seed]` lays copies of a memory at generations `from` to `to`, default 0 to 8, in a row along the x axis of the Mansion, each in its own pocket of structure fill 16 blocks from the next, and `/mansion goto <memory> <g>` stands you in one at its open-space core. Pasting sets blocks in loaded chunks a slice per tick and installs light with the blocks.
- Core tests T-2.1, T-3.2, T-3.4, T-4.1, T-5.1, T-5.4, T-5.5, T-6.1 and T-6.3 pass.
- **Evidence.** A capture of a house from inside, pasted at 0 to 8. Screenshots of each copy from the same place in the room.
- **STOP.** Show the strip. The question for the user is whether generation 0 looks faithful inside and streamlined outside, and whether the decay reads like the teaser.

M0.7 Re-expression

- R10 in full, rooms, cuts, rebuilding and layout, including former outdoor space becoming rooms (O5).
- Core tests T-3.7 and T-6.4 pass, and the no-pocket check passes on every re-expressed copy.
- Run it on the world of absurd builds from the first day, as the risk register says.

M0.8 Light, closure and the hum

- Mixin X3 and the steady-light flags. A pasted lamp keeps its captured state when redstone around it changes.
- R7 in full, both rungs of surfaces and the light of the moment, including floodlight banks and the throw.
- The camera-following tint through `adjustLightmapColors` for dusk copies.
- The default light arrangement in re-expressed and converged rooms.
- Spike S3 and the hum. If S3 fails, stop and ask the user about decision E-9.

M0.9 The answer keys

- In the lab, `/lab fixture <name>` builds each original of chapter 29 in a fresh copy of a flat test world, the teaser's sunlit basement, a suburban street, a house with a porch and hedges, a store with glass front doors, a car park and a basement, a yard with grass and a street lamp, and a plain corridor. Fixtures are lab code and never ship.
- Capture each one from where canon's opening was, the basement, inside the house, inside the store, and also from a real build in a copy of Broville and from the absurd-build world.
- Paste each at 0 to 8 and take matched screenshots.

M0.10 The report

- For each answer key, the canon frame beside the copies, the numbers, and an honest note of where it matches and where it doesn't, in the words chapter 29 suggests, "stage four keeps the window too long".
- The list of tests run and their results, the budgets measured, and every parameter changed from Appendix B with its reason.
- **STOP.** The user decides whether M0 feels like the Complex. If it doesn't, the answer keys say what to change, and M0 continues until it does. Nothing from M1 on is started before the user says yes.

41. Milestones M1 to M10, Step by Step

[DESIGN] Each milestone follows the same pattern as M0. Build in the order given, keep the evidence, stop where marked. Exit tests are those of chapter 32.

M1 Storage and capture

1. Spike S1. Mixin X1 writes and reads a small test compound in level.dat, survives a restart, and T-S.1 shows no new file.
2. The level.dat record of chapter 37 in full, with its pending-then-claimed loading.
3. The `mansion:memories` attachment with its serializer, the 384 kibibyte cap and the spill to neighbouring chunks.
4. Spike S2. Reading memory parts from unloaded chunks through `chunkMap.read`, while the game saves chunks around it, without a race. Measure the cost.
5. The Opening Ledger, passengers with their profiles and traces, and the memory index.
6. Exit with T-1.1, T-4.1, T-S.1 and T-S.2, and a captured house that survives a restart and pastes identically.
7. **STOP.** Report the storage numbers, bytes per memory, level.dat size, read times.

M2 Genesis and lineage

1. Nuclei, territories and seam contracts, all pure functions, tested for order independence first.
2. Growth inside a territory, sea fill, connectivity repair and character, as chapter 11 gives them.
3. Both urns and their epochs, landing sites and the island registry.
4. The rasteriser of chapter 38 inside `fillFromNoise`, returning a future, never blocking.
5. The walker bots, the stranding test and the plan images of chapter 42.
6. Exit with T-7.1, T-8.1, T-8.2 and T-12.1, plus walkers and stranding on three worlds' captures.
7. **STOP.** Send the plan images first, then the walker and stranding results. The user looks at the plans before anyone walks the build.

M3 Openings and crossing

1. Null potential, the `mansion:null` attachment and its payload, and mixin X2.
2. Glitches with canon's only signs, items falling through and long objects widening them.
3. Crossing, the fall, arrival and exits, with live captures firing at each crossing.
4. Rule O1 enforced on every opening, live and retrospective.
5. Exit with a first-fall playtest and live captures passing T-1.1.
6. **STOP.** A recorded first fall from a basement in a copy of a real world.

M4 Observation

1. Hold radii, groups, Observer cameras and beds.
2. Spike S4, the pre-load rewrite through `chunkMap.read` and `chunkMap.write`, and its fallback.
3. Re-remembering with local memories, and change behind a lone player's back.
4. Exit with T-13.1 and T-13.2, fuzzed with several bot players, and a "not what you remembered" playtest.

M5 Atmosphere

1. Light liveness everywhere, cold, food and water, sound leaks and the crossing effect.
2. Exit with T-3.1 and a blind playtest against the original photograph.

M6 Still Lifes

1. Templates from passengers, the renderer on the vanilla player model, distortions by generation.
2. Fidelity, the behaviour echo, dens, edibility, plant and animal Still Lifes with mixin X5.
3. Exit with T-9.1 and a "meet yourself" playtest.

M7 Reach and growth

1. The join field, the doorway ladder in Clark's order, doorways.
2. Growth events with the green glow and sliding walls, within the display budget.
3. Time slips and personal dates.
4. Exit with T-10.1 and a full Clark-sequence playtest.

M8 Lifeform

1. Remains, film, colony and walking stages.
2. The procedural body, hearing through vibrations, staying unseen, the sound lure and the charge.
3. Exit with fair-escape playtests.

M9 Threshold

1. Ask the user to decide E-8, the block pattern, before writing anything.
2. Recognition, tests that fail more often than they succeed, and the consequences for world activity.
3. Exit with a full Threshold playtest.

M10 Polish and release

1. Every budget of chapter 30 measured on every test world, single-player and on a server with twelve players spread far apart.
2. The compatibility matrix of chapter 42.
3. The README, written plainly, and the release jar, checked by `checkHuman` and by reading it through.

C Coloured light

1. Only after M2. Spike C0 first, as chapter 34 sets out.
2. Then the recipe, the data component, the `colouredlamps:colours` attachment, the light field and both renderer hooks.
3. Exit with T-C.1 to T-C.5 and the frame budget.

42. Testing, Evidence and Reporting

Test worlds in practice

Kind from chapter 29	Where it comes from
A flat creative town of ordinary houses	Built by the lab's fixtures in a fresh flat world
A real, long-lived server map	A fresh copy of `Broville v11.4.2`. It's a 1.20.1 save, so 1.21.1 upgrades it on first load, which is why only copies are ever opened
A fresh survival world	Generated new in the lab's run folder from a fixed seed
A redstone-heavy base	Ask the user for one
A heavily modded world	Ask the user before downloading any NeoForge 1.21.1 decoration or furniture mods
Absurd builds	Ask the user which of their worlds to use
An empty world	A flat void world generated by the lab

[YOUR RULE] Copy a world into the lab's run folder under a name with the date and time, use it, and delete it when the run ends. Check free space first and don't start a run with less than 10 gigabytes free.

Walker bots

[DESIGN] Walkers run in the lab on the walk graph of a generated area, so they test the geometry players will meet without needing real players.

1. **The walk graph.** A cell is standable where a player fits, two blocks of open space above a block with collision. Edges join standable cells a player can move between, a step up of one block, a drop of up to three blocks, a ladder or vine, a stair, a door. Drops of more than three blocks are one-way edges.
2. **The walkers.** At least ten, all starting on the same cell, each given a different heading in three dimensions. Each follows the Pledge rule, keeping its heading where it can and wall-following otherwise, counting turns until the count returns to zero.
3. **Spread, then regroup.** For a fixed number of steps they spread out. Then each finds its way back to the group's centre through the graph. The run reports how far each got, how many got stuck, and how many couldn't regroup.
4. **Multiplayer.** The same run is repeated on a lab server with the walkers as fake players, so hold radii, re-remembering and growth events see many people far apart.

The stranding test

[DESIGN] Drop ten thousand simulated players on random standable cells of a generated region. Each must reach some cell at least 500 blocks away horizontally through the walk graph, one-way drops allowed. The result is the share that can't. It must be zero, and every stranded cell is listed with its position and a plan image round it.

Plan images

[YOUR RULE] Every generation test writes full, uncropped plan images, one per stratum, styled like the fan maps of the Complex.

- Top-down, four pixels per block, covering the whole generated area plus a margin, never cropped to the interesting part.

- Walls and solid fill dark, walkable floor coloured by generation from warm for fresh islands to the sea's pale yellow, portals and punched openings marked, pits and chutes marked with their direction, former outdoor space outlined.
- A scale bar, the stratum's height range and the run's seed in the corner.
- Saved as PNG in `lab/results/<run>/plans`, named by stratum.

Screenshots

[DESIGN] The lab client takes screenshots from set camera positions with the game's own screenshot function, at a fixed field of view, with the debug screen hidden, into `lab/results/<run>/shots`. For answer keys, the same camera position in each generation's copy. Look at every one before sending it.

Measuring performance

- **Server.** In the lab only, every Mansion tick handler and worker task is wrapped with a timer. The report gives the mean and the slowest one in a hundred per tick, single-player and with twelve fake players.
- **Garbage collection.** Runs use Java Flight Recorder, and no pause may exceed 20 milliseconds.
- **Client.** Average and slowest-one-in-a-hundred frame times in the Mansion against a plain flat world at the same render distance, within 10 per cent.
- **Chunks.** Generation and lighting times per column against chapter 30.

Compatibility matrix

[YOUR RULE] The mods listed in the project's earlier quality rules were Fabric mods. For NeoForge 1.21.1, test against whichever of these exist as NeoForge 1.21.1 builds at the time, a chunk renderer such as Sodium or Embeddium, Iris with a shader pack, a server optimiser such as Lithium, Create, a dynamic lights mod, a decoration or furniture pack, and a replay or recording mod. Ask the user before downloading anything, keep the jars only in the lab's run folders, and never add them as dependencies.

The report template

```
What I built
How I tested it, on which worlds, with how many players
Results, with numbers, against the exit test and budgets
Images, plans first, then screenshots beside canon frames
What failed or only partly works, plainly
What I changed from the bible, with decision record numbers
What comes next, and anything I need you to decide
```

[YOUR RULE] A result is a pass only if it meets the test as the bible states it. Anything weaker is reported as what it is.

43. Defaults, Resolutions and Traps

Decisions made in version 0.8

[DESIGN] Earlier versions left these open or got them wrong. They're settled here so nothing blocks the build. Each can be changed through a decision record.

Question	Resolution
Is R0 applied at capture or at remembering	At remembering. Memories are stored raw (chapter 38)
R1's parameters	THETA0 minus 1.0, A 1.6, noise 0.6, nothing below gEff 0.05. The earlier values contradicted T-6.2
What stops the flood	The game's own light opacity, open, transmissive or matter, with I_MIN 0.02 and hardness scaling
How big a snapshot is	4, 8 or 12 chunks round the opening, by kind
How roles are decided	The fourteen ordered rules of chapter 38
How a substitute is chosen	One cost of colour, flatness and source tier, with monotony as an exclusion
What a material is	A family, the full block a shape is cut from, found through recipes
Can the red floodlight tint vary across a street in M0	No. The lightmap follows the camera. Per-block colour waits for the coloured light module
How lamps keep their state for any mod	Mixin X3 on block changes, refusing lit and powered flips at steady positions
Is a mixin needed for light tints	No. The dimension's own special effects class handles it
Fixed-point maths	Not needed. Doubles with StrictMath are deterministic on Java 21
Where test tools live	The lab subproject, its own mod, never shipped
Test libraries	JUnit in core's tests only
Where the vanilla colour table lives	In the Mansion's own resources, measured by the lab and committed
How level.dat is read before the server exists	Held as pending, claimed in `ServerStartingEvent`

Traps

- **Files sneak in.** SavedData, level attachments, config specs, crash-report extras and debug dumps all write files. T-S.1 catches them, but check before adding anything that persists.
- **Blocking waits.** `join` or `get` on a future anywhere on the server thread breaks the threading rule. Chain futures, or pick the result up on a later tick.
- **Unsaved chunks.** Changing an attachment without marking the chunk unsaved loses the change.
- **Light without blocks.** Install light in the same update as the blocks, or players beyond the view edge won't see it.
- **Painting held space.** Never rewrite a chunk any player holds. M4's rules decide what's held.
- **Version drift.** Old saves load unknown blocks as air. Check a world's version before using it.
- **Fabric jars.** Nothing in `Mods For Testing` loads on NeoForge.
- **Disk.** The C drive is nearly full. Clean up copies and Gradle run folders.
- **Tuning to a map.** If a number only works on one world, it's wrong. Validate on at least two very different worlds.
- **Comments creeping back.** IDE templates, NeoForge's template mods.toml, generated build files and pasted snippets all bring comments. `checkHuman` catches them, and the fix is to delete them, never to loosen the check.

APPENDICES

Traceability, default parameters, the founder prior, a glossary, open decisions, and what changed since version 0.1.

A. Law to Subsystem Traceability

Law or canon	Implemented by	Tested by
The human-made mandate	Chapter 0, chapter 36, `checkHuman` (chapter 35)	T-H.1
1 Capture	Capture events, the transmittance flood and passengers (chapter 8), the Opening Ledger, landing sites (chapter 12), retrospective captures (chapter 23)	T-1.1
2 Two fidelities	R2 drift, R4 merge, R8 echoes, the identity guard on every operator (chapter 10)	T-2.1, T-6.3, T-13.2
3 Matter, not light	The sky deficit (chapter 8), exposure, the starting generation, R0, the sky drain and outdoor rules O1 to O4 (chapter 9), R7 sky closure (chapter 10), the dimension type (chapter 7)	T-3.1, T-3.2, T-3.3, T-3.5
Your sky-exposure theory	Chapter 9	T-3.2, answer-key review of the blue-box house and the film's street
Your re-expression idea	R10 (chapter 10)	T-6.4, the store and absurd-build answer keys
Captured light state	Chapter 9 and chapter 19	T-3.4
Glass passes the capture	Chapter 9	T-3.6
4 The open space is the centre	The transmittance flood (chapter 8), R6 doors onto brick (chapter 10)	T-4.1
5 Interpretation	The Role Lexicon (chapter 8), R5 fill from the nearest real source and R7 (chapter 10)	T-5.1
6 Worse copies	R1 and R3 (chapter 10), the Lineage Field (chapter 12), re-remembering (chapter 14)	T-6.1, T-6.2
7 The snowball	The material urn and the memory urn with epochs, drift and mutation (chapter 12)	T-7.1
8 Continuation	Genesis, nuclei, territories, seams, growth, repair, pits (chapter 11)	T-8.1, T-8.2
9 Structure before function	Mechanism fidelity (chapter 10), Still Life fidelity, plants and animals (chapter 20)	T-9.1
10 Reaching back	The join field, the doorway ladder, signature objects (chapter 16)	T-10.1
Your stability hypothesis	Hold radii, groups, cameras, the never-in-front rule (chapter 14)	T-13.1
Your floodlight hypothesis	Night closure with low, directional red floodlights (chapter 10)	Answer-key review
Your plant hypothesis	Plant Still Lifes (chapter 20)	Game test on flagged plants
Canon, forward-only time	Chapter 17	Game test on crossings
Canon, the green glow	Chapter 18	T-13.1
Your storage rule	Chapter 6	T-S.1, T-S.2

B. Default Parameters

Gamerules

Gamerule	Default	Meaning
mansionFounderWeight	100	Strength of the founder prior, in per cent. 0 grows the default entirely from the world
mansionGlitchRate	100	Glitch frequency, in per cent of the baseline
mansionForgetDays	3	In-game days unobserved before a chunk is re-remembered
mansionPlayerBuildDrift	57	Drift of player-made objects, as a percentage of normal drift. 0 turns it off
mansionLootFidelity	35	Percentage of copied container stacks kept at capture
mansionRetrospectiveRate	3	Retrospective captures per real hour. 0 turns them off
mansionReachIntensity	100	Frequency of signature objects near joined places, in per cent
mansionOpeningCap	64	The most openings the feedback loop can hold open in a world at once
mansionExitScatter	3000	Blocks from the nearest known opening that an exit may put you
mansionTenchJumps	false	Whether a slip moves the real world clock on a server
mansionDisplayBudget	600	Display entities in view per player

Internal parameters

Parameter	Default	Where
Dimension height	1,024, from minus 512	Chapter 7
MU_AIR and muMatter	0.004 and 0.9 per block, matter scaled by 0.6 plus 0.27 times hardness, between 0.6 and 2	Chapters 8 and 38
I_MIN	0.02	Chapter 38
Transmissive blocks	MU_GLASS plus 0.06 per level of light opacity	Chapter 38
Snapshot radius	4, 8 or 12 chunks, 48, 96 or 128 blocks up and down	Chapter 38
Capture budgets	32 thousand, 400 thousand, 2 million voxels	Chapter 8
THETA0 and A, in R1	Minus 1.0 and 1.6, noise within 0.6, inactive below gEff 0.05	Chapters 10 and 38
SIGMA and BETA, in R2	0.35 blocks and 0.5	Chapters 10 and 38
R0 thresholds	Wear with probability e, one colour by smoothstep 0.2 to 0.8, clutter below scale 2.5 e, plants and edges from e 0.5	Chapter 38
R6 break probability	1 minus 0.9 to the power gEff	Chapter 38
Objects beating the shell in R4	From gEff 3, probability (gEff minus 3) over 6, at most 0.9	Chapter 38
Substitute cost weights	Flatness 10, source tier 1.5 per tier	Chapter 38
Floodlight throw	At most 64 light blocks per chunk, 3 apart	Chapter 38
Room priors in R10	Span 7 spread 2, height 3 or 4, growth 10 per cent to 16 blocks, L below 70 per cent fill	Chapter 38
KMAT, in R3	0.82, so about a fifth of the original material survives eight generations	Chapter 10
Echo probability	0.002 times g, capped at 0.03	Chapter 10
Echo ratios	0.785 at first, shrinking by 0.744, gaps by 0.939	Chapter 10
Convergence	Above generation 7	Chapter 10
Nucleus radius and vertical flattening	64 blocks and 3.5	Chapter 11
Instance overlap	Up to 30 per cent of open volume	Chapter 11

Parameter	Default	Where
Sea generation	9 to 16	Chapter 12
Halo step	0.6 to 1.6 generations	Chapter 12
Founder counts	20,000 per role at weight 100	Chapter 12
Mutation share	About 5 per cent of sea nuclei	Chapter 12
Epoch length	24 in-game hours or 64 captures	Chapter 12
Hold radius	24 blocks plus view to 64, 60 per cent more per extra player up to five	Chapter 14
Observer camera cone	16 blocks	Chapter 14
Glitch baseline	About one per 10 to 20 hours for basement builders	Chapter 15
World activity after a Threshold	Rising from 3 to 6 times	Chapter 15
Exit density	About one per 40 territories	Chapter 15
Ordinary crossing offset	Up to 20 minutes	Chapter 17
Tench slip	20 to 80 days	Chapter 17
Still Life fidelity	e to the power of minus g over 1.6	Chapter 20
Mechanism fidelity	e to the power of minus g over 2	Chapter 10
Lifeform life cycle	1 to 3 days film, 3 to 8 days colony	Chapter 21
Chunk attachment cap	384 kibibytes	Chapter 6
G_SKY, starting generation at full exposure	2.5	Chapter 9
Exposure curve	e to the power 1.5	Chapter 9
K_SKY, the sky drain	0.03 per block at full exposure	Chapter 9
MU_GLASS	0.15 per block	Chapter 9
Opening exposure limit, rule O1	Below 0.4	Chapter 9
Outdoor share of a capture's open space	At most a quarter	Chapter 9
Outdoor-looking share of walkable space, rule O4	Under 2 per cent	Chapter 9
Re-expression	Probability rising from 0 at generation 1 to 1 at generation 3	Chapter 10
Room cut width in R10	3 blocks or less	Chapter 10
Substitute ranking	Same shape first, then colour distance in CIELAB, then flatness	Chapter 10
Monotony threshold	A neighbouring role's block within a colour distance of 10 counts as the same	Chapter 10
Fingerprint size	About 24 bytes per modded material	Chapter 6
Floodlight throw	Invisible light blocks, densest by the banks and thinning across the enclosure	Chapter 10
Coloured lamp brightness	15, the same as a vanilla lamp	Chapter 34
Coloured light frame budget	Under 5 per cent of frame time	Chapter 34

C. The Founder Prior

[YOUR RULE] The founder prior is numbers, never shapes. It holds counts for materials by role, proportions and the light rhythm, all measured from canon, and it only ever acts through the laws, as the starting counts of the urn and as the shape R3 and R9 pull toward. Setting mansionFounderWeight to zero removes it.

Proportions and rhythm

Quantity	Value	Canon basis
Ceiling light rhythm	A line every 5 blocks one way, alternating 4 and 5 the other	The 1.27 by 1.14 metre tile Async measured
Prior ceiling height	3 blocks, sometimes 4	The rooms of the original photograph and the series
Prior room span	Centred on 7 blocks, spread 2	The partitioned bays of the photograph and the film's restaging
Partition spacing	5 blocks, with gaps of 1 to 2	Teaser stage 8, the office partitions
Prior growth bias	Rooms pulled about 10 per cent larger per remembering until the span cap	Teaser stages 4 and 6, "the room is bigger"
Stratum spacing	About 6 blocks, room, floor and cavity	"More than one floor", the space above the tiles

Materials, calibrated not guessed

[DESIGN] Kane and Vermette both note that the famous yellow is partly a white-balance accident, and that the real room was probably closer to beige. So candidates are calibrated.

1. Compute each candidate block's mean sprite colour in CIELAB from the vanilla atlas.
2. Compare against reference targets taken from your own frames of the series and the photograph, which are never redistributed, for wall, carpet, ceiling and light.
3. Render test rooms under the default light rhythm and compare whole-frame colour histograms. Choose by colour distance and by eye.

Role	Candidates in vanilla 1.21.1	Notes
Wall	Smooth sandstone, cut sandstone, sandstone, birch planks, stripped birch, end stone, yellow terracotta	A pattern can come from two close tones alternating. Avoid busy textures at room scale
Skirting	Stripped oak or dark oak slabs and stairs, black concrete	
Floor	Brown carpet over wool, brown wool, packed mud, mud bricks, stripped oak	"Stained beige". Calibrate between darker brown and beige
Ceiling	Calcite, white concrete, smooth quartz, diorite	Calcite's speckle reads as fissured mineral fibre
Light	End rods, sea lanterns, froglights, glowstone, redstone lamps, behind a white stained glass diffuser	Ordered by how closely the lit result matches canon
Structure fill	Smooth stone, grey concrete, tuff	"Cast concrete"
Floodlight housing	The stained glass nearest in colour to the sunlight at capture, from vanilla's sunset colour function	Both canon red copies are lit from low, directional lamps, the dusk theory

D. Glossary

Term	Meaning
Memory	A captured volume of matter at one moment, with its passengers
Instance	One placement of a memory, with a transform, a generation and a seed
Generation	How many times something has been remembered. Lower is fresher
Island, sea, halo	Fresh captures, the old default rooms made of a world's earliest memories, and the gradient of copies between them
Territory, nucleus, seam contract	Genesis's irregular regions, their origin points, and the connections neighbours agree on
Material urn	The counts by role that fill gaps and set the default materials
Memory urn	The weights that choose which memories the sea is built from
Founder prior	Canon's numbers, the starting counts of the material urn and the shape the remembering pulls toward
Opening Ledger	The record in level.dat of every opening that ever touched anything
Home chunk	The chunk whose NBT holds a memory, the chunk its opening was in
Hold radius	The space an observer keeps steady
Re-remembering	Rewriting an unobserved chunk one generation worse before anyone returns
Pre-load rewrite	Re-remembering done on a chunk's saved NBT before it's loaded
Thin place	A wall in the Mansion linked to a live opening, through which sound leaks
Join field	The area around openings in our world where Reach can act
Echo chain	Repeated, shrinking instances of one object pointing toward its fresher source
Fidelity	How much of a Still Life's behaviour survived copying
Exposure	How much sky a captured spot saw, its sky light divided by 15
Starting generation	The generation a part of a capture begins at, from its exposure. Interiors 0, open sky 2.5
Streamlining	What happens to sky-facing parts, one colour per face, simple shapes, no clutter, no wear, a diorama
Sky drain	The extra loss a capture suffers in air open to the sky
Re-expression	Rebuilding a place in the Mansion's image from its learned rooms, materials and objects, operator R10
Captured light state	Whether a lamp was on or off when taken, kept forever
Flatness	How well a block's texture compresses. Flat textures like concrete score highest and are preferred as substitutes
Inert	A block with no behaviour of its own, eligible to be a substitute
Fingerprint	The colour, flatness, shape, properties and role of a modded block the Mansion has used, kept in level.dat so the block can be replaced if its mod is removed
Family	The full block a stair, slab or other shape is cut from, found through recipes. Urn materials are families
Effective generation	An element's own generation, the instance's plus the starting generation from its exposure
Lab	The test mod, mansionlab, holding fixtures, walkers, plan images and measurement, never shipped
Human check	The build task that fails on any comment, dash or other sign of machine-made work

E. Open Decisions for You

Each has a recommended default, so nothing is blocked.

Decision	Recommended default
E-1 The founder prior. Canon's yellow default from the start, or a default grown purely from the world	Weight 100, canon's look. Both are law-driven. The difference is whether canon's numbers seed the urn
E-2 Loot fidelity of copied containers	35 per cent
E-3 Drift of player bases inside	On, gentle, settable to zero
E-4 Retrospective captures of saved builds and generated structures	On, a few per hour, with exclusion areas
E-5 Real clock jumps from slips on servers	Off, personal dates only
E-6 Advancements, and whether they read as dated Async records	A small set, dated in Async's style, no story text
E-7 Dimension height	1,024 blocks
E-8 The Threshold's block pattern	Decided at M9, magnet and coil around a doorway, as chapter 15 requires
E-9 If the synthesised hum stream can't be made to work, allow a buffer generated by the build and packed in the jar	Yes, as a fallback only
E-10 Licence and distribution	Source-available, non-commercial fan project, on Modrinth and CurseForge
E-11 Exit scatter	3,000 blocks
E-12 The crossing effect at all	Yes, following the vanilla Distortion Effects slider
E-13 Strength of streamlining, G_SKY	2.5, tuned at M0 against the blue-box house and the film's street
E-14 How early re-expression starts	From generation 1, always by 3
E-15 Whether the coloured light module ships with the Mansion or as its own mod	Decided. Part of the Mansion for now, nested in its jar, built as its own mod so it can be published alone at any time
E-16 The coloured light module's name and mod id	Coloured Lamps, colouredlamps

F. What Changed Between Versions

Area	Version 0.1	Version 0.2	Why
Dependencies	Optional compile-time integrations with voice chat and other mods	NeoForge only	Your dependency rule
Storage	Cache files, an index file and a journal in a world subfolder	level.dat and chunk NBT only, memories in the chunk they were captured from	Your storage rule
Settings	A server config file in the world and a client config file	Gamerules and vanilla options	Your storage rule
Chunks	Never saved, regenerated on every visit	Saved normally, rewritten before load when re-remembered	Fewer mixins, compatibility, lighting computed once
Infinite space	64-bit address space paged onto one level, with seamless transfers	One ordinary level, 1,024 tall. Vertical travel on a research track	Risk, complexity, and no canon need for an endless fall
Founder Library	Six authored rooms shipped as data	Numbers only, configurable to zero. The sea is a world's own earliest memories	Your rule against presets and templates
Elsewhere	Procedural generators of houses, shops, offices, pools, streets and hotels	Removed. Retrospective captures of the world's own builds and generated structures	A building generator by another name, against your rules
Canon Sites	Hand-built Main Chamber, Division 14, Kelowna Branch and more	Removed. The laws produce canon's situations from what the world supplies (chapter 24)	Authored content
Troffer	A hand-designed assembly with ballast and conduit blocks	A law, lights behind a diffuser on the tile's rhythm, from stock	Authored content
Glitch tells	Compass wobble, stick test, torch shimmer	Only canon's, things falling through and long objects widening the zone	Not shown in canon
Threshold	A fixed multiblock scripted so the sixth test succeeds	Pattern decided later, success rising with distortion, nothing scripted	A gimmick
Storms	Geomagnetic storms with a green aurora in the overworld sky	Removed	Decoration
Voices	Voice chat replayed by the Lifeform and through thin places	Removed. The Lifeform mimics sounds	Dependency rule, privacy and the charter
Doorway signs	A map redrawn as the Complex, standing in for the television	Removed	Not something canon shows
Async	Authored outposts, logs and paperwork	The players are Async. Their own cables, cameras and notes get remembered	Authored content
Dens	A procedurally composed melody	The stock's own captured jukebox or note block, or silence	Every sound from a real source
Display entities	Up to 256 per chunk and 6,000 in view	32 per chunk and 600 in view	Frame rate and tick rate
Acoustics	Room-graph acoustics from the start	Vanilla attenuation, room-graph acoustics later	Scope
Order of work	Infrastructure spikes first	The feel slice first, judged against canon frames	Every earlier version failed on feel, not on infrastructure

Version 0.3

Area	Version 0.2	Version 0.3	Why
Fidelity	Fresh copies faithful everywhere, falling with distance	Faithful where enclosed, streamlined where the capture saw the sky, by a starting generation from Minecraft's sky light	Your sky-exposure theory, which fits all seven canon cases, including the film's street that distance couldn't explain
Capture reach	The same in all open air	Drained in sky-exposed air, passing through glass	The bounce model, and the store's front copied through its glass doors
Streamlining	Not modelled	The diorama operator R0, clean base blocks from the game's recipes, one colour per face, clutter dropped, plants simplified, edges straightened	Canon's stage-set exteriors
Rebuilding	Geometry kept and bent until convergence at generation 7	Re-expression R10 from generation 1, always by 3, rooms rebuilt in the Mansion's image from their own materials and objects	Your re-expression idea, and canon's dark maze copy of Clark's store
Outdoor places	Rare only as a side effect	Four hard rules, openings only inside, the drain and a quarter cap, islands only, a 2 per cent test	Outdoor places filled every earlier version
Retrospective captures	Any built place, villages included	Only from inside a building	The same mistake as earlier versions, caught in review
Lights	Always lit	Keep the state they were captured in	The two dark copies of Clark's store, and canon's wording
Reference frames	None	Frames from canon, one Minecraft build and a fan project, for reference only	To show what the mechanism must reproduce

Version 0.4

Area	Version 0.3	Version 0.4	Why
Closure light	Night closures always red-lit	The light of the moment. The sun replaced by fittings keeping the colour and direction of its light at capture, from the game's time of day and its own sunset colour function	The dusk theory, from your sunset question. Both red copies have stars, low directional red and white street lamps
Floodlights	A few lamps behind red glass	Banks of lamps, invisible light blocks for the throw, a tint volume for colour	No Minecraft block lights beyond fifteen blocks
Coloured lamps	Not planned	A module after M2, a vanilla lamp with a colour component, your crafting recipe, exact coloured light, gated by a spike	Your request, and exact colour for the Complex's closures

Version 0.5

Area	Version 0.4	Version 0.5	Why
Coloured lamps packaging	An optional module, packaging undecided	A mod of its own, colouredlamps, nested in the Mansion's jar, its own namespace, one small API, ready to publish alone	Your decision to keep it in the Mansion for now and detachable later

Version 0.6

Area	Version 0.5	Version 0.6	Why
Outdoor test	Counted only space under a sky-closure surface	Counts all former outdoor space not yet re-expressed, whatever covers it	A street under a plain ceiling still reads as outdoors, as the film's street shows

Area	Version 0.5	Version 0.6	Why
Re-expression of outdoor space	Not specified	Rule O5. Former outdoor space becomes a room, walls of the Complex, a ceiling with the light arrangement, the ground kept as the floor, fronts kept as wall features	The mechanism that actually eats the outdoors, matching Mary's yard and the playground in canon

Version 0.7

Area	Version 0.6	Version 0.7	Why
Substitutes	Matched by role and colour from sprite colours	Same shape, inert, closest colour, then flattest texture, with a measured vanilla table for servers and captured biome colours for tinted blocks	Your points that a carpet changes floor height and that substitutes should be cheap and flat, plus servers having no textures
Monotony	Not handled	Breaking monotony prefers a different close block, from stock, installed mods or vanilla, and allows a merge when nothing fits	Your idea, with loss accepted as canon
Modded blocks	Mods' blocks used only when captured	Every installed mod's blocks are real candidates	Your point that installed mods are part of the world
Removed mods	Missing matter filled by role	Fingerprints in level.dat, palette rewrite before decoding, no holes	Your safeguard
Biome colours	Copies drawn in the Mansion biome's colours	Captured per cell, restored by a client colour hook	Copies must look like what was captured

Version 0.8

Area	Version 0.7	Version 0.8	Why
Human-made work	One standing rule among others	A mandate above every law, stated first, with the forbidden tells listed and a build check that fails on any breach	Your instruction that it's a must imposed above law, always
For a new builder	Assumed the conversations behind it	Chapter 0 with canon in brief, every rule, the history of failures and where everything is, and Part E with setup, formats, algorithms, verified API names and step-by-step milestones	So the bible alone is enough to build the mod
R1	THETA0 1.5, A 0.55	Minus 1.0 and 1.6	The old values contradicted test T-6.2
Floodlight colour in M0	Described as varying across the enclosure	Follows the camera until the coloured light module exists	The lightmap is one table for the whole view, checked in the sources
Lamps keeping their state	A mixin per vanilla lamp	One mixin on block changes, covering modded lamps	Generality
Determinism	Fixed-point maths	Doubles with StrictMath	Strict floating point since Java 17
Storage warnings	SavedData forbidden	Level attachments forbidden too	NeoForge saves them to a file of their own, checked in the sources
Open questions	Flood classes, I_MIN, roles, ports, substitution weights, families, R6 in Minecraft	All settled in chapter 38	Nothing left to guess